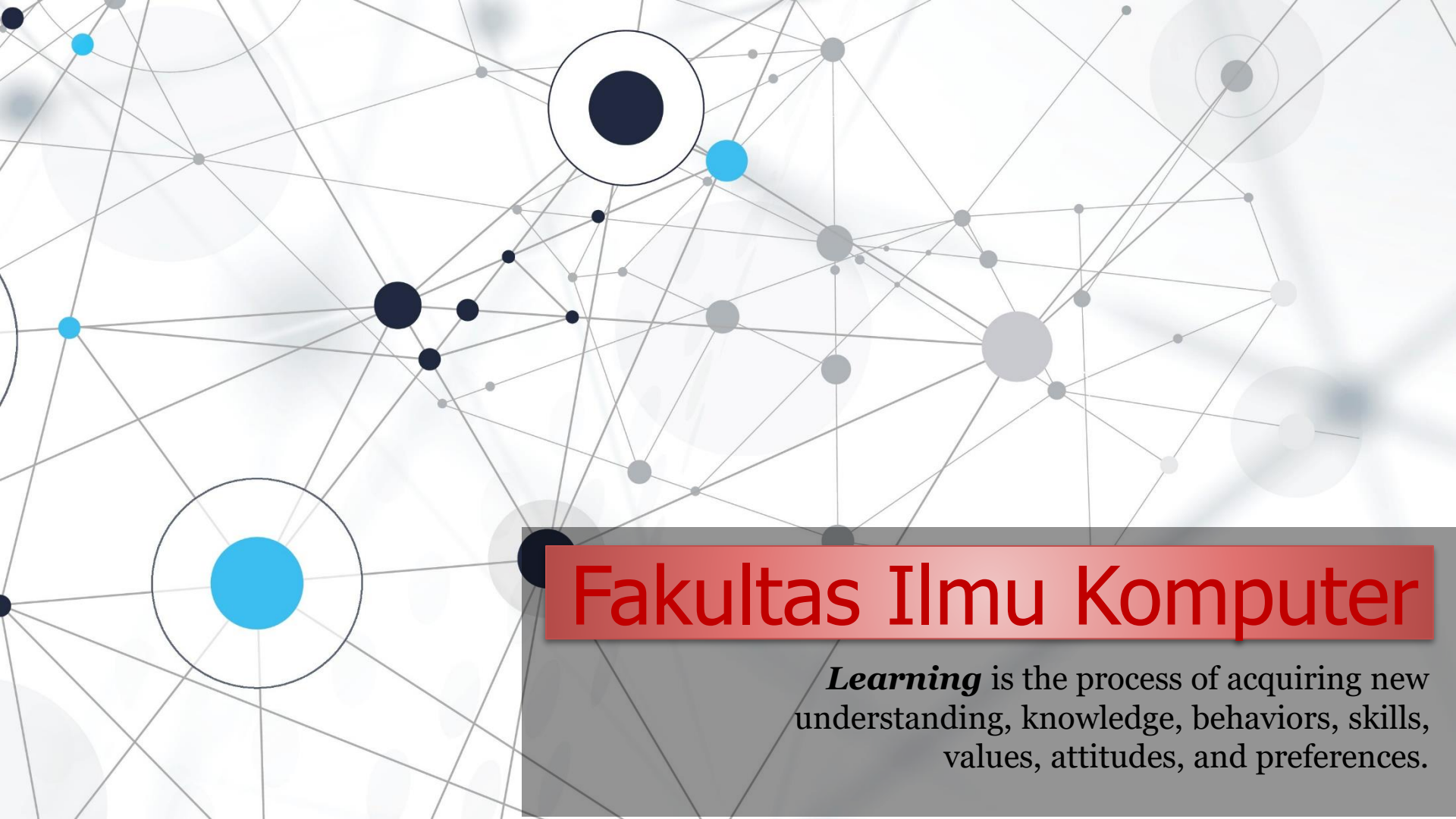


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Server Web in Cloud VPS Linux IdCloudHost

PHP Installation & Configuration in IDCloudHost (Linux, Apache, MySQL, PHP)

(IDCloudHost is a web hosting provider based in Indonesia)



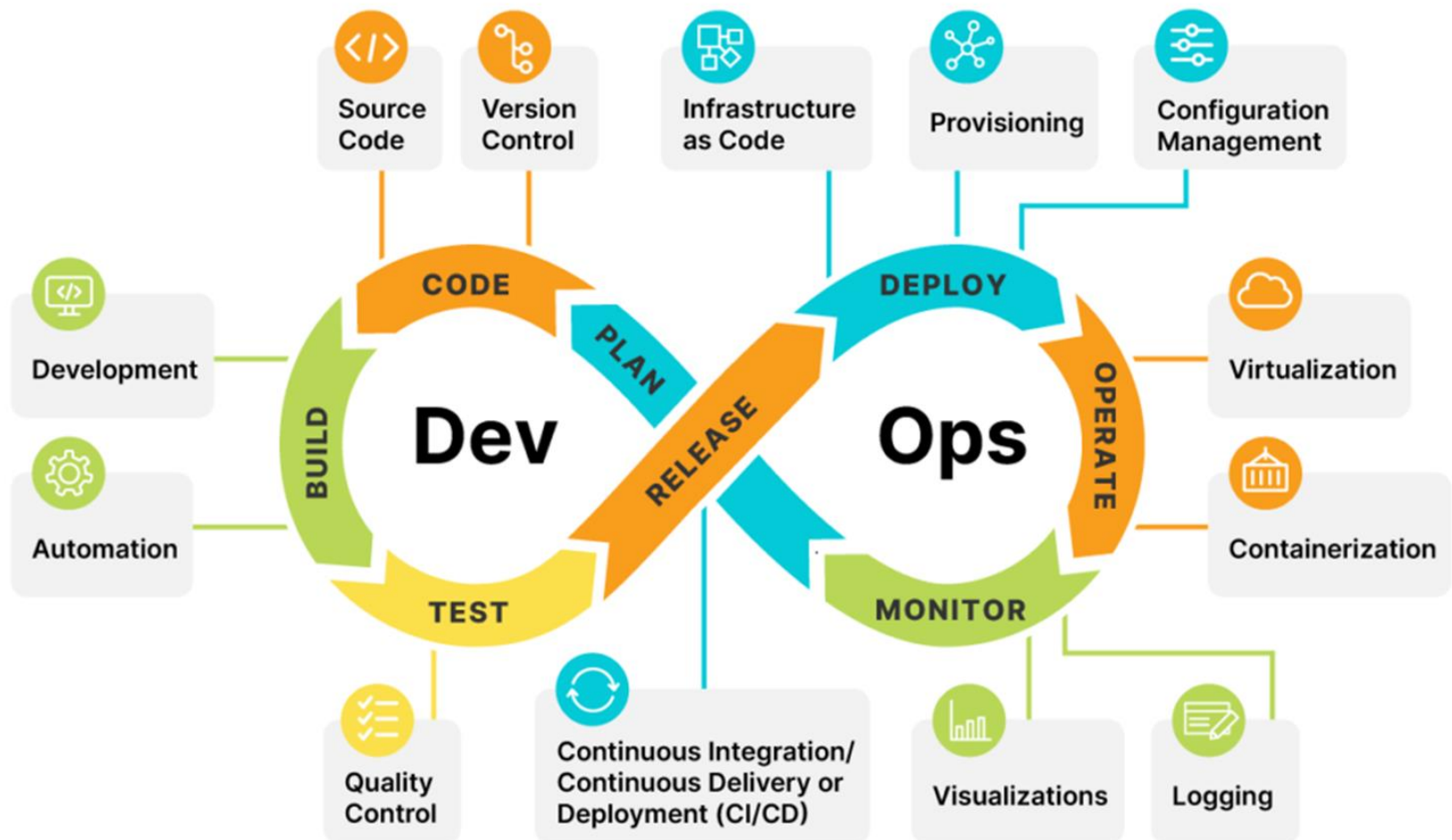
Welcome to *the* Back-end Web Development *course*



Mengapa back-end developer perlu belajar tentang cloud server (contoh: IDcloudHost)?

What is DevOps?

DevOps is a combination of two words, where DevOps stands for **development (Dev)** and **operations (Ops)**. DevOps is the unification (*penyatuan*) of divisions, processes and technologies in the phase of creating software, websites or applications.





Mengapa kita membutuhkan server jika dibandingkan dengan Firebase??? apa manfaat virtual server?

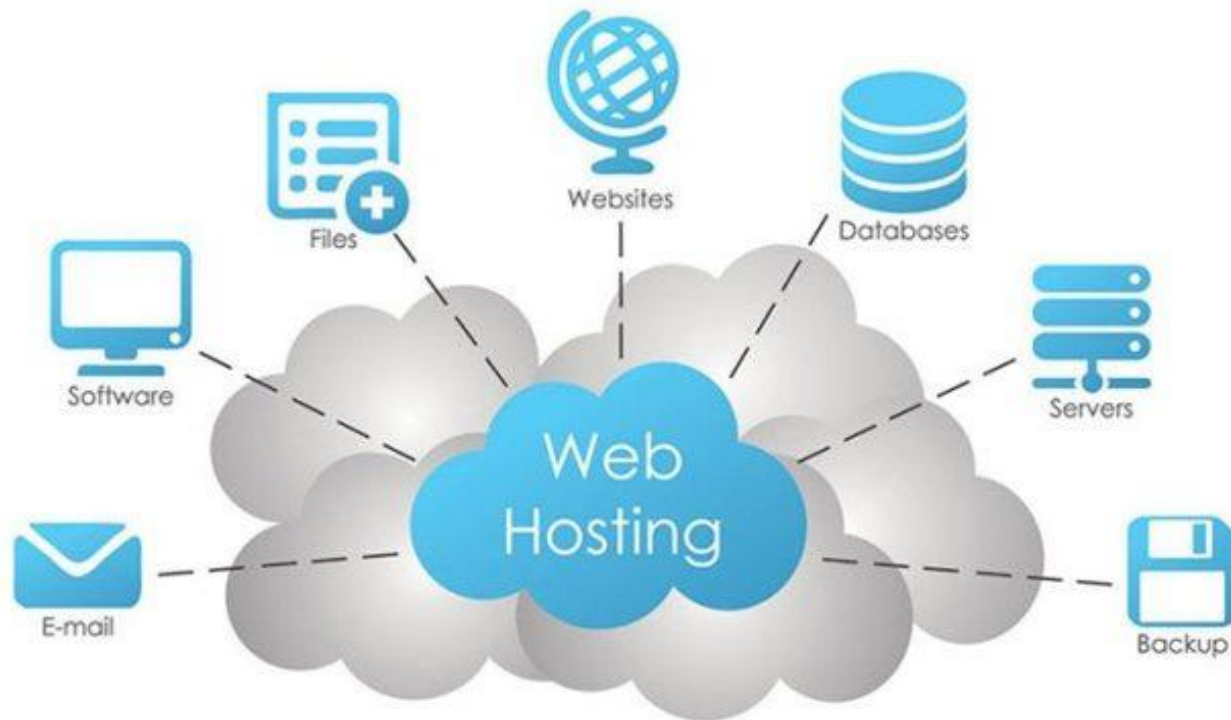
Mengapa kita membutuhkan server jika dibandingkan dengan Firebase??? apa manfaat virtual server?

- ❖ Server and Firebase have different roles and uses in application or website development.
- ❖ Firebase is an application development platform that allows users to develop applications easily without having to build their own server infrastructure.
- ❖ Firebase offers various services such as user authentication, database, file storage, and others that can be used in applications.
- ❖ **Firestore dirancang agar mudah digunakan dan dikonfigurasi, sehingga pengembang dapat fokus pada pengembangan aplikasi mereka tanpa perlu memikirkan infrastruktur server.**

Mengapa kita membutuhkan server jika dibandingkan dengan Firebase??? apa manfaat virtual server?

- ❖ Virtual server (*cloud server*) is a *physical server* that is divided into multiple virtual servers that are isolated from each other. The benefit of virtual servers is that we allow users to have greater **control over their hosting environment**, as users have *root access* to virtual servers. This allows users to install and configure software independently and allows applications to run in a more efficient manner.
- ❖ Virtual servers are typically used by **organizations that require a high level of control and security**, or by organizations that require greater server resources and are more focused on performance and scalability.
- ❖ Virtual servers and Firebase have different benefits and uses depending on user and organisational needs.
- ❖ Firebase is suitable for users who want to focus on application development without thinking about server infrastructure, while virtual servers are suitable for organisations that require a greater level of control over the hosting environment.

Server & Hosting Concepts



Servers and **hosting** are two important concepts in the world of information technology related to storing and presenting information on the internet. A computer system that stores data, applications, or services and makes them accessible to other users over the internet is called a server. On the other hand, hosting is the process of renting out space on a server to store websites and make them accessible to users over the internet.

Server & Hosting Concept



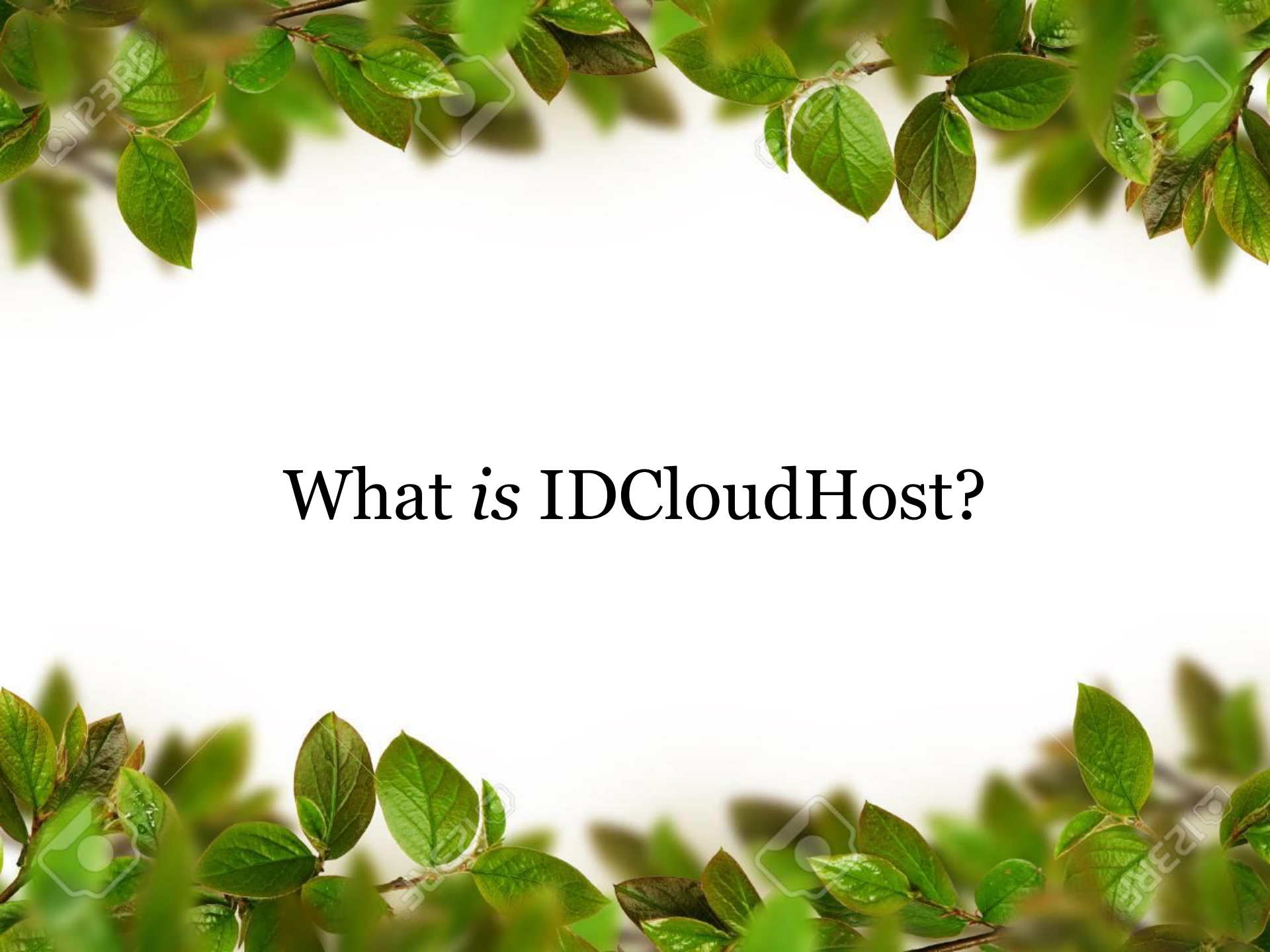
- ❖ **Servers** can be placed in various places such as data centers, corporate offices, or even in someone's home.
- ❖ Servers can be operated independently or can be used as part of a network.
- ❖ Some types of servers include web servers, email servers, database servers, game servers, and so on.



- ❖ **Hosting** is the process of rent out space on a server to store a website.
- ❖ In this process, users pay a monthly or annual fee to access the hosting service.
- ❖ Hosting can be shared hosting, where multiple websites are placed on the same server, or it can be dedicated hosting, where a single server is used exclusively for a single website or application.



- ❖ In the world of online business, choosing the right hosting is critical to maintaining availability, security, and speed of access to a website or application.
- ❖ Choosing the right server is also important to ensure the availability, security, and performance of the applications or services stored on it.

The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

What *is* IDCloudHost?

Login IDCloudHost

Sistem Presensi Referral Link:

<https://console.idcloudhost.com/referral/d2tzy4>

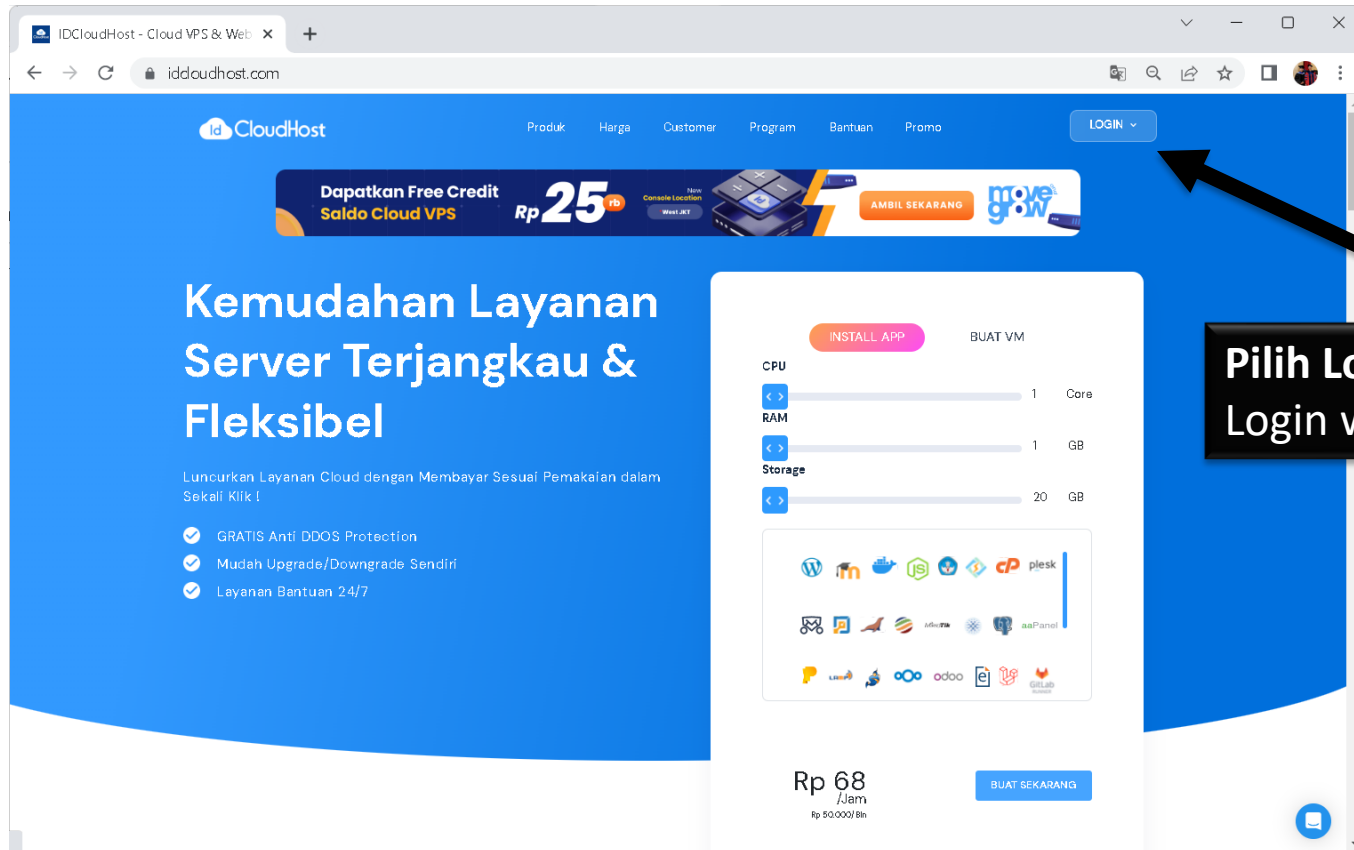
Referral Program

You can earn and give free credits by sharing your unique referral code.

- ✓ For every new user who signs up through the referral program and tops up at least Rp 0, you will receive Rp 25.000 worth of free credits.
- ✓ Every new user who signs up through the referral program and tops up at least Rp 0 will receive a bonus worth of Rp 25.000 worth credits.

Official Website 2023

Link: <https://console.idcloudhost.com/referral/d2tzy4>



Pilih Login User:
Login with “**Console Platform**”

IDCloudHost adalah perusahaan penyedia layanan *hosting web* yang berkantor pusat di Jakarta, Indonesia. Perusahaan ini menyediakan berbagai layanan hosting, termasuk **shared hosting**, **VPS hosting**, dan **dedicated server hosting**.

Signin *with* Google Account

Link: <https://console.idcloudhost.com/referral/d2tzy4>

Powerful Control Panel & API

One-Click Deploy

Powerful API

Easy Management

Id CloudHost

Log in

Email

Password [Show](#)

Log in

OR

Log in with Google

[Forgot password?](#)

[Create an account](#)

ISO/IEC 27001:2013 Certified

ISO 9001 Quality Management

Welcome to IDCloudHost.com

The screenshot displays the IDCloudHost.com dashboard in a web browser. The browser's address bar shows the URL `console.idcloudhost.com/user/semmy@unklab.ac.id/lab?redirects=1`. The dashboard header includes the IDCloudHost logo, a 'Dashboard' tab, and user information: 'Remaining credit Rp 120.925', 'Top Up Now' button, a notification bell, and the user name 'Semmy Taju'.

The left sidebar contains navigation links for COMPUTE, STORAGE, NETWORK, SERVICES, ACCESS, and BILLING, each with a corresponding icon. A chat bubble icon is located at the bottom of the sidebar.

The main content area features a 'Welcome to IDCloudHost.com' message with a paragraph: 'You are now part of a revolution by choosing a regional cloud provider collaborating globally with others to offer an easy-to-use alternative to a public cloud oligopoly. Keep your data local and secure. Fast and personal support. Clear pricing. Start your best project today!'. To the right of this message are links for 'API' and 'API Documentation', and a 'Manage API Keys' button.

On the right side of the dashboard, there is a promotional banner for 'High availability load balancer' with the text 'AVAILABLE NOW' and an arrow pointing right. The banner includes a 3D diagram of a load balancer distributing traffic to multiple application servers.

Below the welcome message, there is a 'Start Something New' section with three tabs: 'Core Services', 'App Catalog', and 'Managed Services'. The 'Core Services' tab is active. It displays a grid of service cards, each with an icon, a title, and a right-pointing arrow:

- Virtual Machine Compute
- Object Storage Storage
- Floating IP Network
- VPC Network Network
- Load Balancer Network (marked with a 'NEW' badge)
- Access Delegation Access
- API Token Access

A 'More Details' link with a dropdown arrow is located to the right of the 'Managed Services' tab.

Menu IDCloudHost.com

Terdapat beberapa menu yang ada di **IDCloudHost.com** di antaranya yaitu:

- 1) **Compute menu** at IDCloudHost.com contains options to select the type of hosting such as Shared Hosting, VPS Hosting, and Dedicated Server Hosting. Within this menu, we can choose the storage capacity, number of CPUs, RAM, and other resources that suit our needs.
- 2) **Storage menu** at IDCloudHost.com contains options to select storage types such as Cloud Storage, Backup Storage, and Object Storage. In this menu, we can choose the storage capacity and other storage features that suit our needs.
- 3) **Network menu** at IDCloudHost.com contains options to select network types such as Load Balancer, Firewall, and Public IP. In this menu, we can choose the network features that suit our needs.
- 4) **Services menu** at IDCloudHost.com contains options to select services such as Domain Registration, SSL Certificate, Email Hosting, and Website Builder. In this menu, we can choose the service that suits our needs.
- 5) **Access menu** at IDCloudHost.com contains options to access the control panel, SSH, FTP, and others. Within this menu, we can manage access and control over our hosting environment.
- 6) **Billings menu** at IDCloudHost.com contains billing and payment information for the hosting and other services we have chosen. Within this menu, we can manage the payments and access the billing information.



Follow the steps on the next slide!!!

What is LAMP Stack?



LAMP stack is a popular open-source web development platform. LAMP Stack development is popular web development methodology that allow web developers to build projects quickly, efficiently, and cost-effectively.

The LAMP Stack is a group of open-source software technologies that work together to run dynamic websites and applications. LAMP, which stands for **Linux**, **Apache**, **MySQL**, and **PHP**, allows to build websites quickly and easily without buying expensive third-party software licenses.

LAMP Stack Alternatives:

- ✓ **MEAN** stack (MongoDB, Express, Angular, Node. js)
- ✓ **LEMP** stack (Linux, NGINX, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **LAPP** stack (Linux, Apache, PostgreSQL, PHP)
- ✓ **LEAP** stack (Linux, Eucalyptus, AppScale, Python)
- ✓ **LLMP** stack (Linux, Lighttpd, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **XAMPP** stack (Cross-platform, Apache, MariaDB, PHP, Perl)

HTTP Status Codes Server

HTTP STATUS CODES

2xx Success

200 Success / OK

3xx Redirection

301 Permanent Redirect

302 Temporary Redirect

304 Not Modified

4xx Client Error

401 Unauthorized Error

403 Forbidden

404 Not Found

405 Method Not Allowed

5xx Server Error

501 Not Implemented

502 Bad Gateway

503 Service Unavailable

504 Gateway Timeout

HTTP Status Codes

Level 200

200: OK
201: Created
202: Accepted
203: Non-Authoritative Information
204: No content

Level 400

400: Bad Request
401: Unauthorized
403: Forbidden
404: Not Found
408: Conflict

Level 500

500: Internal Server Error
501: Not Implemented
502: Bad Gateway
503: Service Unavailable
504: Gateway Timeout
599: Network Timeout

An HTTP status code is an error code displayed by the server regarding the status of a client's request when accessing a web page. If the browser displays it, it means there is an issue with the originating server while attempting to process the request.

Apache MySQL PHP in Ubuntu 22.04

- ❖ **LAMP** stack stands for Linux, Apache, MySQL and PHP. LAMP is a collection of opensource applications that are commonly used together to create a Web Server or dynamic web hosting.
- ❖ With **LAMP**, we use a *server* with Linux as the operating system, Apache as the web server, MySQL to store the database and dynamic content created with the PHP programming language.



Key Differences of XAMPP vs LAMP???

The XAMPP full form is for (X) cross-platform, Apache, MySQL's MariaDB, Pearl, and PHP while the full form of LAMP is Linux, Apache, MySQL, and PHP. Both of them act as a local server where we can run our application and test its working.

A decorative border of green leaves and branches at the top of the slide.

1 - Linux as Operating System (*Virtual Private Server (VPS)*)

A decorative border of green leaves and branches at the bottom of the slide.

Create New Virtual Machine (2020)

Berikut adalah langkah-langkah untuk membuat Virtual Machine baru di IDCloudHost.com:

- 1) Login ke akun IDCloudHost.com.
- 2) Pilih menu **Compute** dan klik Virtual Machine.
- 3) Klik tombol Create Virtual Machine.
- 4) Pilih jenis Virtual Machine yang ingin kita buat, seperti Ubuntu atau CentOS, dan versi yang diinginkan.
- 5) Atur ukuran dan kapasitas penyimpanan yang kita inginkan.
- 6) Pilih jaringan yang akan digunakan dan pilih opsi IP Address. Kita dapat memilih IP Public atau IP Private.
- 7) Buat nama host dan pilih lokasi server yang akan digunakan.
- 8) Atur konfigurasi Virtual Machine seperti *username* dan *password*, *firewall*, dan akses SSH.

Setelah mengkonfigurasi Virtual Machine, klik tombol **Create Virtual Machine**. Selanjutnya, IDCloudHost.com akan memproses permintaan dan membuat Virtual Machine baru sesuai dengan konfigurasi yang ditentukan. Setelah selesai, kita dapat mengakses Virtual Machine dan menggunakannya untuk tujuan hosting atau pengembangan web.

Create New Virtual Machine (2023)

The screenshot shows the IDCloudHost console interface. The left sidebar contains navigation menus for COMPUTE, STORAGE, NETWORK, SERVICES, ACCESS, and BILLING. The main area displays the 'Create New Resource' page. The 'Type*' dropdown is set to 'Virtual Machine'. The 'OS*' dropdown is set to 'Ubuntu 22.04'. The 'Location*' dropdown is set to 'Indonesia'. The 'App Catalog' is also visible. The top bar shows the user's profile and remaining credit.

COMPUTE

myserver
103.59.95.245 ...
Monthly cost
Rp 100.000

+ Create New Resource

Create New Resource

Create a new Virtual Machine or choose something from the App Catalog to get your services running in minutes.

Type*

Virtual Machine

App Catalog

OS*

Ubuntu
22.04

Debian

Almalinux

Centos

Fedora

Opensuse

CloudLinux

Rocky Linux

Virtuozzo

VzLinux

RHEL

Windows

Location*

Indonesia

Singapore

Selected OS: Ubuntu 22.04

Cloud Server Size 2022

Rp 50.000 / month Rp 68 / hour 1 CPU 1 GB RAM 20 GB DISK	Rp 56.000 / month Rp 77 / hour 1 CPU 1 GB RAM 30 GB DISK	Rp 100.000 / month Rp 137 / hour 2 CPU 2 GB RAM 40 GB DISK	Rp 225.000 / month Rp 308 / hour 2 CPU 4 GB RAM 60 GB DISK	Rp 50.000 / month Rp 68 / hour CPU 1 1-16 GB RAM 1 1-8 GB DISK 20 20-1000
--	--	--	--	---

Selected Size: 2 CPU, 2 GB RAM, 40 GB Disk

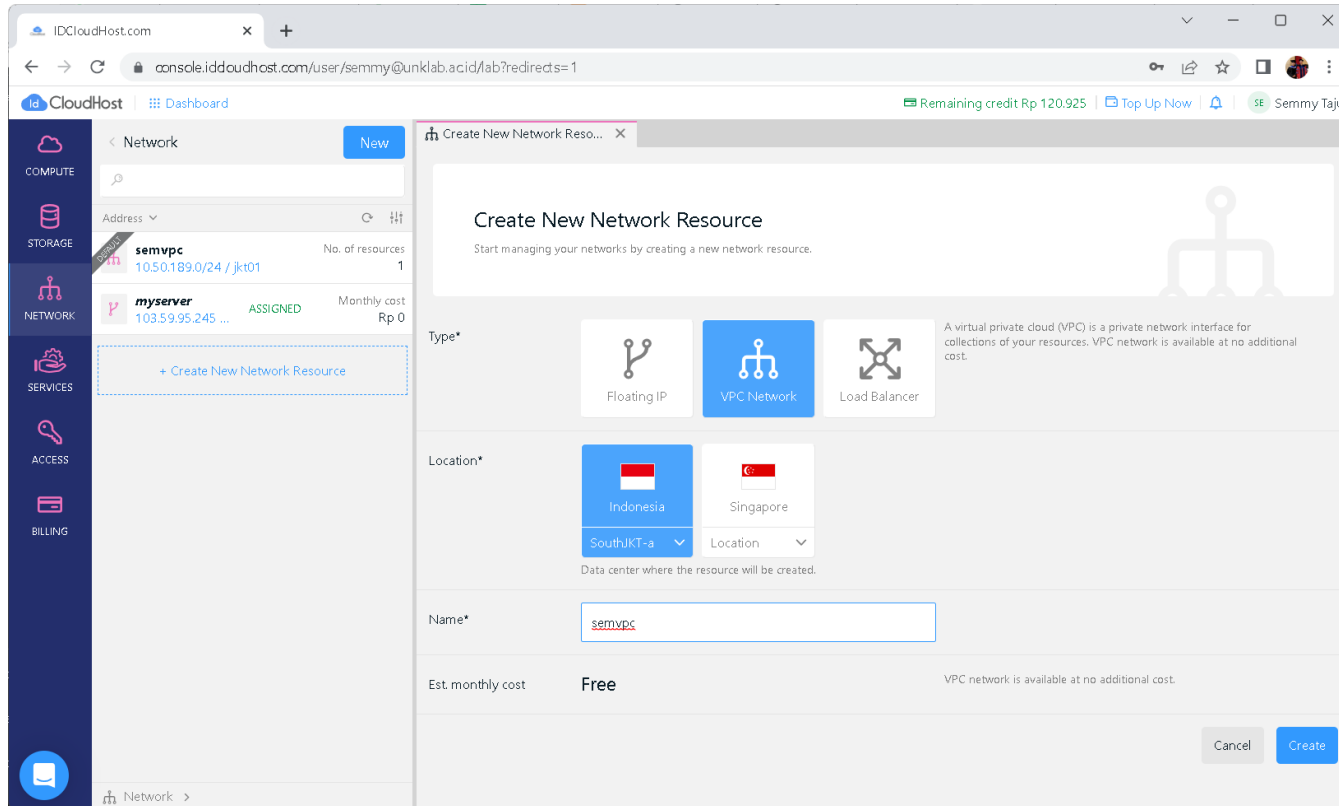
Size (*cloud server capacity*) di IDCloudHost.com merujuk pada ukuran Virtual Machine atau *cloud server* yang tersedia untuk dipilih saat membuat atau mengonfigurasi server. *Size* menentukan jumlah sumber daya yang akan dialokasikan untuk *server*, seperti jumlah CPU, memori, penyimpanan (*storage capacity*), dan *bandwidth*. Semakin besar *size*, semakin banyak sumber daya yang tersedia untuk digunakan oleh *server*.

Apakah membangun VPS di IDCloudHost dengan 2 cpu, 2 GB RAM, 40 GB storage cocok untuk aplikasi presensi di Unklab?
JIKA terdapat 3000 active students yang akan mengambil absen secara *online*?

Tergantung pada **kompleksitas** dan **intensitas** dari aplikasi absensi online yang digunakan.



Create Free VPC Network



Virtual Private Cloud (VPC) di IDCloudHost.com adalah layanan yang menyediakan jaringan *cloud virtual* yang eksklusif dan aman untuk digunakan oleh pelanggan. Dengan menggunakan VPC, pelanggan dapat membuat jaringan sendiri di *cloud* yang terpisah dari jaringan publik dan jaringan internal yang digunakan oleh IDCloudHost.com. VPC memungkinkan *users* untuk mengendalikan dan mengatur aliran data antara sumber daya *cloud* mereka dan internet, dan antara sumber daya *cloud* yang berbeda.

Top Up Your Account (DANA)

The screenshot shows the IDCloudHost console interface. A red arrow points from the 'Top Up Now' button in the top right corner to the 'Top Up Your Account' modal. The modal is titled 'Top Up Your Account' and contains the following fields:

- Amount (min. Rp 55.000):** A text input field with the value '100000' entered and underlined in red.
- Payment Method:** A dropdown menu with 'DANA' selected and underlined in red.
- Buttons:** 'Cancel' and 'Top Up' buttons at the bottom right.

Additional information in the modal includes: 'Your total charge will be Rp 111.000 (Rp 100.000 + 11% VAT)'. The background shows the user's account details for 'Semmy Taju' with a remaining credit of 'Rp 120.925'.

The 'Choose method' dialog is shown, listing various payment methods. A toggle switch for 'Only automated methods' is visible at the top right. The list includes:

- BNI Virtual Account
- BRIVA
- Maybank Virtual Account
- Ritel
- QRIS By ShopeePay
- OVO
- DANA

Each method is accompanied by its respective logo.

Create Virtual Machine (VPS)

The screenshot displays the IDCloudHost console interface for creating a new Virtual Machine (VPS). The main window shows the 'Create New Resource' form, which is partially filled out. The form includes the following fields and options:

- Public IP:** A checkbox labeled 'Create a public IPv4 address for the resource' is checked.
- VPC network*:** A dropdown menu shows 'semvpc' with the IP address '10.50.189.0/24'.
- Username*:** The text 'unklab' is entered.
- Password*:** The text 'abc123ABC' is entered, with a 'Hide' button next to it.
- Confirm password*:** The text 'abc123ABC' is entered, with a 'Hide' button next to it.
- SSH:** A text area for the SSH key pair is empty.
- Resource name*:** The text 'myserver' is entered.
- Est. monthly cost:** The cost is displayed as 'Rp 56.000 / month' and 'Rp 77 / hour'.
- Billing account*:** A dropdown menu shows 'Semmy Taju' with the payment method 'Pre-payment'.

At the bottom right of the form, there are 'Cancel' and 'Create' buttons. A progress bar at the bottom of the console indicates the VM is 'Building (myserver)... 4%'.

Virtual Server Ready !!!

The screenshot shows the IDCloudHost console interface. The browser address bar displays `console.idcloudhost.com/user/semmy@unklab.acid/lab`. The top navigation bar includes the IDCloudHost logo, a 'Dashboard' link, and user information: 'Remaining credit Rp 120.925', 'Top Up Now', a notification bell, and the user 'Semmy Taju'.

The left sidebar contains navigation links for COMPUTE, STORAGE, NETWORK, and SERVICES. The main content area is titled 'Compute' and features a 'New' button. Below the search bar, a resource named 'myserver' is listed with the IP address '103.59.95.245 ...' and a 'Monthly cost Rp 100.000'. A '+ Create New Resource' button is visible below the resource list.

The right sidebar, titled 'Dashboard', shows a 'Recently Modified' section with three items:

- myserver**: an hour ago • Virtual machine (STATUS: RUNNING)
- 103.59.95.245**: 2 days ago • Floating IP (STATUS: ASSIGNED)
- semvpc**: 2 days ago • VPC Network

This thumbnail screenshot shows the 'Compute' section of the IDCloudHost console. It displays a resource named 'Building (myserver)...' with a progress bar indicating 4% completion. The '+ Create New Resource' button is also visible below the resource list.

Cloud VPS Details

Perhatikan di bagian yang saya highlight warna merah.

The screenshot displays the IDCloudHost console interface. The left sidebar contains navigation links for COMPUTE, STORAGE, NETWORK, SERVICES, ACCESS, and BILLING. The main content area shows the details of a VPS named 'myserver'. A red box highlights the top action bar with buttons: Start, Stop, Force Stop, Reboot, Clone, SSH, Virtual Console, Resize, Change Password, Rescue Mode, Reinstall, and Delete. Another red box highlights the VPS configuration details, including Name (myserver), Type (Virtual Machine), OS (Ubuntu 22.04 LTS), Status (RUNNING), Public IPv4 (103.59.95.245), Private IPv4 (10.50.189.240), VPC network (semvpc), Processor (2 VCPU), Memory (2048 MB), and Boot disk (40 GB). A black arrow points to the Public IPv4 address. To the right, metadata is listed: UUID (455097a6-ae03-4821-9e01-cf42...), Username (semmy), MAC (52:54:00:ad:c2:d1), Location (SouthJKT-a (jkt01)), Created (19-03-2023 00:19:46 WITA), Last modified (21-03-2023 11:07:55 WITA), Billing account (Semmy Taju), and Est. monthly cost (Rp 100.000). The bottom section shows the Metrics tab with four graphs: CPU (%), Memory (MB), Disk I/O (MB/s), and Bandwidth (Mbit/s).

CloudHost | Dashboard

Remaining credit Rp 120.925 | Top Up Now | Semmy Taju

Compute

myserver

Monthly cost Rp 100.000

+ Create New Resource

myserver

Name: myserver

Type: Virtual Machine

OS: Ubuntu 22.04 LTS

Status: RUNNING

Public IPv4: 103.59.95.245

Private IPv4: 10.50.189.240

VPC network: semvpc

Processor: 2 VCPU

Memory: 2048 MB

Boot disk: 40 GB

UUID: 455097a6-ae03-4821-9e01-cf42...

Username: semmy

MAC: 52:54:00:ad:c2:d1

Location: SouthJKT-a (jkt01)

Created: 19-03-2023 00:19:46 WITA

Last modified: 21-03-2023 11:07:55 WITA

Billing account: Semmy Taju

Est. monthly cost: Rp 100.000

All Metrics Disks Networking Backups Snapshots

Metrics

CPU (%)

Memory (MB)

Disk I/O (MB/s)

Bandwidth (Mbit/s)

Access VPS *via* Virtual Console

The screenshot displays the IDCloudHost management interface in a web browser. The browser's address bar shows the URL `console.idcloudhost.com/user/semmy@unklab.acid/lab`. The page header includes the IDCloudHost logo, a 'Dashboard' link, and user information: 'Remaining credit Rp 120.925', a 'Top Up Now' button, a notification bell, and the user name 'Semmy Taju'.

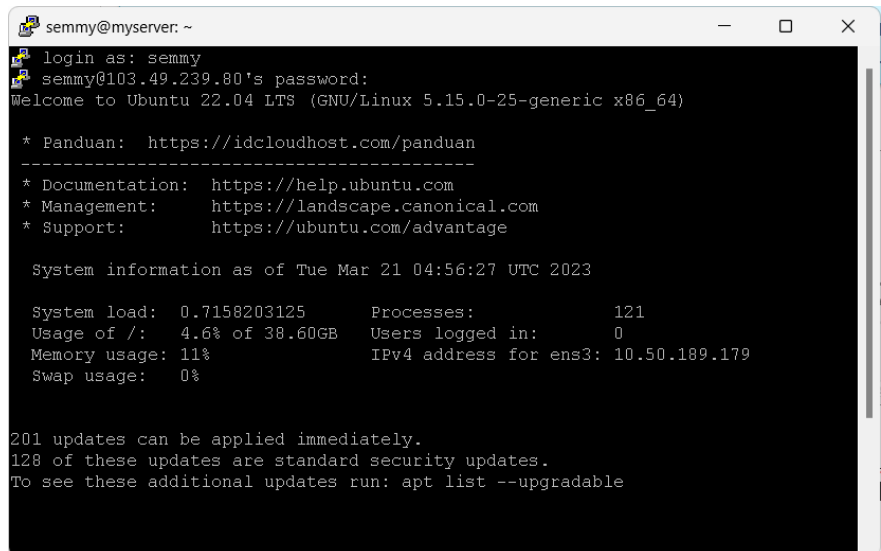
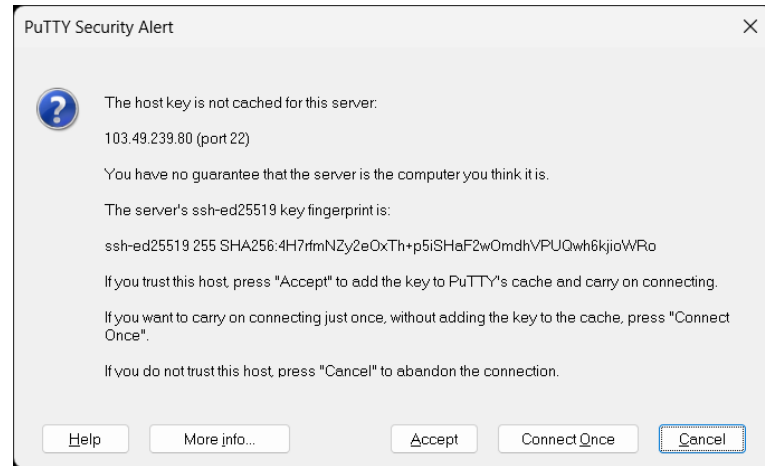
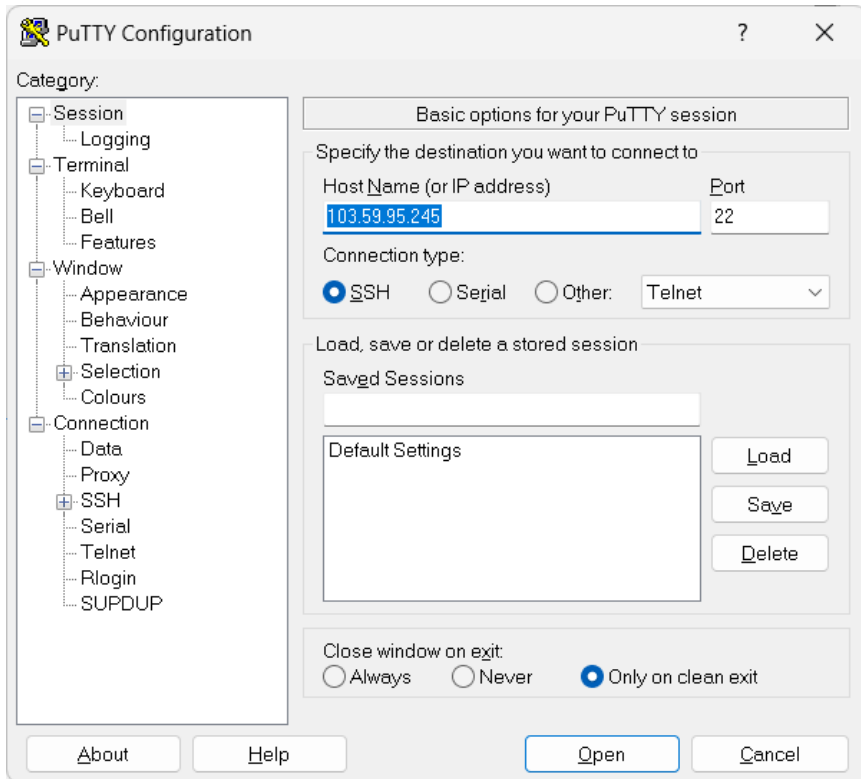
The left sidebar contains navigation links for COMPUTE, STORAGE, NETWORK, SERVICES, ACCESS, and BILLING. The main content area is titled 'Compute' and features a 'New' button. Below this, a list of resources is shown, with 'myserver' (IP: 103.59.95.245) selected. Its monthly cost is listed as 'Rp 100.000'. A '+ Create New Resource' button is visible below the resource list.

On the right, the 'myserver Console' window is open, showing a terminal session. The terminal output is as follows:

```
Ubuntu 22.04 LTS myserver tty1
myserver login: semmy
Password: _
```


Access VPS *via* Putty

Download Putty: <https://www.putty.org/>

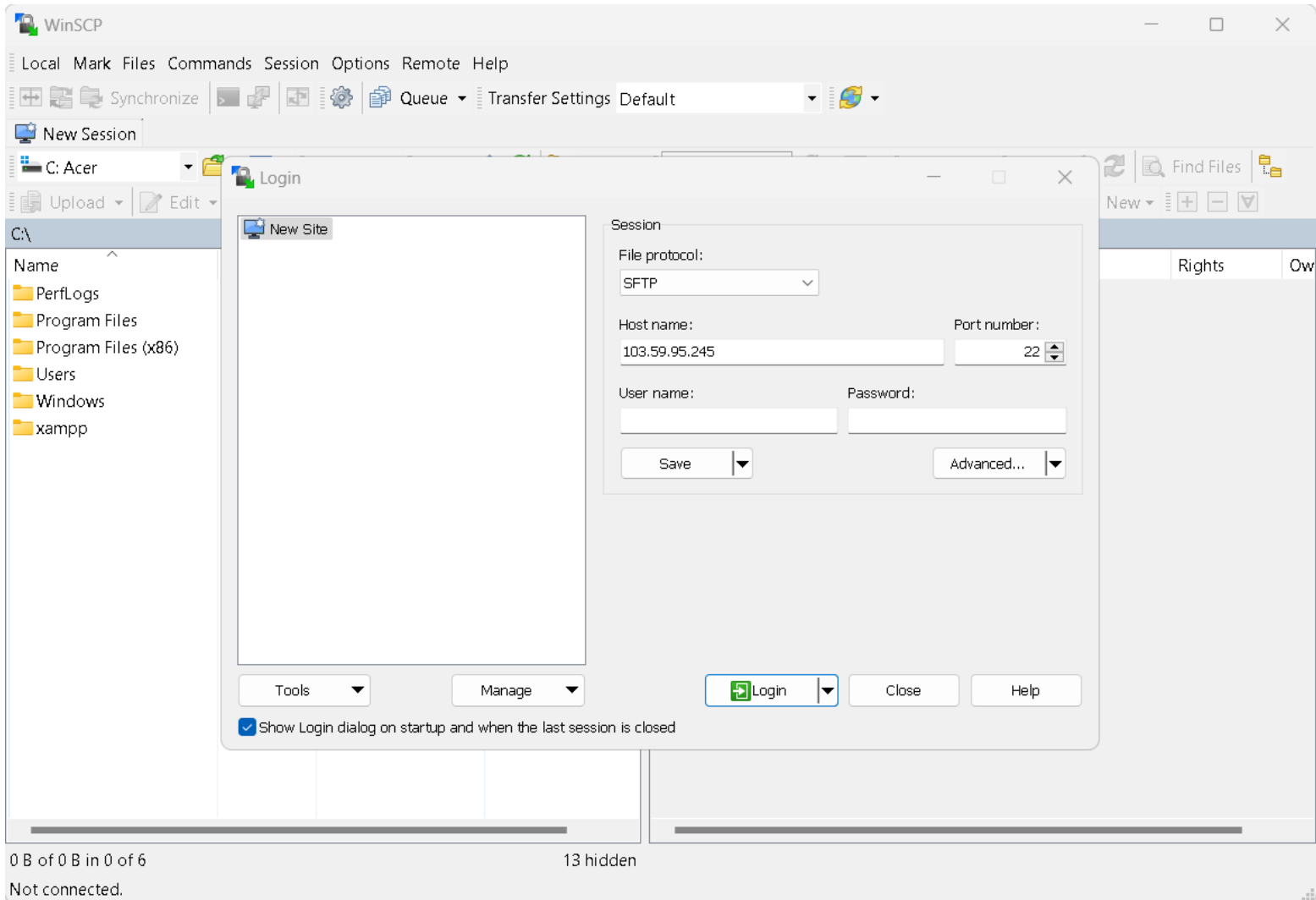


Port 22 pada Putty adalah nomor port default untuk protokol SSH (Secure Shell).

SSH adalah protokol jaringan yang aman untuk mengakses dan mengontrol sebuah server dari jarak jauh.

Access VPS *via* WinSCP

Download WinSCP: <https://winscp.net/eng/download.php>



Tujuan penggunaan winSCP & Putty?

WinSCP dan Putty adalah dua aplikasi yang umum digunakan dalam pengelolaan server dan akses ke jaringan komputer.

- 1) **WinSCP** adalah *software* client FTP (File Transfer Protocol) dan SFTP (Secure File Transfer Protocol) yang digunakan untuk mengirim dan menerima file antara komputer lokal dan server jarak jauh dengan cara yang aman. WinSCP menyediakan antarmuka grafis yang intuitif dan mudah digunakan untuk memudahkan pengguna dalam mengelola dan mengirim file ke server. WinSCP juga menyediakan fitur enkripsi yang aman untuk memastikan keamanan file saat dikirim.
- 2) **Putty** adalah *software* yang digunakan untuk mengelola *remote server* dengan protokol SSH (Secure Shell) dan Telnet. Putty menyediakan koneksi jarak jauh ke server dengan aman dan memungkinkan pengguna untuk melakukan perintah dan mengakses aplikasi di server dari komputer lokal. Putty juga menyediakan koneksi serial untuk mengakses perangkat jaringan seperti router dan switch.

The purpose of using WinSCP and Putty is **to manage and access remote servers in a secure and efficient manner.**

WinSCP and Putty provide an easy-to-use UI and powerful security features, so users can easily send and receive files and access servers remotely safely and quickly.



2 – Apache *as* Web Server

What is Apache2 ?



- ❖ **Apache2** adalah sebuah perangkat lunak *server web* yang bersifat **open-source** dan gratis.
- ❖ Perangkat lunak ini **digunakan untuk menghosting dan mengelola situs web di server.**
- ❖ Apache2 is becoming one of the **most popular** and widely used **web server** software worldwide, with many being used to manage websites and web applications on *servers*.
- ❖ Apache2 is developed by the Apache Software Foundation and is available for various platforms, including Linux, macOS, and Windows.

Install Apache2 Using Putty

[1] Change hak akses *superuser* atau *root*:

➤ ***sudo su***

[2] Perbaharui paket yang tersedia ke versi terbaru ubuntu dengan perintah:

➤ ***sudo apt update***

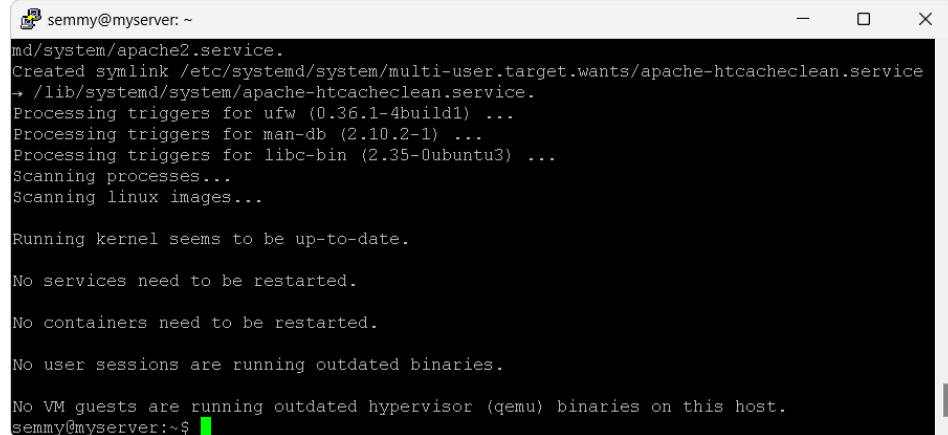
➤ ***sudo apt upgrade***

[3] Install Apache di linux ubuntu 22.04, dengan menjalankan perintah berikut:

➤ ***sudo apt install apache2***

Keterangan:

- 1) Perintah ***sudo*** adalah singkatan dari "*Superuser Do*". Perintah ini digunakan untuk memberikan akses root atau superuser pada pengguna yang tidak memiliki izin tersebut.
- 2) Perintah ***su***, di sisi lain, adalah singkatan dari "*Substitute User*". Perintah ini digunakan untuk beralih ke akun pengguna lain tanpa harus keluar dari sesi terminal saat ini.
- 3) Perintah ***apt*** adalah singkatan dari "*Advanced Packaging Tool*". Perintah *apt* merupakan sebuah tool pada sistem operasi Debian dan turunannya, seperti Ubuntu, yang digunakan untuk mengelola paket software, yaitu menginstal, menghapus, atau memperbarui paket software pada sistem operasi.



```
semmy@myserver: ~  
md/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service  
→ /lib/systemd/system/apache-htcacheclean.service.  
Processing triggers for ufw (0.36.1-4build1) ...  
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for libc-bin (2.35-0ubuntu3) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
semmy@myserver:~$
```

Possible Error [#1]

Could not get lock
/var/lib/dpkg/lock-frontend.
It is held by process 5044 (*unattended-upgr*)

Hapus penguncian paket secara manual dengan menjalankan perintah:

➤ *sudo rm /var/lib/dpkg/lock-frontend*

Safely remove the lock files using the commands below:

- *sudo kill -9 <process_id>*
- *sudo rm /var/lib/apt/lists/lock*
- *sudo rm /var/cache/apt/archives/lock*
- *sudo rm /var/lib/dpkg/lock*

After that, reconfigure the packages:

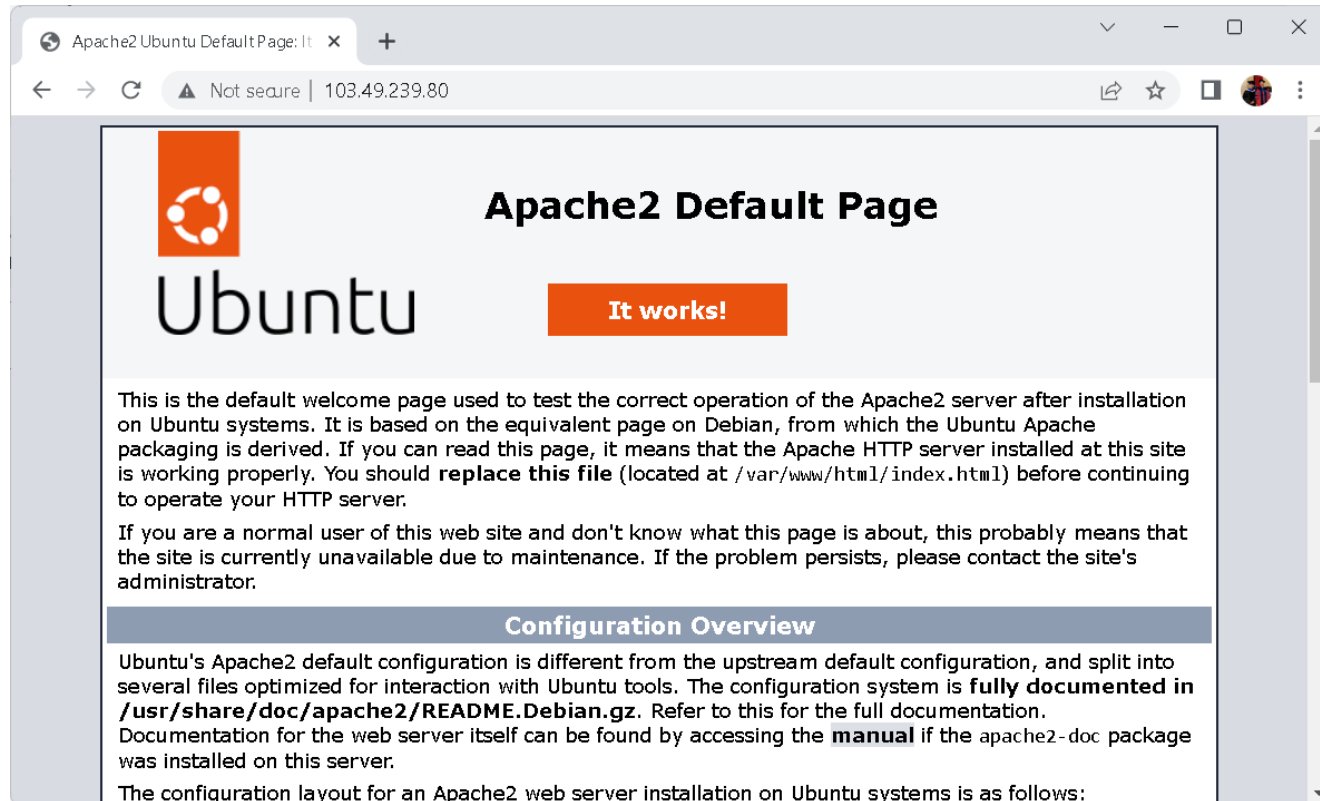
➤ *sudo dpkg --configure -a*

Possible Error [#2]

Could not connect to mirrors.idcloudhost.com:80 (103.15.226.185). - connect (113: No route to host)

- 1) Pastikan bahwa koneksi internet pada server berfungsi dengan baik.
- 2) Jalankan perintah *ping mirrors.idcloudhost.com* untuk memastikan bahwa server dapat dicapai dari server yang kita buat. Jika tidak ada respons, itu mungkin menandakan masalah jaringan pada sisi server atau masalah konfigurasi DNS.
> **ping mirrors.idcloudhost.com**
- 3) Jika server dapat diakses melalui ping, jalankan perintah *sudo apt-get update* untuk memperbarui daftar paket *apt*.
> **sudo apt-get update**
> **sudo apt-get install apache2**
- 4) Jika masih terjadi kesalahan, coba tambahkan opsi *--fix-missing*, seperti yang disarankan oleh pesan kesalahan.
> **sudo apt-get install apache2 --fix-missing**

Open Browser & Public IP Address



Replace this file (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

Untuk memeriksa apakah Apache2 sudah berjalan, ketikkan alamat **http://public_ip_address/** di browser. Jika muncul halaman "**It works!**", maka Apache2 sudah berhasil diinstall dan berjalan.

Check Apache Error

```
sudo nano /var/log/apache2/error.log
```

How To Install Linux, Apache, MySQL, PHP (LAMP) stack on Ubuntu 20.04

<https://www.digitalocean.com/community/tutorials/how-to-install-linux-apache-mysql-php-lamp-stack-on-ubuntu-20-04>

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3 - Directory “**/var/www/html/**”

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`/var/www/html/` vs `public_html`

- ❖ Dalam konteks Apache2 pada sistem Linux, `/var/www/html/` adalah direktori default yang digunakan untuk menyimpan konten website yang di-hosting di server. Sedangkan, `public_html` adalah sebuah direktori spesifik yang terletak dalam direktori home (berada di `/home/username/`) dari pengguna yang memiliki akun di server, dan digunakan sebagai lokasi default untuk menyimpan konten website yang terkait dengan pengguna tersebut.
- ❖ Jadi, perbedaan utama antara `/var/www/html/` dan `public_html` adalah lokasi direktorinya dan siapa yang memiliki akses ke direktori tersebut. Direktori `/var/www/html/` biasanya dikelola oleh administrator server dan diakses oleh pengguna dengan hak akses administrator, sedangkan `public_html` dimiliki oleh pengguna dengan akun di server dan dapat diakses oleh pengguna tersebut. Hal ini memungkinkan pengguna untuk memiliki akses eksklusif ke konten website mereka tanpa mempengaruhi konten website pengguna lain.
- ❖ Namun, pada dasarnya keduanya memiliki fungsi yang sama yaitu sebagai **lokasi untuk menyimpan konten website**. Pengguna dapat memilih untuk menggunakan salah satu dari direktori tersebut tergantung pada kebutuhan dan preferensi mereka. Beberapa aplikasi web atau CMS mungkin memiliki konfigurasi khusus yang memerlukan pengguna untuk menyimpan konten website di direktori `public_html`.

Hak Akses Direktori /var/www/html/

Disarankan agar hanya pengguna dengan hak akses administrator (biasanya dengan akun "*root*") yang dapat mengakses direktori **/var/www/html/** pada sistem Linux/Ubuntu yang menjalankan Apache2. Hal ini karena direktori tersebut merupakan bagian dari **root directory** atau direktori induk yang memerlukan hak akses administrator untuk diakses dan dimodifikasi.



Create User “public_html” Directory

- ❖ Perintah ***sudo a2enmod userdir*** digunakan untuk mengaktifkan modul "userdir" pada server Apache2. Modul ini memungkinkan pengguna untuk membuat direktori ***public_html*** di direktori *home* mereka, dan menggunakan direktori tersebut sebagai lokasi *default* untuk menyimpan konten *website* yang terkait dengan pengguna tersebut.
- ❖ Dengan mengaktifkan modul "userdir", pengguna dapat mengakses konten *website* mereka dengan alamat ***http://[alamat-ip-server]/~[nama-pengguna]/*** atau ***http://localhost/~[nama-pengguna]/***.
- ❖ Namun, sebelum menggunakan perintah ***sudo a2enmod userdir***, pastikan bahwa modul "userdir" belum diaktifkan pada *server* Apache2. Untuk memeriksa apakah modul "userdir" sudah aktif, pengguna dapat menggunakan perintah ***sudo a2query -m userdir***.
- ❖ Jika modul "userdir" belum aktif, maka pengguna dapat menggunakan perintah ***sudo a2enmod userdir*** untuk mengaktifkannya. Setelah modul "userdir" aktif, pengguna dapat membuat direktori ***public_html*** di direktori *home* mereka dan menambahkan konten *website* yang diinginkan ke direktori tersebut.
- ❖ Namun, perlu diingat bahwa pengguna harus berhati-hati saat mengaktifkan modul "userdir" pada *server* Apache2, karena hal ini dapat meningkatkan risiko keamanan *server*. Oleh karena itu, sangat disarankan untuk **melakukan konfigurasi keamanan yang tepat dan membatasi akses ke direktori *public_html*** agar hanya dapat diakses oleh pengguna yang berwenang saja.

Create User “public_html” Directory

Jalan perintah di bawah ini untuk membuat User “public_html” Directory:

Change hak akses superuser atau root:

➤ *sudo su*

Check modul "userdir" sudah aktif pada server Apache2:

➤ *sudo a2query -m userdir*

Mengaktifkan modul "userdir" pada server Apache2:

➤ *sudo a2enmod userdir*

Me-restart atau memulai ulang service Apache2:

➤ *sudo systemctl restart apache2*

Create direktori *public_html* di direktori home pengguna "semmy":

➤ *sudo mkdir /home/**semmy**/public_html*

Mengubah hak akses dan kepemilikan direktori:

➤ *sudo chown **semmy:semmy** /home/**semmy**/public_html*

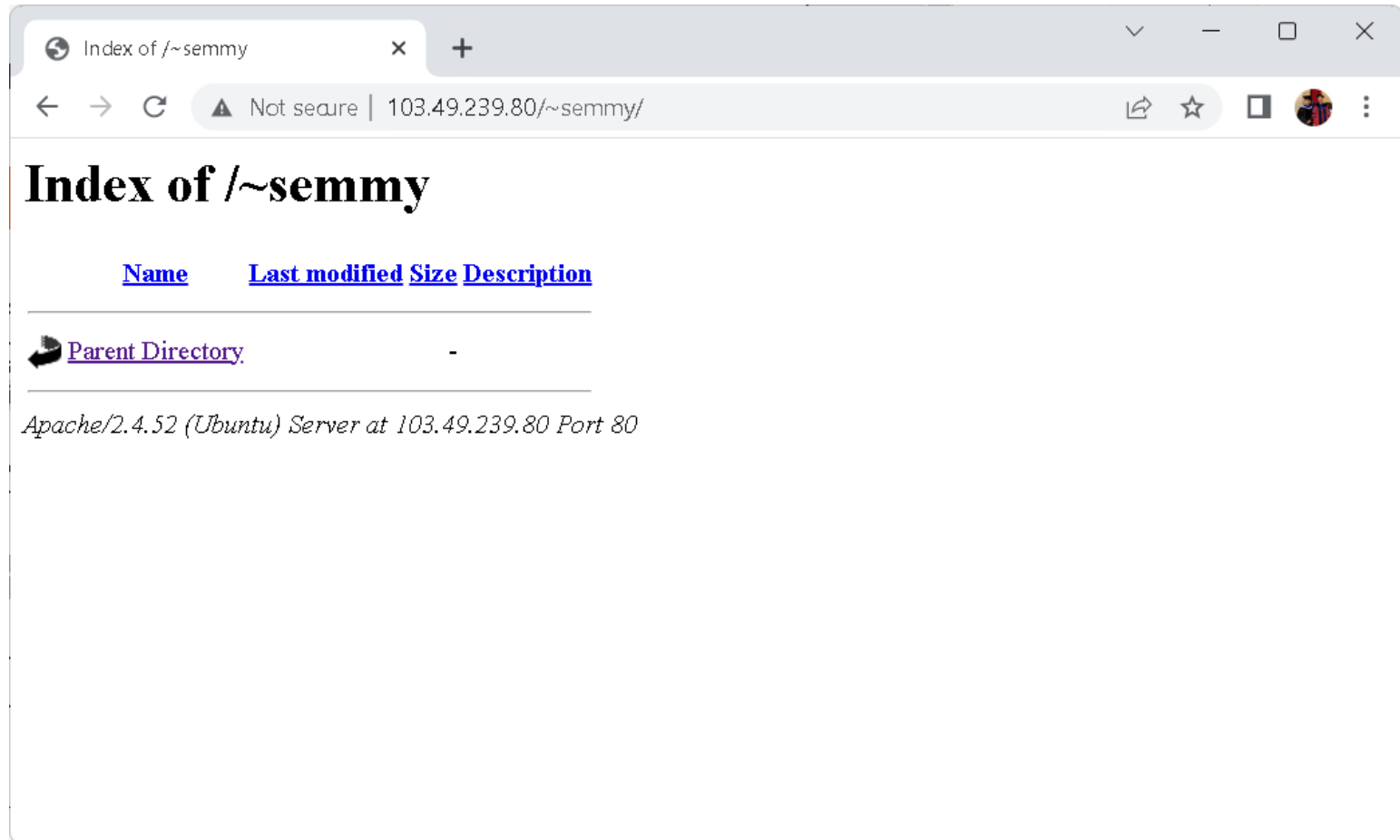
➤ *sudo chmod -R 755 /home/**semmy**/public_html*

➤ *sudo chmod -R 755 /home/**semmy***

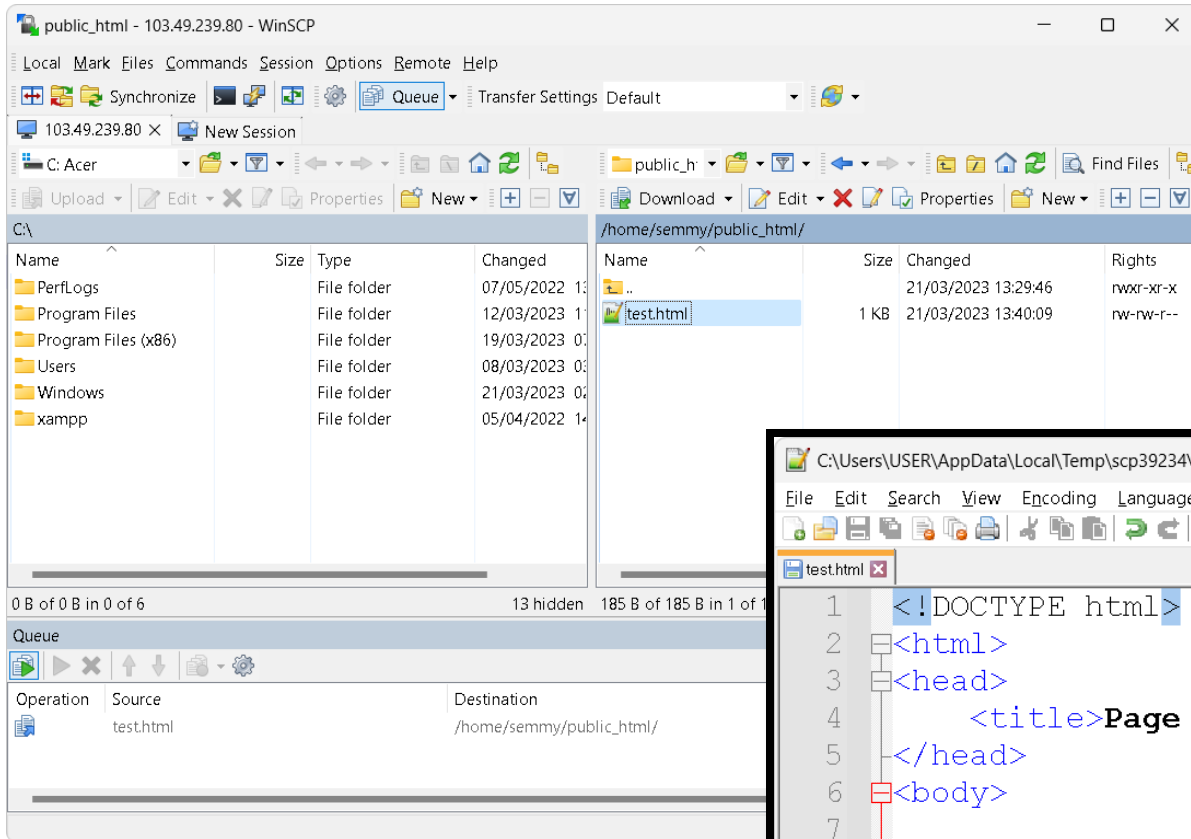
Keterangan:

- 1) Perintah **sudo** adalah singkatan dari "Superuser Do".
- 2) Perintah **su**, di sisi lain, adalah singkatan dari "Substitute User".

Testting Public_Html *in* User Dir



Create HTML File in “public_html”



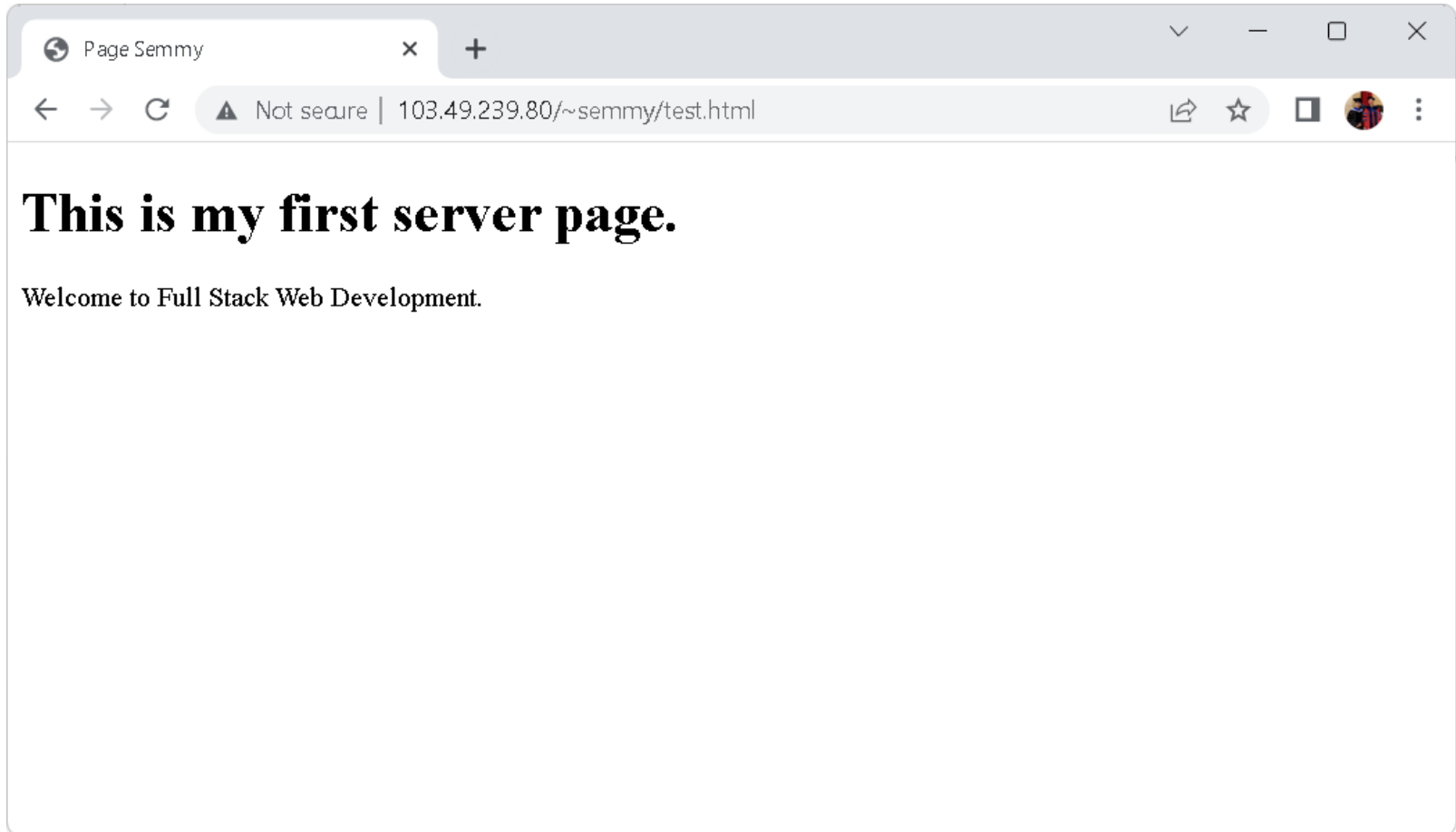
Buka WinSCP dan create sebuah file HTML dengan code seperti dibawah ini.

The screenshot shows the Notepad++ editor with the title bar "C:\Users\USER\AppData\Local\Temp\scp39234\home\semmy\public_html\test.html - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and Help. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Page Semmy</title>
5 </head>
6 <body>
7
8     <h1>This is my first server page.</h1>
9     <p>Welcome to Full Stack Web Development.</p>
10
11 </body>
12 </html>
```

The status bar at the bottom shows "length : 185 lines : 12 Ln : 9 Col : 41 Pos : 161 Windows (CR LF) UTF-8 INS".

Access File *via* Browser





4 - MySQL as Database Server

(Install MySQL di Ubuntu Server 22.04)

What is MySQL ?

MySQL adalah aplikasi untuk manajemen database RDBMS (Relational Database Management System) yang menggunakan bahasa SQL. MySQL digunakan untuk menambah, mengakses, dan mengolah data yang disimpan dalam database komputer.

Install MySQL in Ubuntu Server

1. Untuk memulai instal MySQL bisa dimulai dengan perintah:

➤ ***sudo apt install mysql-server***

2. Selanjutnya aktifkan service MySQL di Ubuntu 22.04 dengan perintah:

➤ ***sudo systemctl start mysql.service***

3. Untuk menambahkan keamanan pada MySQL database, kita bisa menjalankan perintah ***mysql_secure_installation*** yang akan menambahkan beberapa pilihan seperti validasi ***keamanan password***, tidak memberikan akses untuk *remote login* dan membuang sample user.

Disini ada beberapa tahap yang harus kita lakukan, dimulai dari masuk ke MySQL prompt dengan mengetikkan perintah berikut:

➤ ***sudo mysql***

➤ **mysql>** ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '***ganti_password_kita***';

Kemudian keluar dari MySQL prompt

mysql> exit

Install MySQL in Ubuntu Server

[4] Setelah metode autentikasi untuk *user root* kita ubah, kemudian kita jalankan perintah berikut:

➤ **sudo mysql_secure_installation**

NOTE:

ADA BEBERAPA PERTANYAAN YANG AKAN DI JAWAB, BAGIAN INI HARUS DILAKUKAN KARENA USER ROOT'S PASSWORD HARUS DI GANTI PADA BAGIAN INI.

[5] Setelah proses **mysql_secure_installation** selesai. Selanjutnya, kita bisa mengembalikan metoda autentikasi *user root* ke semula yaitu *auth_socket* dengan beberapa perintah seperti berikut:

> *sudo su* (masuk ke server)

> **mysql -u root -p**

..(masukan password anda)

mysql> ALTER USER 'root'@'localhost' IDENTIFIED WITH auth_socket;

[6] Selanjutnya, kita bisa melihat status MySQL service dengan perintah:

➤ **systemctl status mysql.service**

Expected Output

 semmy@myserver: ~[illegible]

Pada tahap awal silahkan masuk ke MySQL server:

- sudo mysql
- mysql -u root -p

```
mysql> CREATE DATABASE db_unklab;
mysql> CREATE DATABASE db_unklab;
Query OK, 1 row affected (0.78 sec)

mysql> |
```

mysql> SHOW DATABASES;

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| db_attendance |
| db_unklab |
| information_schema |
| mysql |
| performance_schema |
| sys |
| testadmin |
+-----+
7 rows in set (0.00 sec)
```

mysql> use db_unklab;

```
mysql> use db_unklab;
Database changed
mysql> |
```

Show struktur table:

mysql> DESCRIBE tbl_customer;

```
CREATE TABLE nama_tabel (
    id INT AUTO_INCREMENT PRIMARY KEY,
    nama VARCHAR(50),
    umur INT,
    alamat VARCHAR(255)
);
```

SETELAH ITU BUAT TABEL DAN ISI DATA PADA TABEL YANG TELAH DIBUAT.

mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));

mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),

-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),

-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),

-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),

-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```
mysql> CREATE TABLE tbl_students (reg_number VARCHAR(255), nim_number VARCHAR(255), email VARCHAR(255), fullname VARCHAR(255), password VARCHAR(255), created_at VARCHAR(255), updated_at VARCHAR(255));
Query OK, 0 rows affected (2.46 sec)
```

mysql> INSERT INTO tbl_students (reg_number, nim_number, email, fullname, password, created_at, updated_at) VALUES

-> ('1001', '1234567890', 'john@example.com', 'John Doe', 'password123', NOW(), NOW()),
-> ('1002', '0987654321', 'jane@example.com', 'Jane Smith', 'password456', NOW(), NOW()),
-> ('1003', '2468013579', 'bob@example.com', 'Bob Johnson', 'password789', NOW(), NOW()),
-> ('1004', '1357902468', 'alice@example.com', 'Alice Brown', 'passwordabc', NOW(), NOW()),
-> ('1005', '3141592653', 'jim@example.com', 'Jim Black', 'passworddef', NOW(), NOW());

```
Query OK, 5 rows affected (0.28 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

mysql> |

Insert satu baris data ke dalam tabel

```
INSERT INTO nama_tabel (nama_kolom1, nama_kolom2, nama_kolom3)  
VALUES ('nilai1', 'nilai2', 'nilai3');
```

Contoh insert table

```
INSERT INTO tbl_mahasiswa (nim, nama, jurusan) VALUES ('123456', 'John  
Doe', 'Teknik Informatika');
```

```
MySQL 8.0 Command Line Client  
mysql> CREATE TABLE Student (  
->   Stud_ID int AUTO_INCREMENT PRIMARY KEY,  
->   Name varchar(45) DEFAULT NULL,  
->   Email varchar(45) NOT NULL UNIQUE,  
->   City varchar(25) DEFAULT NULL  
-> );  
Query OK, 0 rows affected (0.88 sec)  
  
mysql> INSERT INTO Student(Stud_ID, Name, Email, City)  
-> VALUES (1,'Stephen', 'stephen@javatpoint.com', 'Texax'),  
-> (2, 'Joseph', 'Joseph@javatpoint.com', 'Alaska'),  
-> (3, 'Peter', 'Peter@javatpoint.com', 'california');  
Query OK, 3 rows affected (0.18 sec)  
Records: 3  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Student;  
+-----+-----+-----+-----+  
| Stud_ID | Name   | Email                               | City      |  
+-----+-----+-----+-----+  
|      1 | Stephen | stephen@javatpoint.com | Texax    |  
|      2 | Joseph  | Joseph@javatpoint.com  | Alaska   |  
|      3 | Peter   | Peter@javatpoint.com   | california |  
+-----+-----+-----+-----+
```

Uninstall MySQL Server

1. Hapus MySQL Server

- `sudo apt-get purge mysql-server mysql-client mysql-common mysql-server-core-* mysql-client-core-*`

2. Hapus Semua Direktori dan File Terkait MySQL

- `sudo rm -rf /etc/mysql`
- `sudo rm -rf /var/lib/mysql`
- `sudo rm -rf /var/log/mysql*`
- `sudo rm -rf /var/log/mysql`

(Optional) Jika pernah menjalankan mysql di user root atau user lain:

- `sudo rm -rf ~/mysql`

4. Cek Apakah Masih Ada Proses MySQL Berjalan (Optional)

- `ps aux | grep mysql`

Kalau ada proses sisa, kill:

- `sudo kill -9 <PID>`

6. (Optional) Reboot

- `sudo reboot`

Possible Error

Jika tidak terjadi error, skip bagian ini.

Could not get lock
/var/lib/dpkg/lock-frontend.
It is held by process 5044
(*unattended-upgr*)

Hapus penguncian paket secara manual dengan menjalankan perintah:

➤ *sudo rm /var/lib/dpkg/lock-frontend*

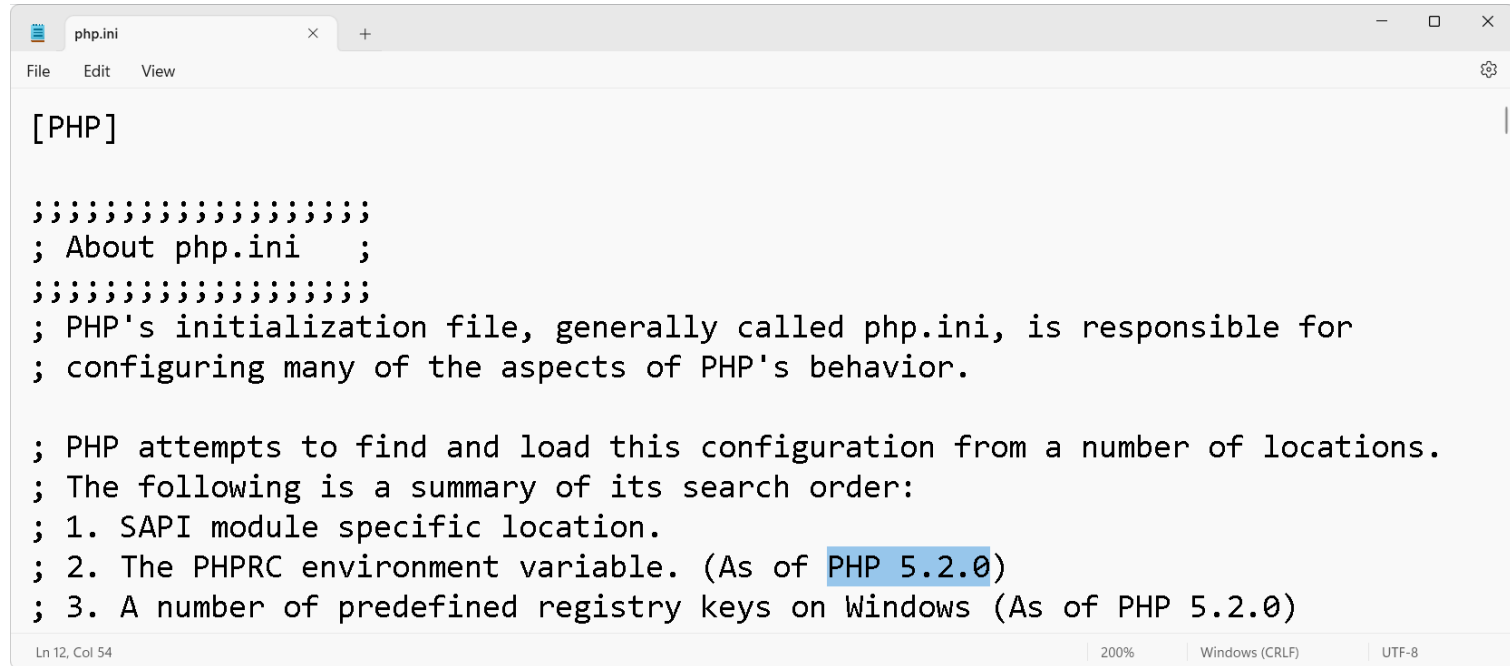
A decorative border of green leaves and branches at the top of the page.

5 – PHP *as* Programming Language

(Install PHP di Ubuntu Server 22.04)

A decorative border of green leaves and branches at the bottom of the page.

XAMPP PHP VERSION

A screenshot of a text editor window titled 'php.ini'. The window has a menu bar with 'File', 'Edit', and 'View'. The content of the file is as follows:

```
[PHP]

;;;;;;;;;;;;;;;;;;;;;;;;;
; About php.ini        ;
;;;;;;;;;;;;;;;;;;;;;;;;;
; PHP's initialization file, generally called php.ini, is responsible for
; configuring many of the aspects of PHP's behavior.

; PHP attempts to find and load this configuration from a number of locations.
; The following is a summary of its search order:
; 1. SAPI module specific location.
; 2. The PHPRC environment variable. (As of PHP 5.2.0)
; 3. A number of predefined registry keys on Windows (As of PHP 5.2.0)
```

The status bar at the bottom shows 'Ln 12, Col 54', '200%', 'Windows (CRLF)', and 'UTF-8'.

PHP adalah singkatan dari Hypertext Preprocessor, merupakan bahasa pemrograman atau skrip open source yang terutama digunakan dalam pembuatan website atau aplikasi lainnya yang berbasis web.

Saat ini versi PHP terbaru adalah PHP 8.2 yang diluncurkan pada 8 December 2022 lalu. Tetapi dalam kesempatan kita hanya akan menginstal PHP versi 7.4, JIKA BISA DI INSTALL.

Install PHP 7.4 di Ubuntu Server

[1] Tambahkan repositori PPA (Personal Package Archive) bernama Ondrej

➤ *sudo apt install software-properties-common*

..

➤ *sudo add-apt-repository ppa:ondrej/php -y*

[2] Selanjutnya instal PHP 7.4 dengan perintah:

➤ *sudo apt update*

➤ *sudo apt install php7.4*

[3] Untuk menambahkan beberapa ekstensi, kita menggunakan perintah :

➤ *sudo apt install php7.4-{cli,common,curl,zip,gd,mysql,xl,mbstring,json,intl}*

[4] Check PHP version yang terinstall dengan perintah:

➤ *php --version*

➤ *php -v*

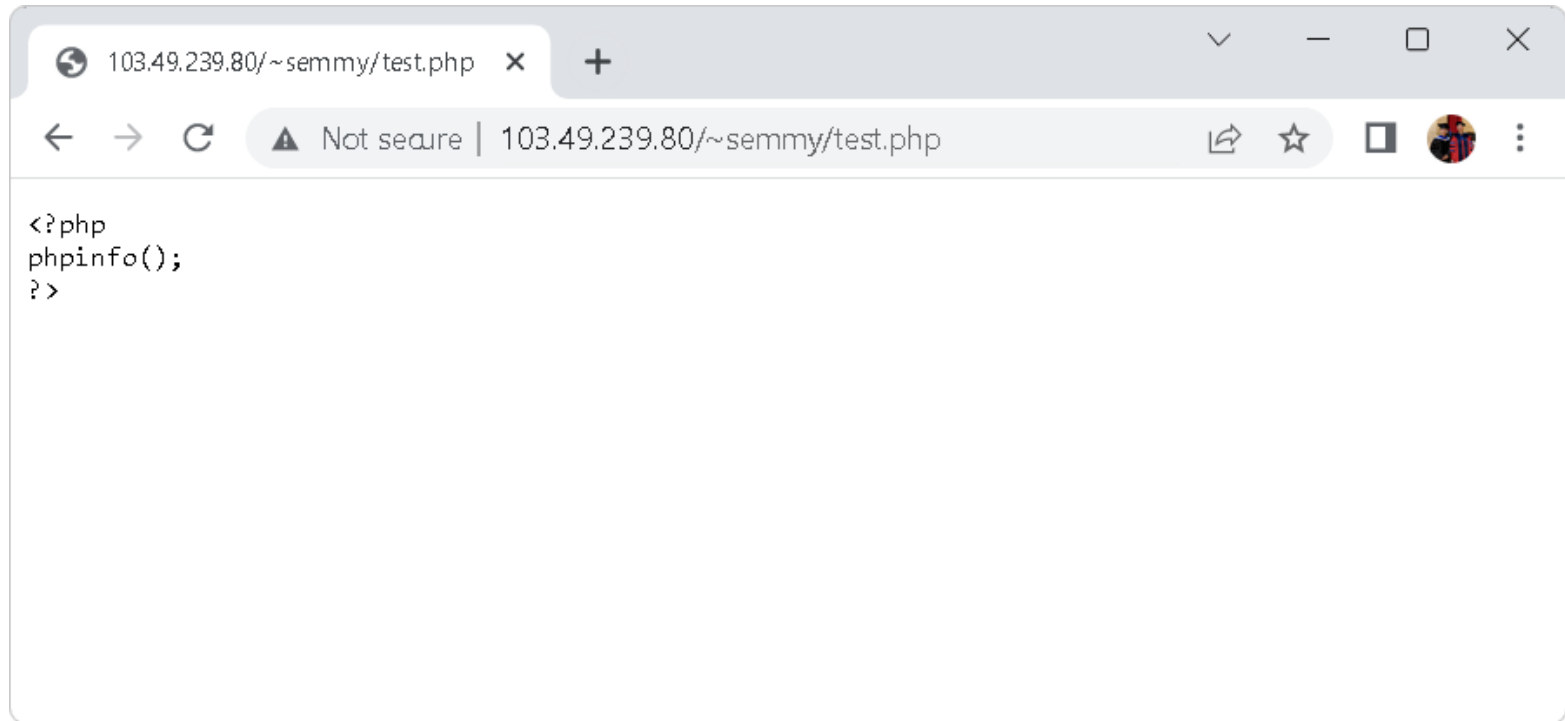
Possible Error

Could not get lock
/var/lib/dpkg/lock-frontend.
It is held by process 5044
(unattended-upgr)

Hapus penguncian paket secara manual dengan menjalankan perintah:

➤ *sudo rm /var/lib/dpkg/lock-frontend*

PHP Not Recognized



Jika hasilnya seperti ini, berarti file dengan extensi .PHP belum bisa di recognize. Kita akan coba uninstall dan install kembali versi terbaru PHP.

Uninstall PHP *in* Cloud Server

Untuk menguninstall PHP dari Ubuntu 22.04 melalui terminal:

[1] Menghapus paket PHP dan semua dependensi:

➤ *sudo apt-get remove php**

[2] Membersihkan paket yang tidak diperlukan dan sisa file konfigurasi:

➤ *sudo apt-get autoremove*

[3] Menghapus konfigurasi dan file-file lain yang terkait dengan PHP:

➤ *sudo apt-get purge php**

[4] Pengecek kembali, seharusnya sudah tidak ada respons:

➤ *php -v*

[1] Install Latest Version of PHP

Opsi #1: menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:

➤ *sudo apt install php libapache2-mod-php php-mbstring php-xmlrpc php-soap php-gd php-xml php-cli php-zip php-bcmath php-tokenizer php-json php-pear*

[2] Buat “*test.php*” file guna memastikan PHP sudah terinstall di server:

➤ *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php  
phpinfo();  
?>
```

[3] Akhiri dengan klik tombol Ctrl + X dan type Y, lalu Enter secara berurutan.

[4] Silakan kembali ke browser dan ketikan “*http://72.168.19.80/test.php*”.
Browser akan memunculkan tampilan PHP version. Berarti kita sudah berhasil.

[2] Install Latest Version of PHP

Opsi #2: jika terjadi error pada cara pertama, gunakan cara yang kedua untuk menginstall PHP versi terbaru secara otomatis:

[1] Menginstall PHP sekaligus modul pendukung lainnya:

➤ *sudo apt install php libapache2-mod-php php-mysql*

[2] Buat “*test.php*” file guna memastikan PHP sudah terinstall di server:

➤ *sudo nano /var/www/html/test.php*

Dan ketikan:

```
<?php  
phpinfo();  
?>
```

[3] Akhiri dengan klik tombol Ctrl + X dan type Y, lalu Enter secara berurutan.

[4] Silakan kembali ke browser dan ketikan “*http://72.168.19.80/test.php*”. Browser akan memunculkan tampilan PHP version. Berarti kita sudah berhasil.

Create Project Directory

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/unkpresent***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/unkpresent***

➤ ***sudo chown -R semmy:semmy /var/www/html/unkpresent***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

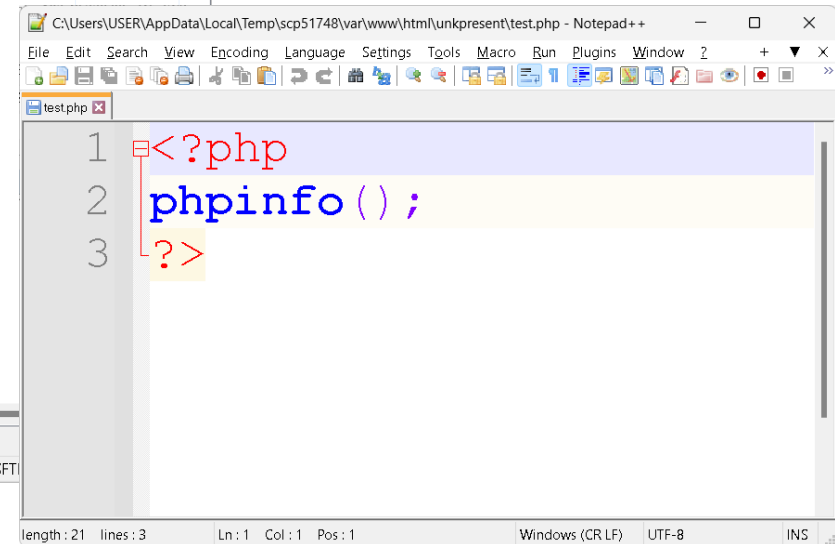
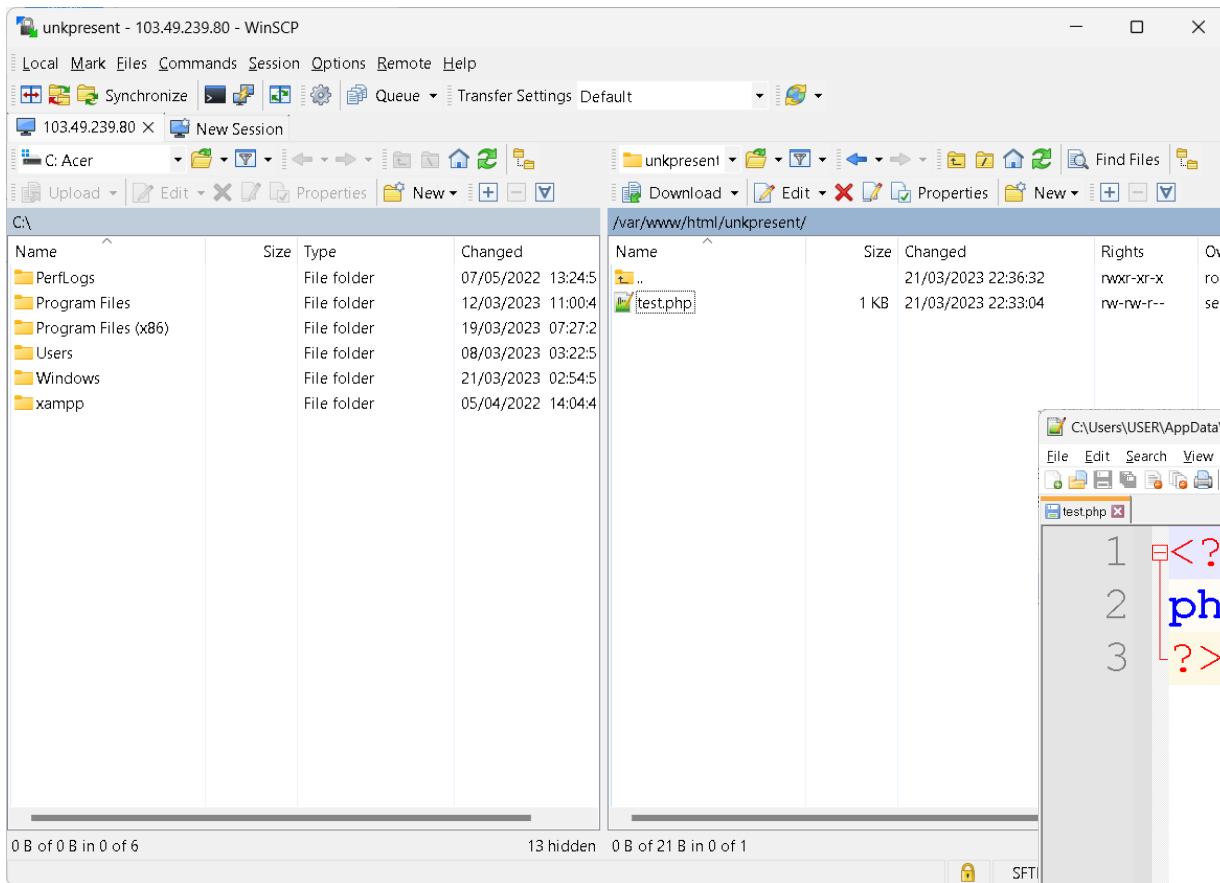
Note:

Ubah nama user “*semmy*” dengan nama *user* access ubuntu kalian.

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Open WinSCP Client FTP

- 1) Buka WinSCP dan arahkan ke “*/var/www/html/unkpresent*”.
- 2) Click kanan dan buat sebuah *file* di dalam folder ***unkpresent*** dan pastikan *file* tersebut bisa di simpan.
- 3) Edit file di local PC dan tuliskan *script* di bawah.
- 4) Buka via Browser.



Expected Output



PHP 8.2.4 - phpinfo()

Not secure | 103.49.239.80/unkpresent/test.php

PHP Version 8.2.4

System	Linux myserver 5.15.0-25-generic #25-Ubuntu SMP Wed Mar 30 15:54:22 UTC 2022 x86_64
Build Date	Mar 16 2023 14:34:48
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.2/apache2
Loaded Configuration File	/etc/php/8.2/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.2/apache2/conf.d
Additional .ini files parsed	/etc/php/8.2/apache2/conf.d/10-opcache.ini, /etc/php/8.2/apache2/conf.d/10-pdo.ini, /etc/php/8.2/xml.ini, /etc/php/8.2/apache2/conf.d/20-bcmath.ini, /etc/php/8.2/apache2/conf.d/20-calendar.ini, /etc/php/8.2/apache2/conf.d/20-ctype.ini, /etc/php/8.2/apache2/conf.d/20-dom.ini, /etc/php/8.2/apexif.ini, /etc/php/8.2/apache2/conf.d/20-ffi.ini, /etc/php/8.2/apache2/conf.d/20-fileinfo.ini, /etc/php/8.2/apache2/conf.d/20-ftp.ini, /etc/php/8.2/apache2/conf.d/20-gd.ini, /etc/php/8.2/apache2/conf.d/20-iconv.ini, /etc/php/8.2/apache2/conf.d/20-mbstring.ini, /etc/php/8.2/apache2/conf.d/20-mysql.ini, /etc/php/8.2/apache2/conf.d/20-pdo_mysql.ini, /etc/php/8.2/apache2/conf.d/20-posix.ini, /etc/php/8.2/apache2/conf.d/20-readline.ini

A decorative border of green leaves and branches at the top of the slide.

6 - Mengaktifkan file *.htaccess* di Apache2

A decorative border of green leaves and branches at the bottom of the slide.

File .htaccess di Apache2

Untuk mengaktifkan file *.htaccess* di Apache2 pada Ubuntu Server, ikuti langkah-langkah berikut:

[1] Pastikan modul *rewrite* sudah diaktifkan. Anda dapat memeriksa apakah modul ini sudah diaktifkan dengan menjalankan perintah:

➤ ***sudo a2enmod rewrite***

[2] Setelah modul *rewrite* diaktifkan, buka file konfigurasi untuk situs web yang ingin Anda aktifkan *.htaccess*. Misalnya, jika Anda ingin mengaktifkan *.htaccess* untuk situs web default, buka file konfigurasinya dengan menjalankan perintah:

➤ ***sudo nano /etc/apache2/sites-available/000-default.conf***

[3] Di dalam blok konfigurasi untuk situs web, tambahkan konfigurasi berikut di bawah direktif **<VirtualHost>**:

<Directory "/var/www/html">

AllowOverride All

</Directory>

[4] **OR** Jika Anda ingin mengaktifkan *.htaccess* di direktori */home/user/public_html*, maka pada langkah ke-3, gantilah konfigurasi sebagai berikut:

<Directory "/home/semmy/public_html">

AllowOverride All

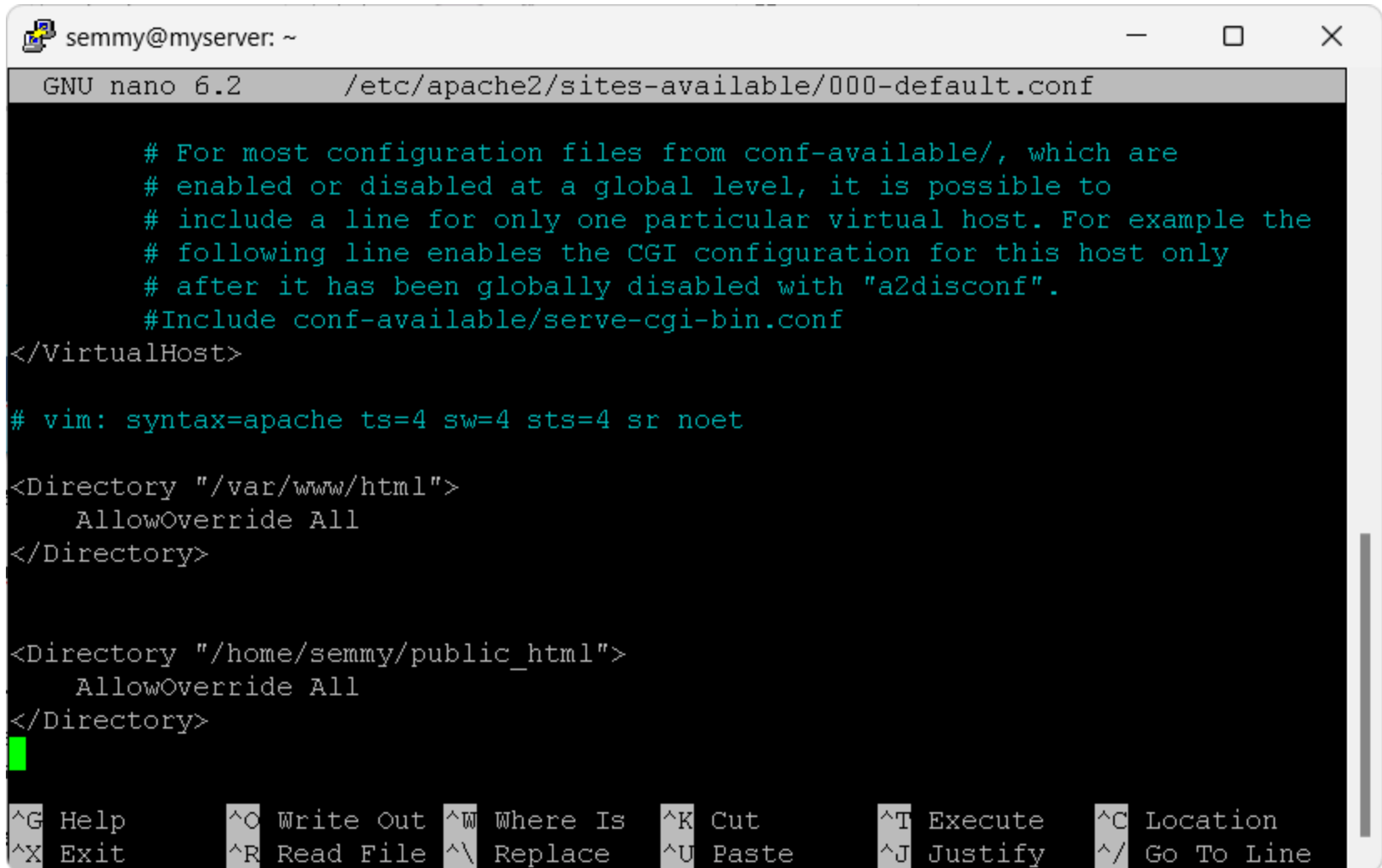
</Directory>

[5] Simpan dan tutup file konfigurasi dengan menekan **Ctrl+X**, lalu ketik **Y** untuk menyimpan perubahan dan **Enter** untuk menutup editor.

[6] Restart apache web server:

➤ ***sudo systemctl restart apache2***

Nano Texteditor Default Config File



The screenshot shows a terminal window with the Nano text editor open. The title bar indicates the user is 'semmy@myserver' and the file being edited is '/etc/apache2/sites-available/000-default.conf'. The editor's status bar at the top shows 'GNU nano 6.2'. The main text area contains the following configuration:

```
# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

<Directory "/var/www/html">
    AllowOverride All
</Directory>

<Directory "/home/semmy/public_html">
    AllowOverride All
</Directory>
```

The bottom status bar displays various Nano editor shortcuts:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line

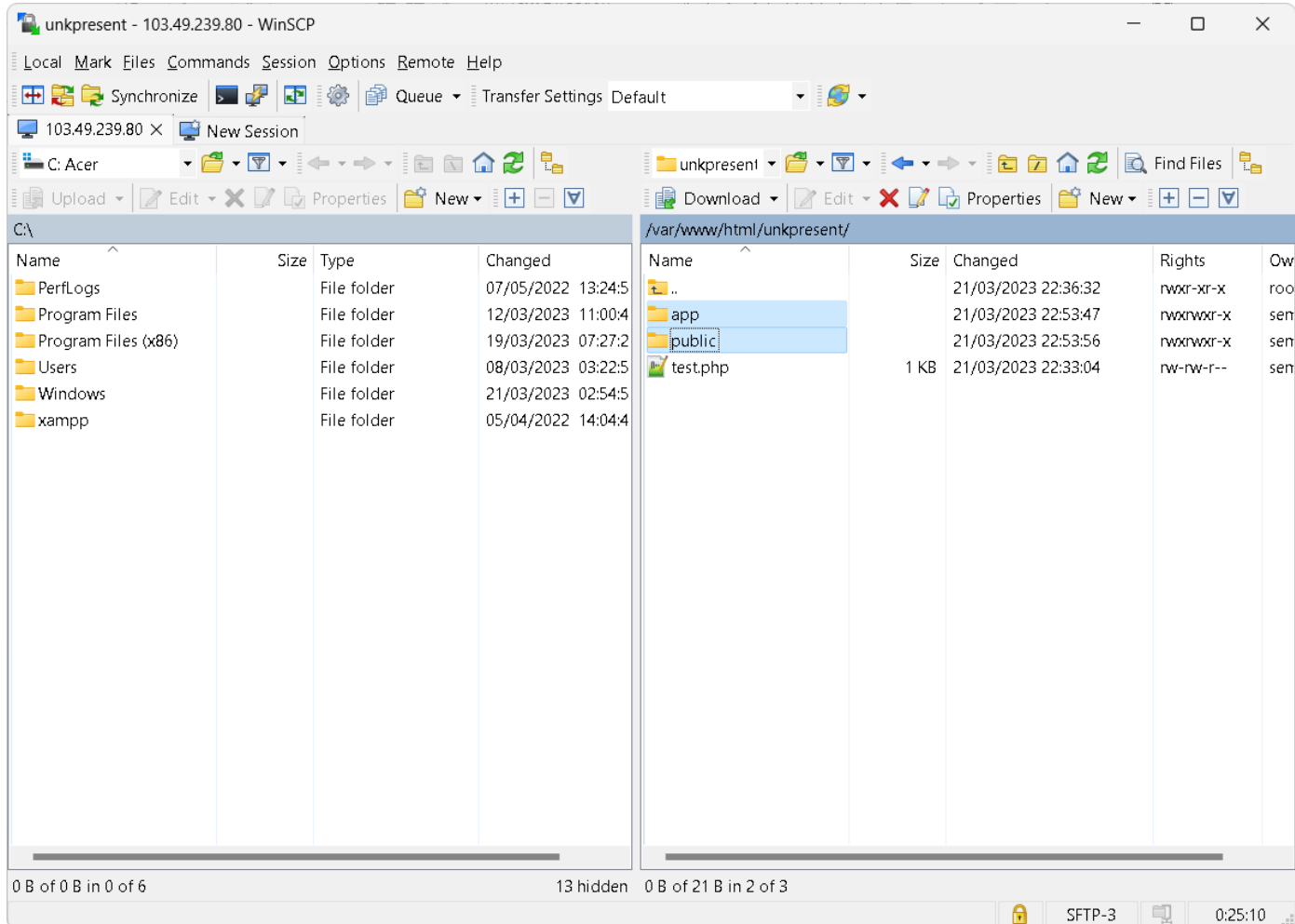
A decorative border of green leaves and branches at the top of the slide.

7 – Upload Project AIMVC Framework

A decorative border of green leaves and branches at the bottom of the slide.

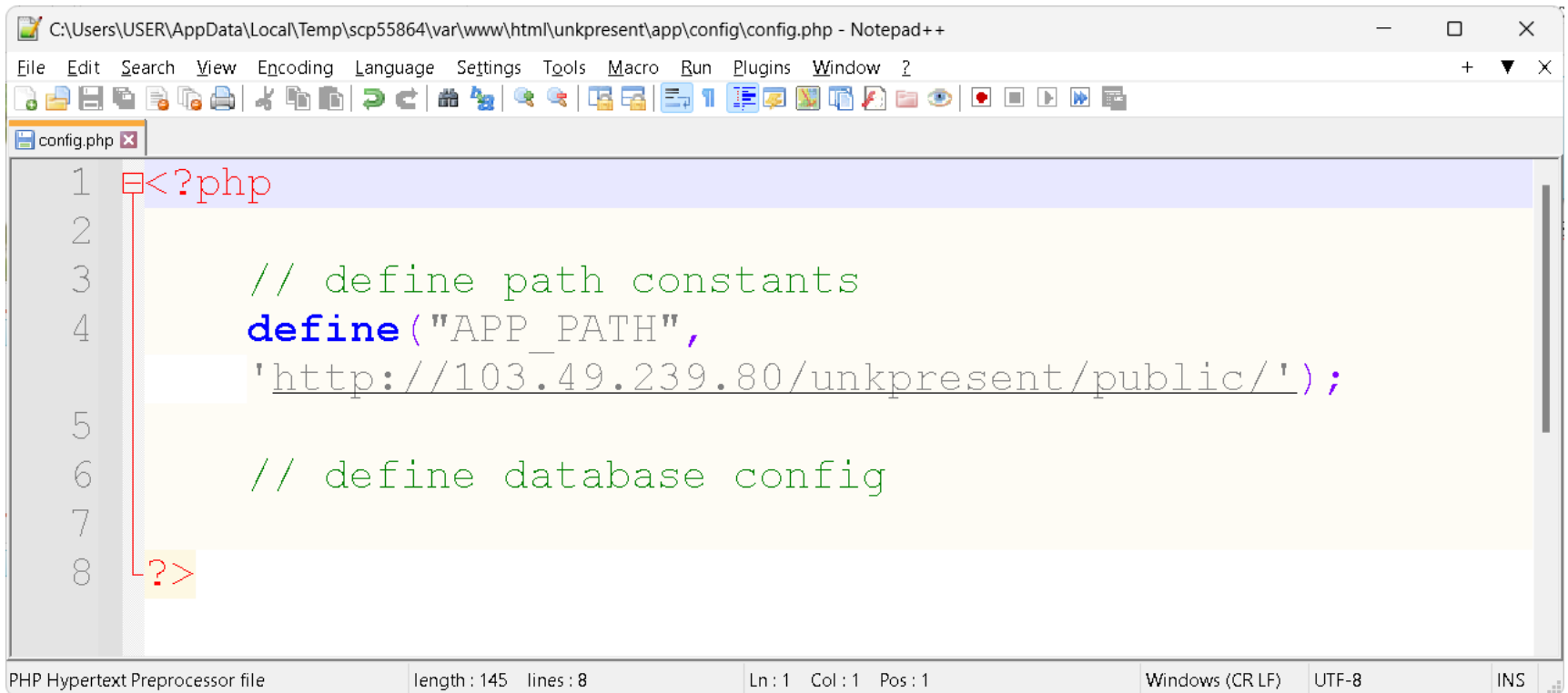
Copy Folder “app” & “public”

Students diminta untuk upload dua *folder* yang sudah dibuat yaitu ***app folder*** dan ***public folder*** ke dalam *cloud server*.



Edit File “config.php”

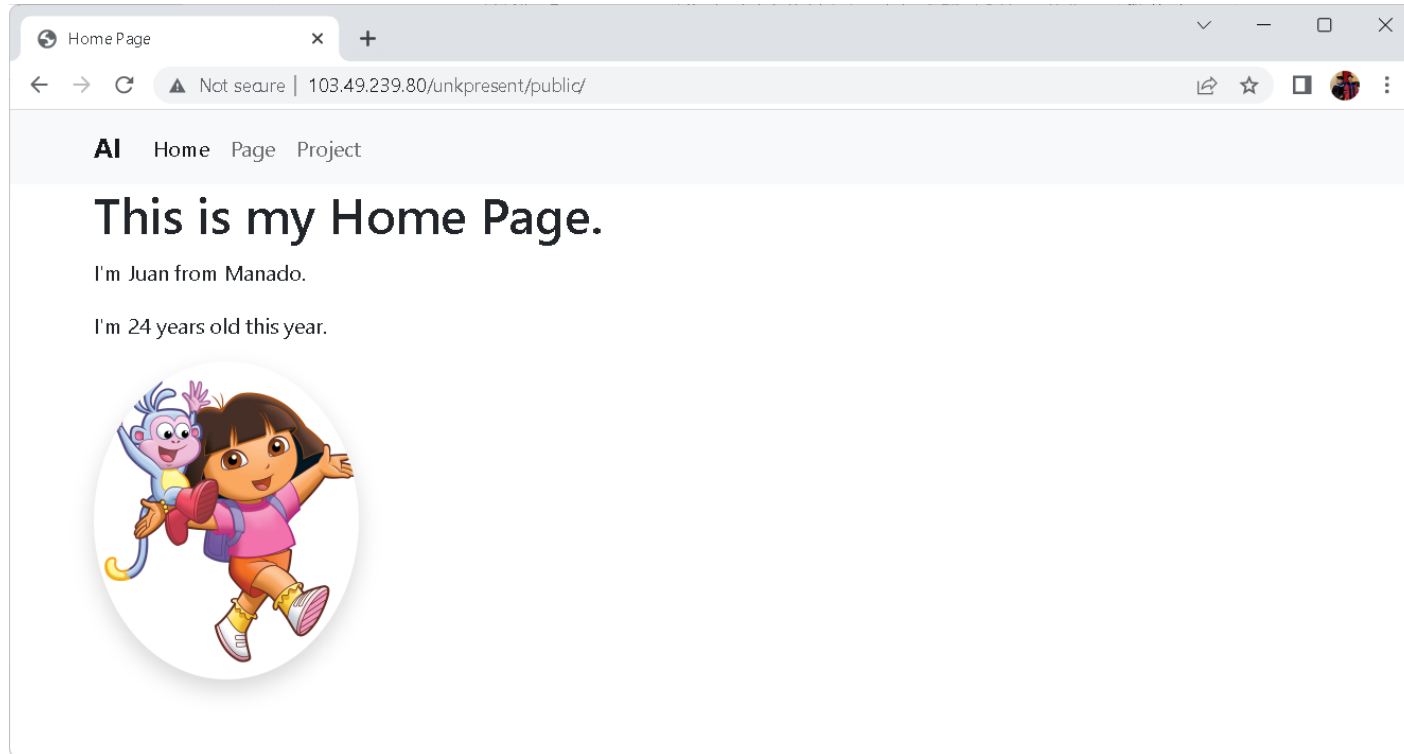
- 1) Edit file “*/var/www/html/unkpresent/app/config/config.php*”.
- 2) Pastikan public IP address sesuai dengan yang ada di *virtual server*.



```
1 <?php
2
3 // define path constants
4 define("APP_PATH",
5     'http://103.49.239.80/unkpresent/public/');
6
7 // define database config
8 ?>
```

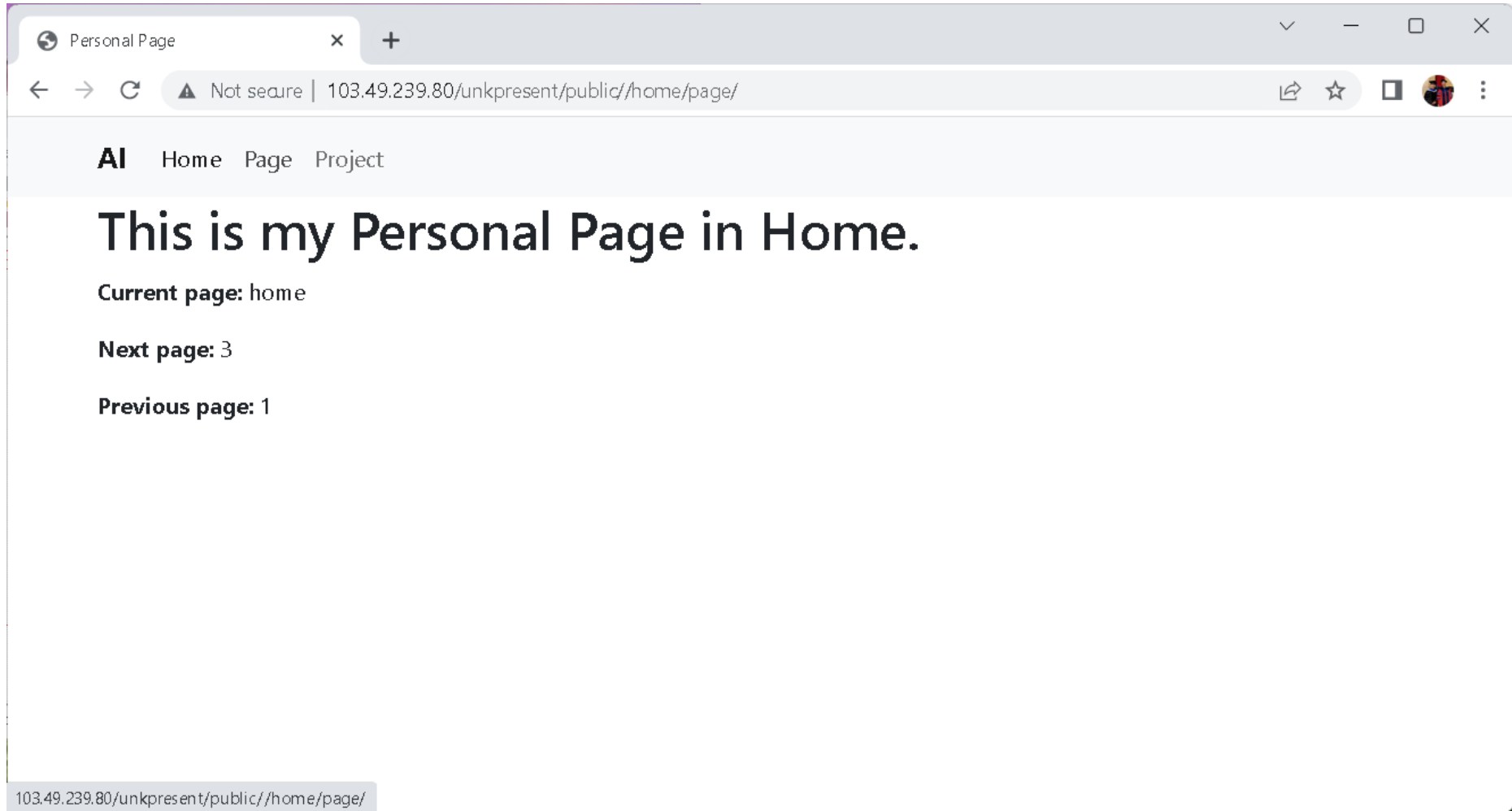
PHP Hypertext Preprocessor file length : 145 lines : 8 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS

Expected Output

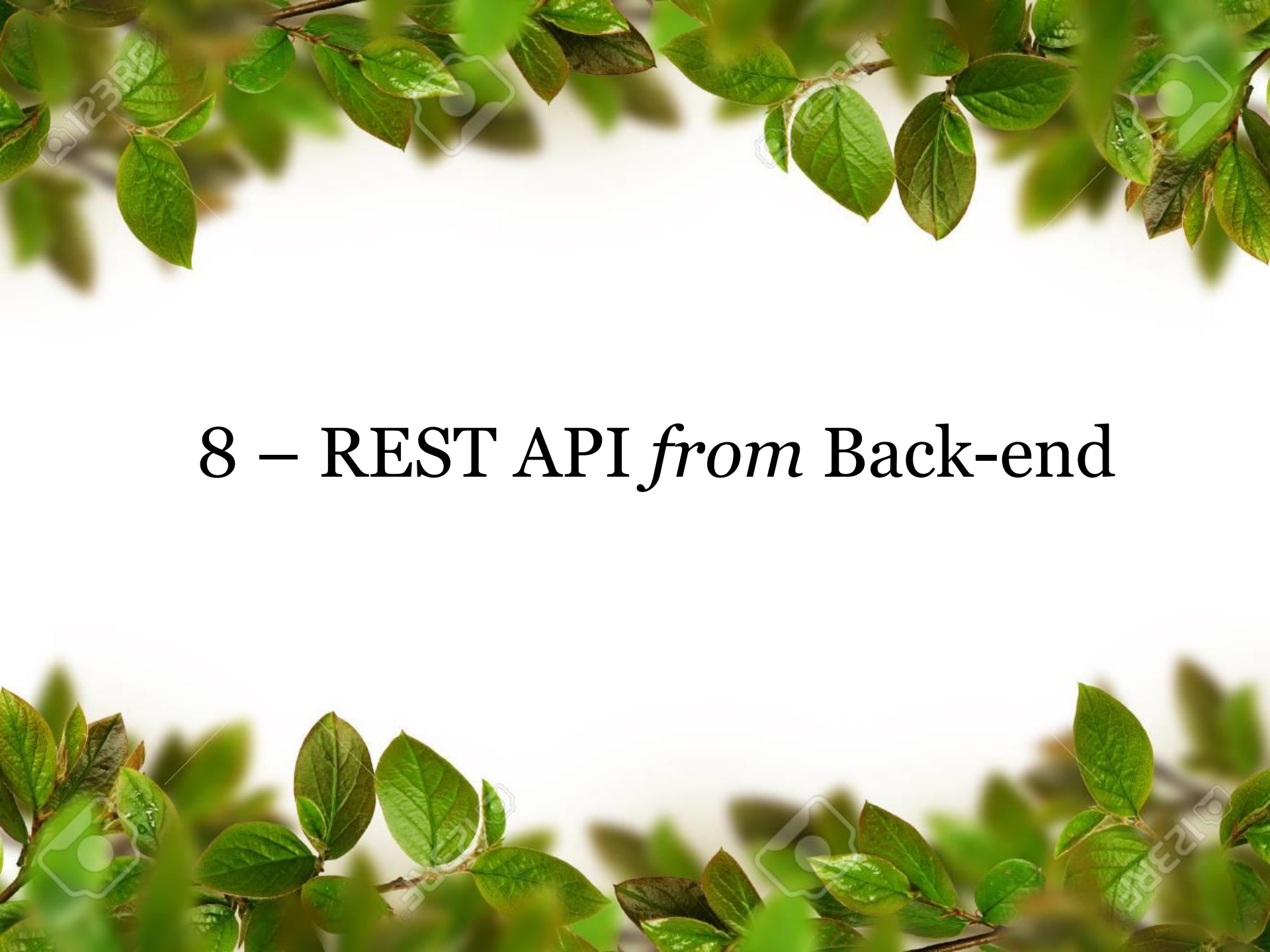


Jika kalian sudah sampai pada tahapan ini, makaproject kalian sudah bisa diakses secara global dari setiap web browser menggunakan public IP address.

Expected Output

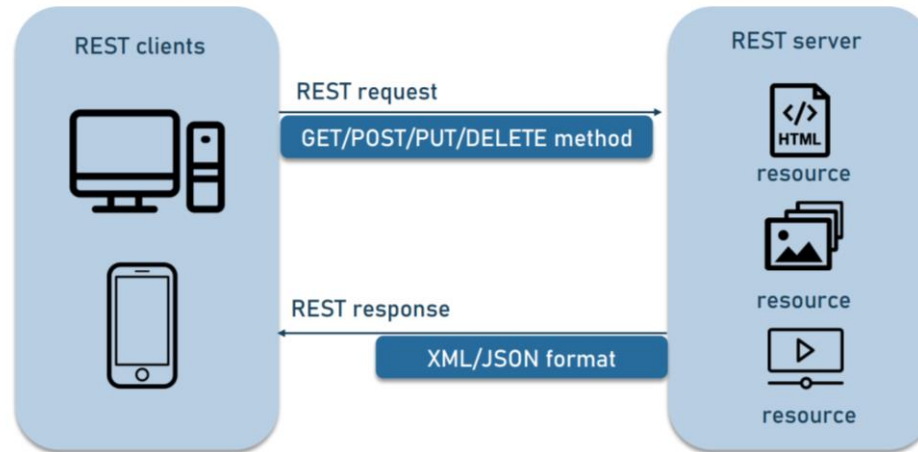


**Saya sudah banyak kali install
dan konfigurasi “cloud server”
dan ikut step-by-step yang dibuat.
Jangan ragu untuk install lebih
dari satu virtual server.**

The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub, with some showing signs of aging or damage. The text is centered in the middle of the slide.

8 – REST API *from* Back-end

API Architecture



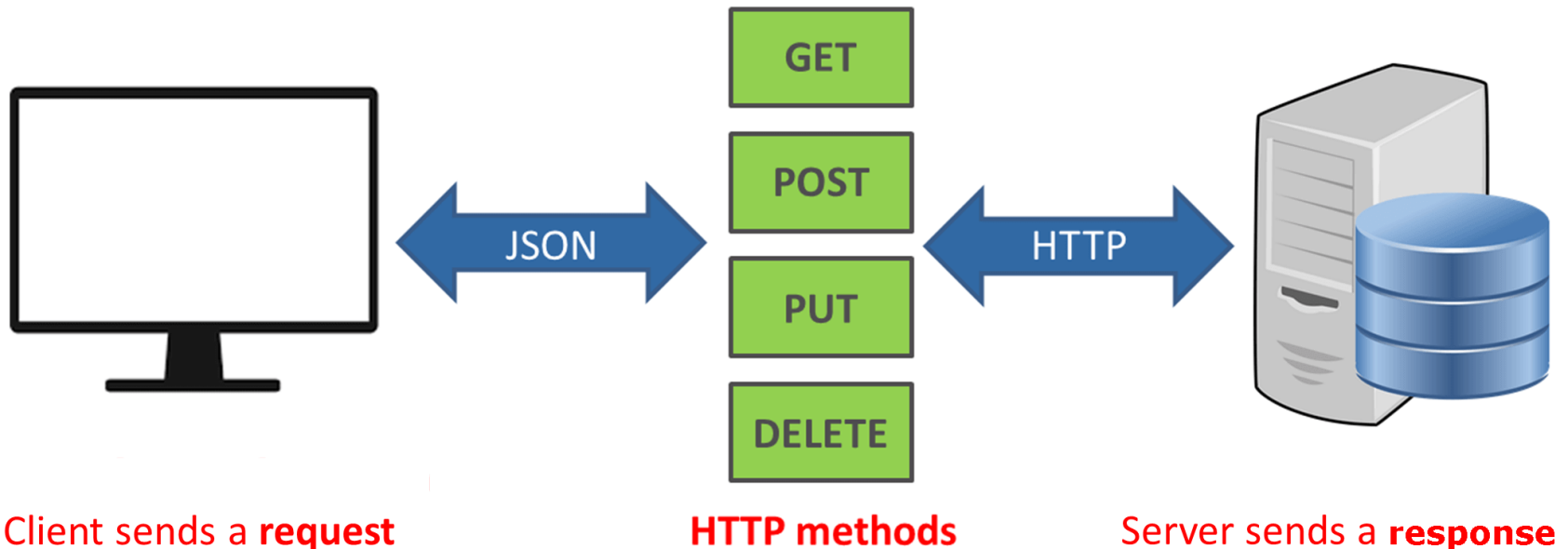
- 1) Dalam konteks REST (Representational State Transfer) API, **sebuah "endpoint"** adalah **URL tertentu** yang diekspos oleh API tersebut untuk berinteraksi dengan sumber daya-sumber dayanya.
- 2) **Endpoint digunakan untuk melakukan berbagai operasi pada sumber daya**, seperti mengambil data, membuat sumber daya baru, memperbarui sumber daya yang sudah ada, atau menghapus sumber daya.
- 3) Setiap **endpoint** biasanya sesuai dengan sumber daya tertentu atau kumpulan sumber daya dalam API.
- 4) An API endpoint is a specific location within an API that **accepts requests and sends back responses**. It's a way for different systems and applications to communicate with each other, by sending and receiving information and instructions via the endpoint.

HTTP Methods

Hypertext Transfer Protocol (HTTP) adalah sebuah jalan atau cara komunikasi dapat terjalin antara client dengan server.

Dalam HTTP Method REST yang sering digunakan antara lain:

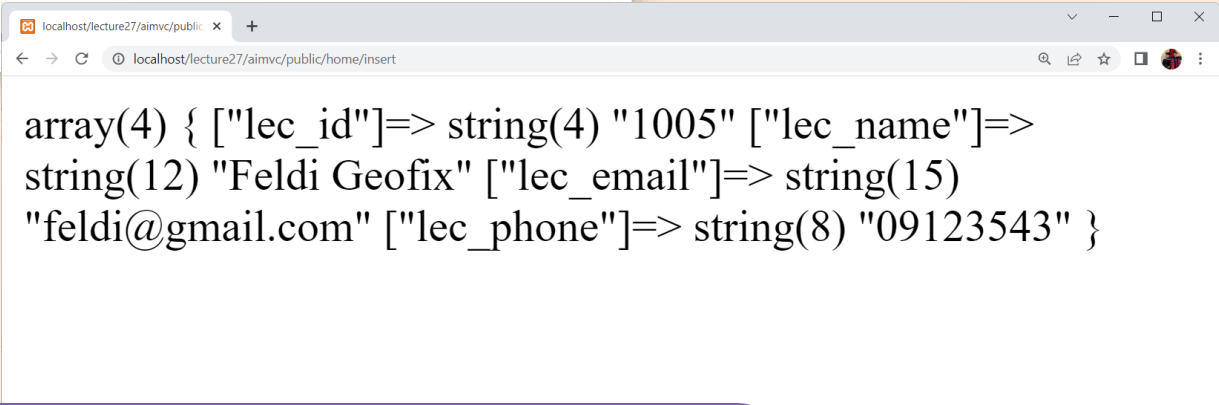
- ❖ **GET** : Fungsi ini digunakan untuk perintah menampilkan data.
- ❖ **POST**: Fungsi ini biasanya digunakan untuk perintah dalam menambahkan data (*secure*).
- ❖ **PUT**: Fungsi ini biasanya digunakan untuk perintah edit data.
- ❖ **DELETE**: Fungsi ini biasanya digunakan untuk perintah dalam menghapus data.



Parameter Sent *after* Click Button Insert

Cari fungsi `insert()` di bawah ini, dan pastikan `var_dump` tidak di comment supaya bisa melakukan *testing* untuk melihat apakah data yang tepat sudah di kirim ke server.

```
public function insert(){  
    var_dump($_POST);  
    //if($this->logic("Lecturer_model")->insertDataLecturer($_POST) == true){  
    //    header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.  
    //    exit;  
    //}
```



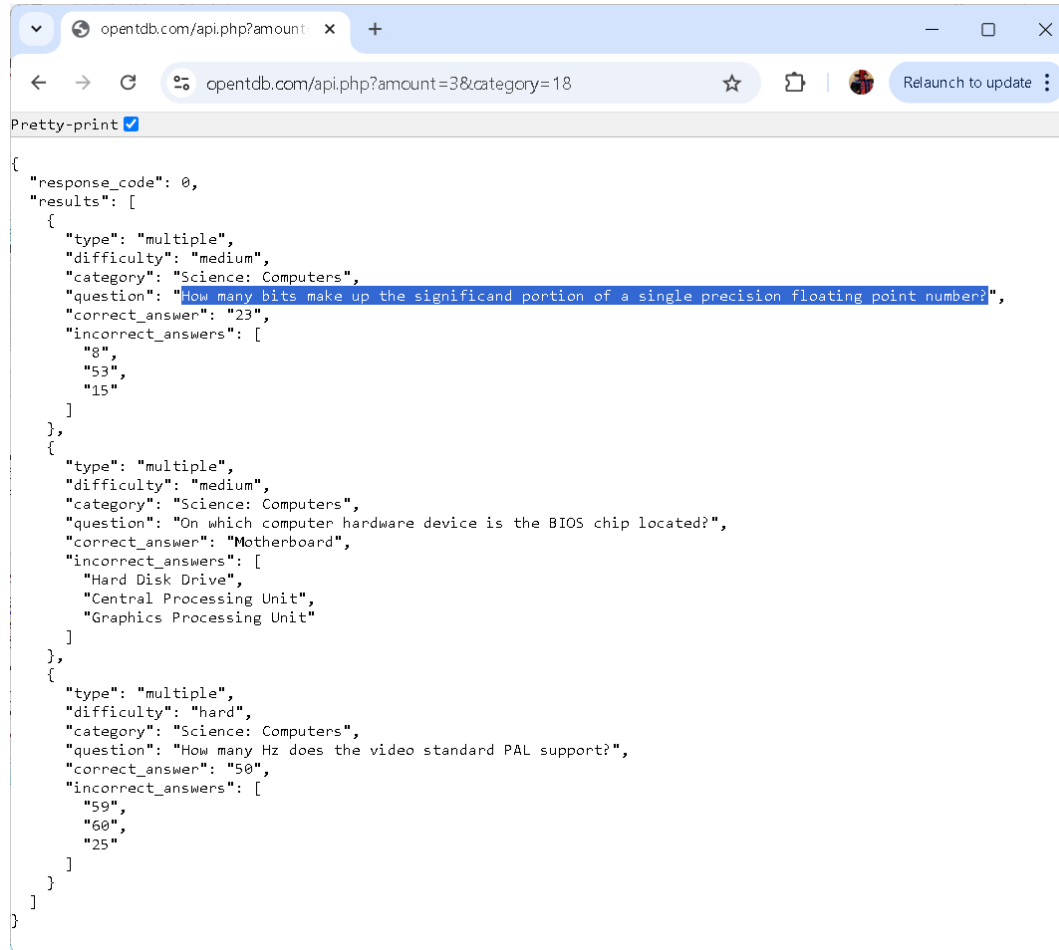
```
array(4) { ["lec_id"]=> string(4) "1005" ["lec_name"]=>  
string(12) "Feldi Geofix" ["lec_email"]=> string(15)  
"feldi@gmail.com" ["lec_phone"]=> string(8) "09123543" }
```

Ini adalah parameter yang dikirim dengan menggunakan method POST.

```
["lec_id"]=> string(4) "1005"  
["lec_name"]=> string(12) "Feldi Geofix"  
["lec_email"]=> string(15) "feldi@gmail.com"  
["lec_phone"]=> string(8) "09123543"
```

RESTful API Web Service

Layanan web RESTful seperti di bawah ini digunakan untuk menyediakan **API web public**.



```
{
  "response_code": 0,
  "results": [
    {
      "type": "multiple",
      "difficulty": "medium",
      "category": "Science: Computers",
      "question": "How many bits make up the significand portion of a single precision floating point number?",
      "correct_answer": "23",
      "incorrect_answers": [
        "8",
        "53",
        "15"
      ]
    },
    {
      "type": "multiple",
      "difficulty": "medium",
      "category": "Science: Computers",
      "question": "On which computer hardware device is the BIOS chip located?",
      "correct_answer": "Motherboard",
      "incorrect_answers": [
        "Hard Disk Drive",
        "Central Processing Unit",
        "Graphics Processing Unit"
      ]
    },
    {
      "type": "multiple",
      "difficulty": "hard",
      "category": "Science: Computers",
      "question": "How many Hz does the video standard PAL support?",
      "correct_answer": "50",
      "incorrect_answers": [
        "59",
        "60",
        "25"
      ]
    }
  ]
}
```

This API is part of the **Open Trivia Database (OpenTDB) API**, which is used to retrieve trivia questions.

The query parameters used in this API call are:

- 1) “amount=1” specifies the number of trivia questions to retrieve (**in this case, 1 question**).
- 2) “category=18” specifies the category of the trivia question. Category 18 corresponds to **“Science: Computers”**.

Keamanan API REST :

- ❖ Manfaatkan HTTPS
- ❖ Gunakan mekanisme otentikasi yang kuat

Pretty-print ☒

```
{
  "response_code": 0,
  "results": [
    {
      "type": "multiple",
      "difficulty": "hard",
      "category": "Science: Computers",
      "question": "Which of these names was an actual codename for a cancelled Microsoft project?",
      "correct_answer": "Neptune",
      "incorrect_answers": [
        "Enceladus",
        "Pollux",
        "Saturn"
      ]
    },
    {
      "type": "multiple",
      "difficulty": "hard",
      "category": "Science: Computers",
      "question": "Which data structure does FIFO apply to?",
      "correct_answer": "Stack",
      "incorrect_answers": [
        "Queue",
        "Heap",
        "Tree"
      ]
    }
  ]
}
```

Four query parameters

RESTful API OpenWeatherMap

`api.openweathermap.org/data/2.5/forecast?q={city name}&appid={API key}`

```
{
  "cod": "200",
  "message": 0,
  "cnt": 40,
  "list": [
    {
      "dt": 1611381600,
      "main": {
        "temp": 303.07,
        "feels_like": 305.04,
        "temp_min": 303.07,
        "temp_max": 304.01,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 993,
        "humidity": 61,
        "temp_kf": -0.94,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 98
        },
        "wind": {
          "speed": 3.57,
          "deg": 338
        },
        "visibility": 10000,
        "pop": 0.33,
        "rain": {
          "3h": 0.25
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-23 06:00:00"
      },
      "dt": 1611392400,
      "main": {
        "temp": 301.12,
        "feels_like": 304.48,
        "temp_min": 300.83,
        "temp_max": 301.12,
        "pressure": 1008,
        "sea_level": 1008,
        "grnd_level": 992,
        "humidity": 74,
        "temp_kf": 0.29,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 2.6,
          "deg": 343
        },
        "visibility": 10000,
        "pop": 0.97,
        "rain": {
          "3h": 4.28
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-23 09:00:00"
      },
      "dt": 1611403200,
      "main": {
        "temp": 298.61,
        "feels_like": 303.68,
        "temp_min": 298.37,
        "temp_max": 298.61,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 86,
        "temp_kf": 0.24,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.2,
          "deg": 23
        },
        "visibility": 10000,
        "pop": 0.98,
        "rain": {
          "3h": 5.34
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 12:00:00"
      },
      "dt": 1611414000,
      "main": {
        "temp": 297.94,
        "feels_like": 302.78,
        "temp_min": 297.91,
        "temp_max": 297.94,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 996,
        "humidity": 89,
        "temp_kf": 0.03,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.46,
          "deg": 26
        },
        "visibility": 5779,
        "pop": 0.94,
        "rain": {
          "3h": 6.37
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 15:00:00"
      },
      "dt": 1611424800,
      "main": {
        "temp": 297.42,
        "feels_like": 302.23,
        "temp_min": 297.42,
        "temp_max": 297.42,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 90,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.25,
          "deg": 218
        },
        "visibility": 6163,
        "pop": 0.99,
        "rain": {
          "3h": 3.66
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 18:00:00"
      },
      "dt": 1611435600,
      "main": {
        "temp": 297.21,
        "feels_like": 301.35,
        "temp_min": 297.21,
        "temp_max": 297.21,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 87,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 99
        },
        "wind": {
          "speed": 0.62,
          "deg": 189
        },
        "visibility": 10000,
        "pop": 0.94,
        "rain": {
          "3h": 0.92
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-23 21:00:00"
      },
      "dt": 1611446400,
      "main": {
        "temp": 298.34,
        "feels_like": 302.84,
        "temp_min": 298.34,
        "temp_max": 298.34,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 82,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 98
        },
        "wind": {
          "speed": 0.2,
          "deg": 355
        },
        "visibility": 10000,
        "pop": 0.9,
        "rain": {
          "3h": 0.78
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 00:00:00"
      },
      "dt": 1611457200,
      "main": {
        "temp": 302.07,
        "feels_like": 305.59,
        "temp_min": 302.07,
        "temp_max": 302.07,
        "pressure": 1011,
        "sea_level": 1011,
        "grnd_level": 996,
        "humidity": 68,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 47
        },
        "wind": {
          "speed": 1.99,
          "deg": 327
        },
        "visibility": 10000,
        "pop": 0.35,
        "rain": {
          "3h": 0.25
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 03:00:00"
      },
      "dt": 1611468000,
      "main": {
        "temp": 302.85,
        "feels_like": 305.88,
        "temp_min": 302.85,
        "temp_max": 302.85,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 67,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 63
        },
        "wind": {
          "speed": 3.08,
          "deg": 342
        },
        "visibility": 10000,
        "pop": 0.69,
        "rain": {
          "3h": 1.39
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 06:00:00"
      },
      "dt": 1611478800,
      "main": {
        "temp": 298.49,
        "feels_like": 302.89,
        "temp_min": 298.49,
        "temp_max": 298.49,
        "pressure": 1008,
        "sea_level": 1008,
        "grnd_level": 993,
        "humidity": 86,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10d"
          }
        ],
        "clouds": {
          "all": 75
        },
        "wind": {
          "speed": 1.06,
          "deg": 46
        },
        "visibility": 2949,
        "pop": 1,
        "rain": {
          "3h": 8.43
        },
        "sys": {
          "pod": "d"
        },
        "dt_txt": "2021-01-24 09:00:00"
      },
      "dt": 1611489600,
      "main": {
        "temp": 297.27,
        "feels_like": 301.25,
        "temp_min": 297.27,
        "temp_max": 297.27,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 88,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 88
        },
        "wind": {
          "speed": 1.03,
          "deg": 170
        },
        "visibility": 10000,
        "pop": 1,
        "rain": {
          "3h": 10.03
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 12:00:00"
      },
      "dt": 1611500400,
      "main": {
        "temp": 296.48,
        "feels_like": 299.76,
        "temp_min": 296.48,
        "temp_max": 296.48,
        "pressure": 1012,
        "sea_level": 1012,
        "grnd_level": 997,
        "humidity": 86,
        "temp_kf": 0,
        "weather": [
          {
            "id": 501,
            "main": "Rain",
            "description": "moderate rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 1.18,
          "deg": 207
        },
        "visibility": 10000,
        "pop": 0.96,
        "rain": {
          "3h": 4.08
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 15:00:00"
      },
      "dt": 1611511200,
      "main": {
        "temp": 296.52,
        "feels_like": 300,
        "temp_min": 296.52,
        "temp_max": 296.52,
        "pressure": 1010,
        "sea_level": 1010,
        "grnd_level": 995,
        "humidity": 85,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 0.8,
          "deg": 159
        },
        "visibility": 10000,
        "pop": 0.93,
        "rain": {
          "3h": 0.68
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 18:00:00"
      },
      "dt": 1611522000,
      "main": {
        "temp": 295.85,
        "feels_like": 298.27,
        "temp_min": 295.85,
        "temp_max": 295.85,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 994,
        "humidity": 82,
        "temp_kf": 0,
        "weather": [
          {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
          }
        ],
        "clouds": {
          "all": 100
        },
        "wind": {
          "speed": 1.46,
          "deg": 195
        },
        "visibility": 10000,
        "pop": 0.4,
        "rain": {
          "3h": 0.91
        },
        "sys": {
          "pod": "n"
        },
        "dt_txt": "2021-01-24 21:00:00"
      },
      "dt": 1611532800,
      "main": {

```

Postman adalah platform kolaborasi untuk pengembangan API. Fitur Postman menyederhanakan setiap langkah pembuatan API dan menyederhanakan kolaborasi sehingga kita dapat membuat API yang lebih baik dan lebih cepat.

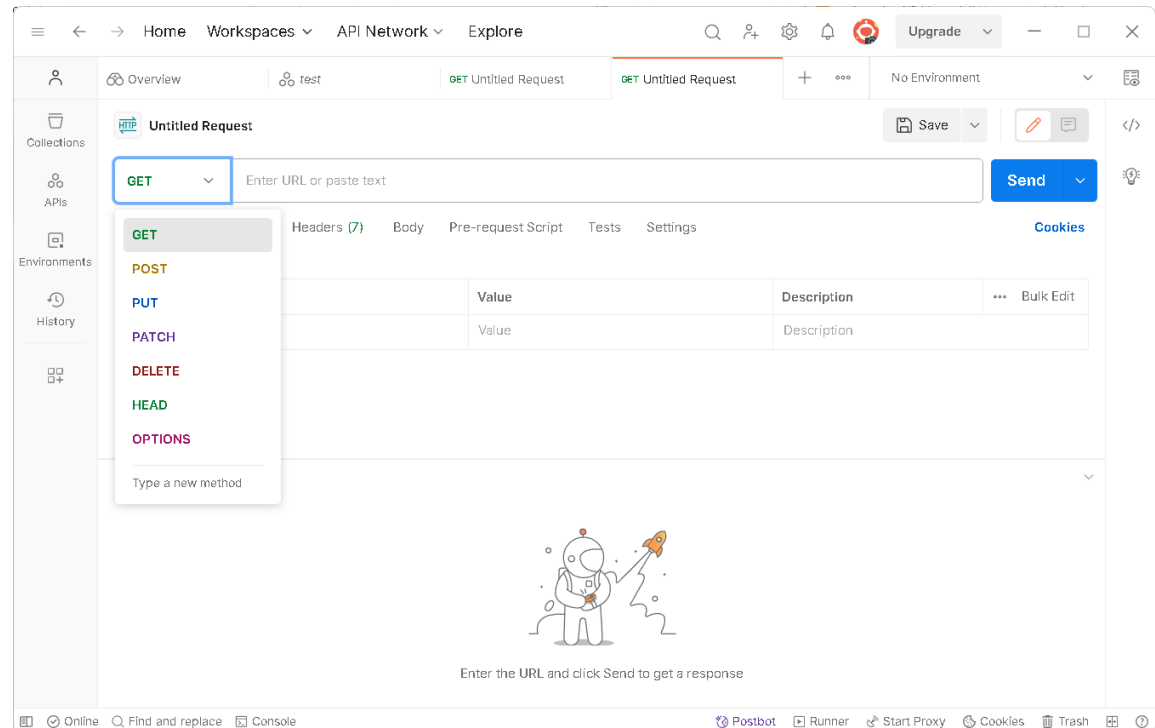
Gunakan URL API yang sudah dibuat (Contoh):

<https://opentdb.com/api.php?amount=2&category=18&difficulty=hard&type=multiple>

Cara download Postman:

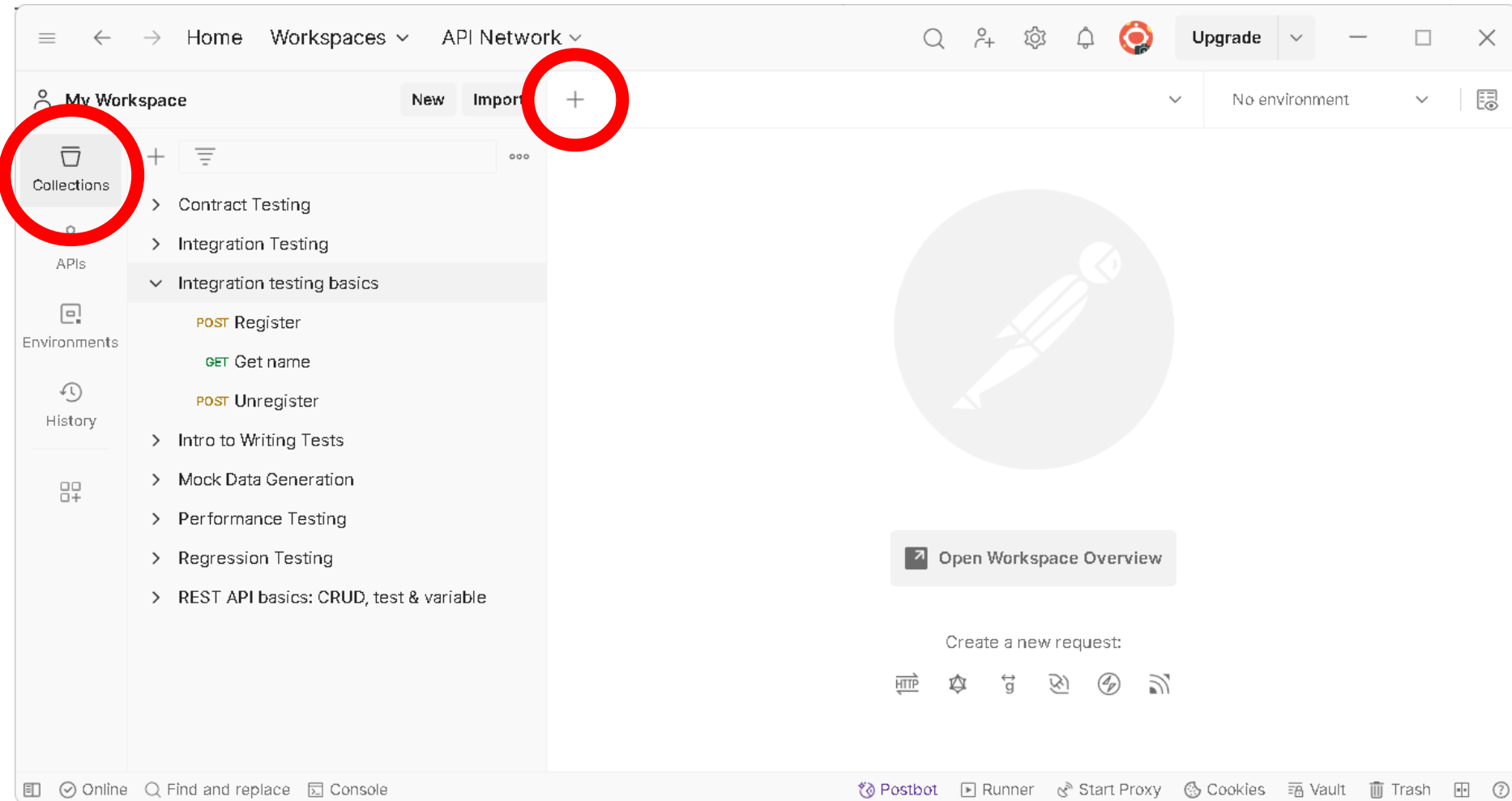
- 1) Buka official website postman di <https://www.postman.com/downloads/>
- 2) Pilih Download App
- 3) Selesai Download Double klik dan pilih install -> Tunggu sampai proses selesai.
- 4) Jika sudah selesai silahkan klik tombol Launch untuk menjalankan
- 5) Silahkan Registrasi menggunakan email aktif

**Sign in dan
Silahkan
Demo cara
menggunakan
an Postman:**



Buka Aplikasi Postman

Buat request API Baru dengan cara di bagian atas Postman, klik tombol **+ New Tab** (tab baru). Ini akan membuka jendela baru untuk membuat request.



Pilih Jenis Server Request

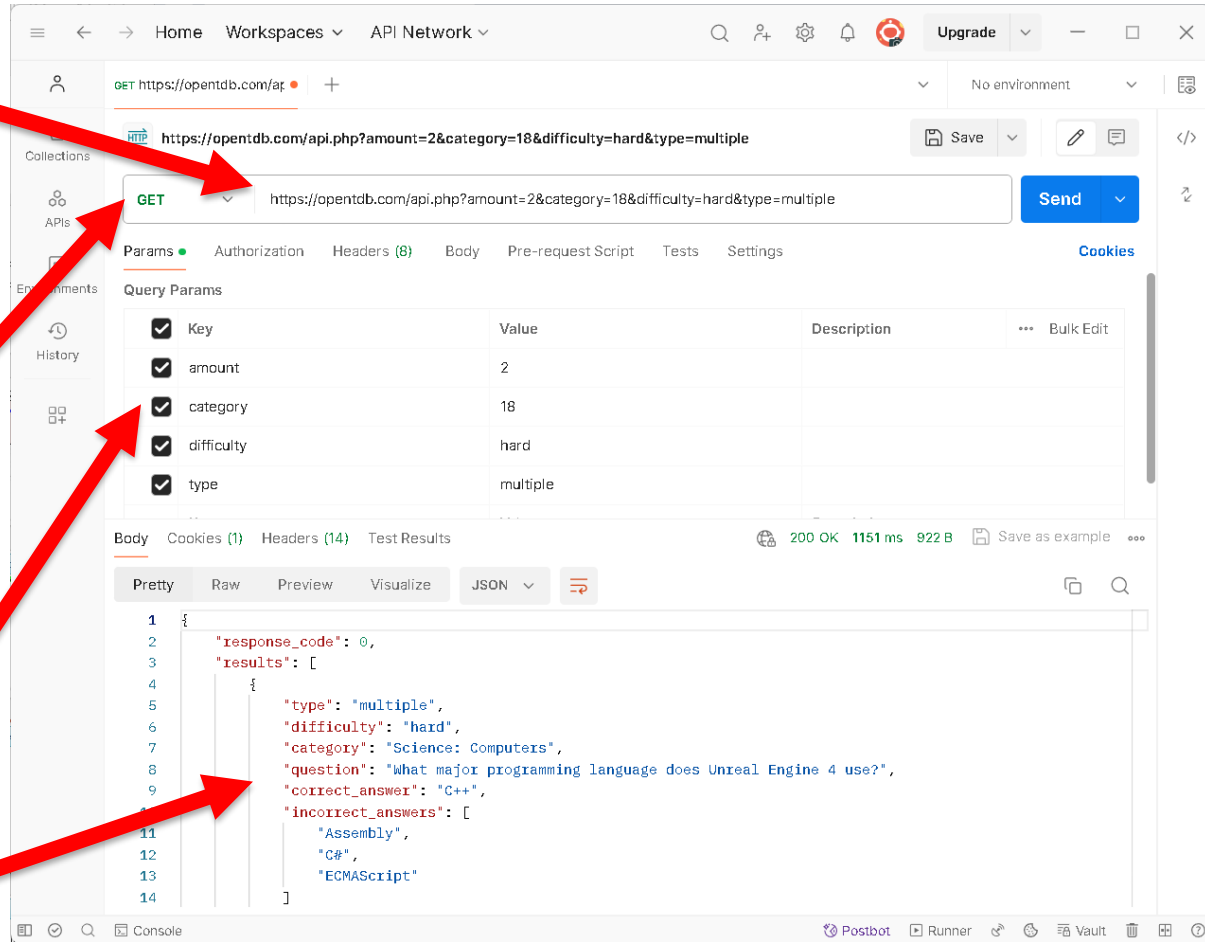
Masukkan **URL Endpoint API**, di sebelah dropdown menu jenis request, masukkan URL endpoint API yang ingin kamu uji.

Klik dropdown dan pilih jenis request yang sesuai:

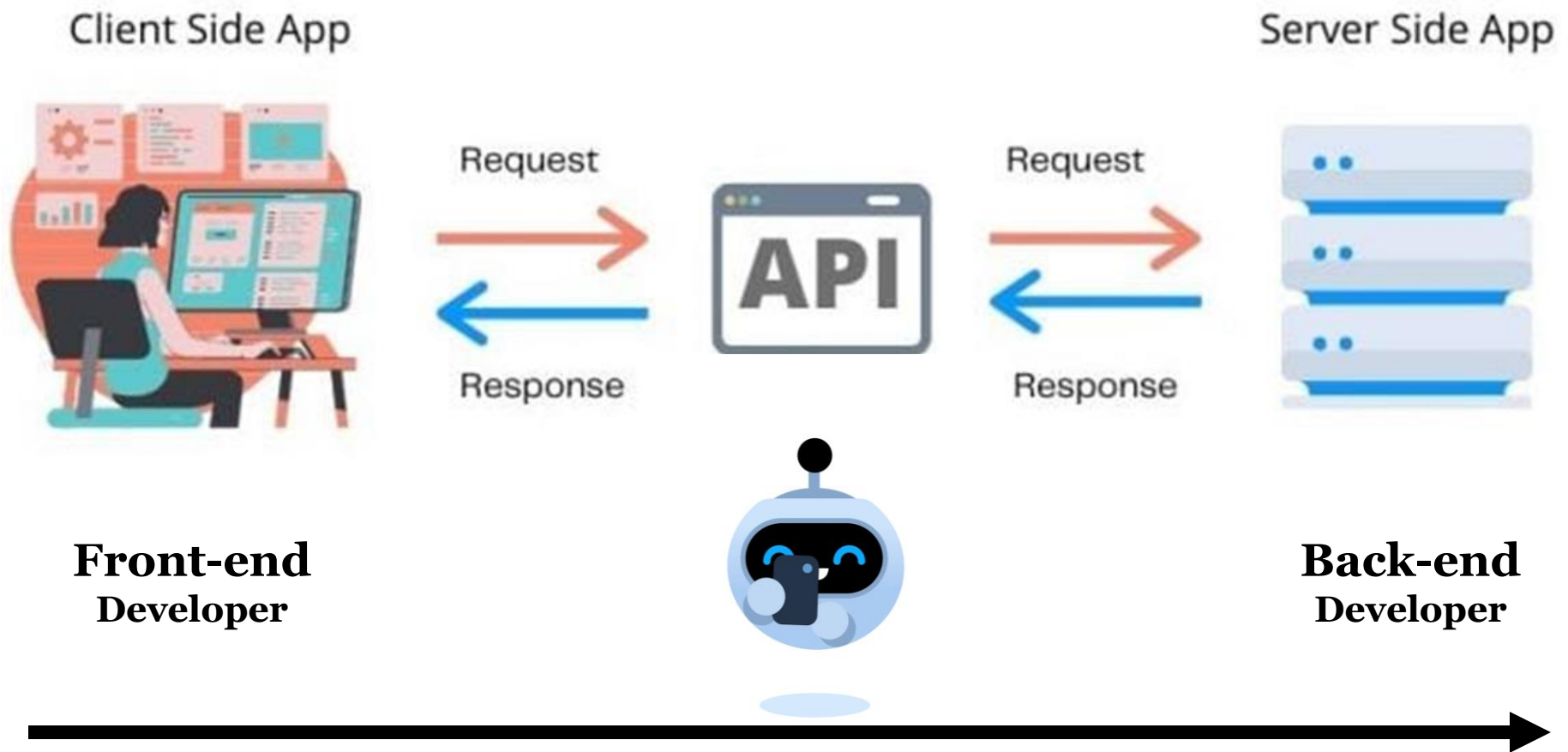
- 1) GET (mengambil data dari server)
- 2) POST (mengirim data ke server)
- 3) PUT (mengubah data di server)
- 4) DELETE (menghapus data di server)

Masukkan **query parameter data** berupa request yang akan dikirimkan ke server. Method **GET** akan secara otomatis jika sudah terdapat key di URL, sedangkan method **POST** harus diinputkan secara manual.

Output Json format adalah respons yang di return oleh server di sisi back-end.



Kapan kita akan menggunakan method POST dan method GET untuk request?



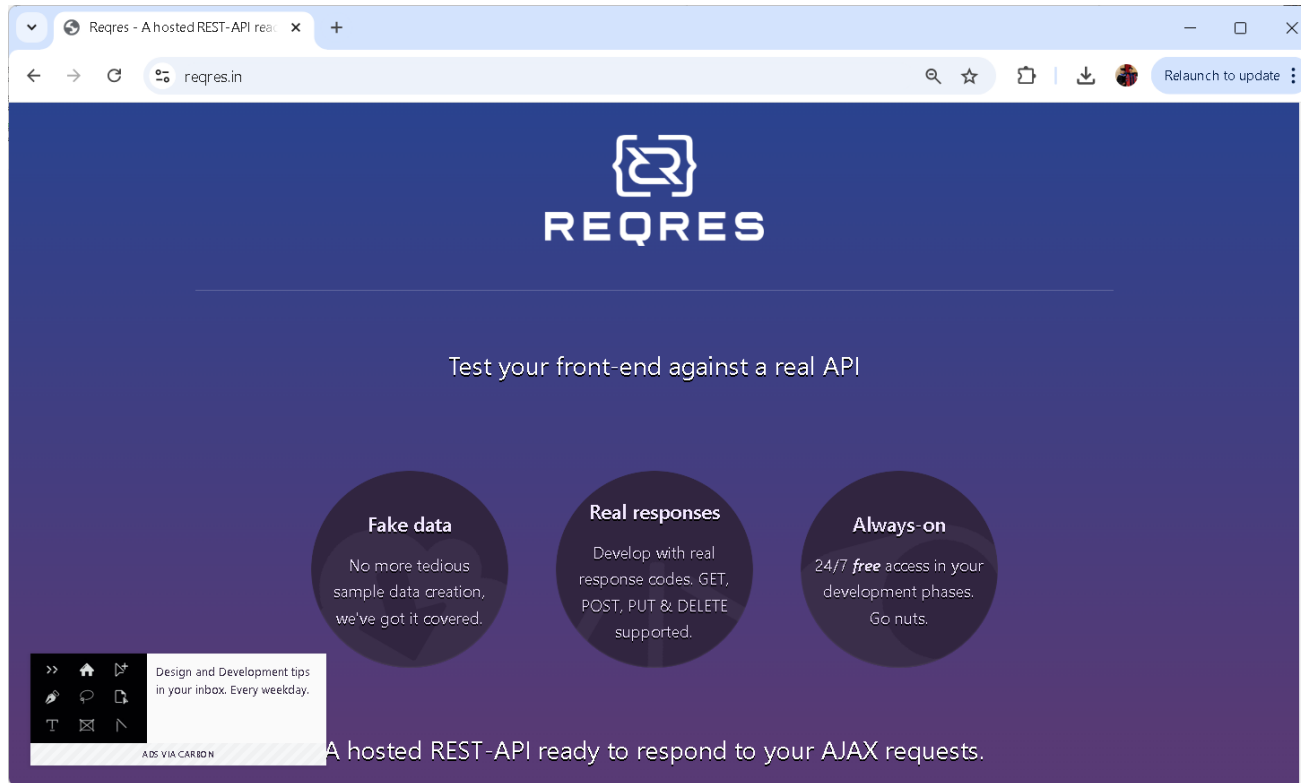
Setiap kali **back-end developer** memberikan URL Endpoint API yang mereka buat untuk front-end, tugas dari **seorang front-end developer** yaitu memastikan setiap **jenis request**, **URL endpoint API** yang ingin diuji dan **data parameter** yang akan di request ke server sesuai atau tidak.



REST API Demo

(Login menggunakan URL Endpoint API)

Link Reqres: <https://reqres.in/>

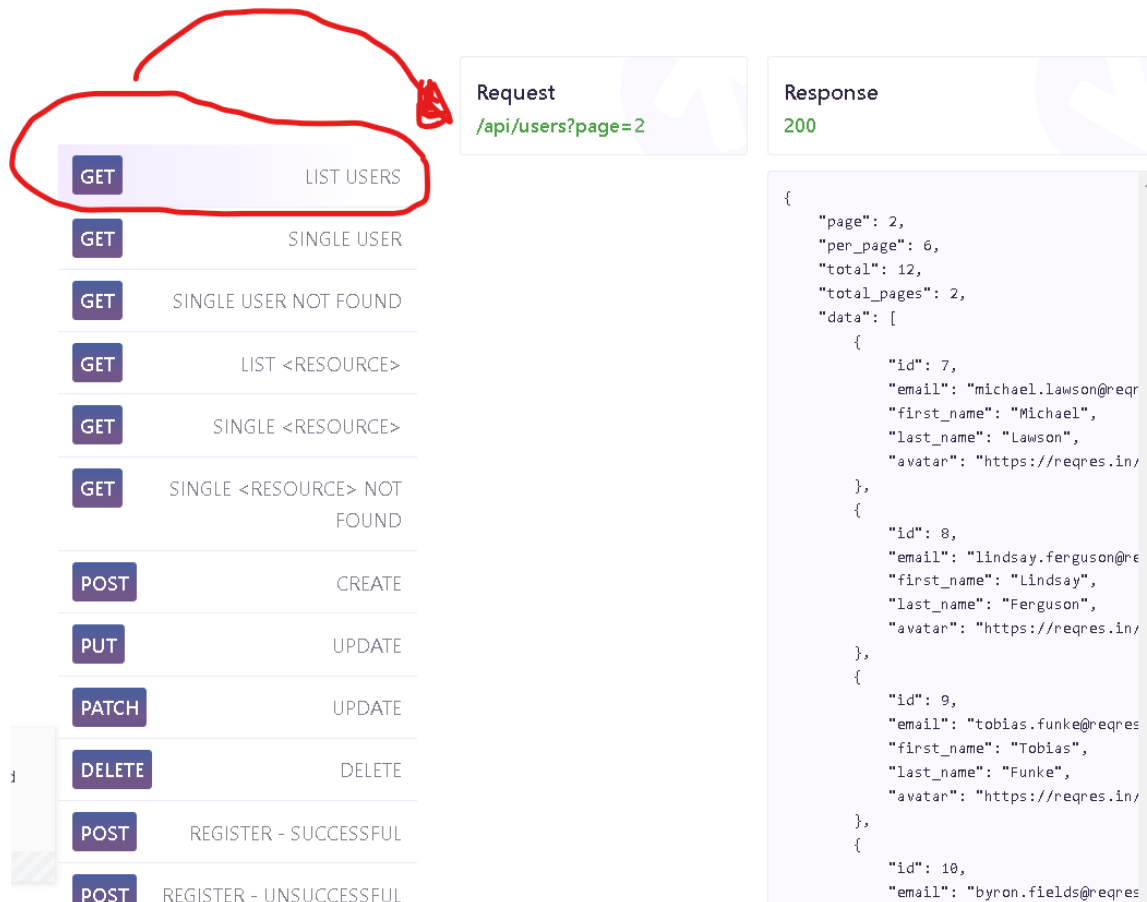


Reqres is a hosted REST-API service designed for testing and prototyping APIs. It allows developers to simulate and test AJAX requests by providing pre-configured API endpoints that return static data. This is particularly useful for front-end developers who want to practice making API calls without needing to build a back-end server.

URL Endpoint API *for* Login

Request URL Endpoint API: <https://reqres.in/api/users?page=2>

Parameter request: page



The screenshot displays the ReqRes API Explorer interface. The 'LIST USERS' endpoint is selected, and the 'GET' method is highlighted with a red circle. A red arrow points from the circle to the 'Request' box. The 'Request' box shows the URL `/api/users?page=2`. The 'Response' box shows a 200 status code and a JSON response.

Method	Endpoint
GET	LIST USERS
GET	SINGLE USER
GET	SINGLE USER NOT FOUND
GET	LIST <RESOURCE>
GET	SINGLE <RESOURCE>
GET	SINGLE <RESOURCE> NOT FOUND
POST	CREATE
PUT	UPDATE
PATCH	UPDATE
DELETE	DELETE
POST	REGISTER - SUCCESSFUL
POST	REGISTER - UNSUCCESSFUL

Request
`/api/users?page=2`

Response
200

```
{
  "page": 2,
  "per_page": 6,
  "total": 12,
  "total_pages": 2,
  "data": [
    {
      "id": 7,
      "email": "michael.lawson@reqres.in",
      "first_name": "Michael",
      "last_name": "Lawson",
      "avatar": "https://reqres.in/img/avatar/7.jpg"
    },
    {
      "id": 8,
      "email": "lindsay.ferguson@reqres.in",
      "first_name": "Lindsay",
      "last_name": "Ferguson",
      "avatar": "https://reqres.in/img/avatar/8.jpg"
    },
    {
      "id": 9,
      "email": "tobias.funke@reqres.in",
      "first_name": "Tobias",
      "last_name": "Funke",
      "avatar": "https://reqres.in/img/avatar/9.jpg"
    },
    {
      "id": 10,
      "email": "byron.fields@reqres.in",
      "first_name": "Byron",
      "last_name": "Fields",
      "avatar": "https://reqres.in/img/avatar/10.jpg"
    }
  ]
}
```

View Data *from* Postman

URL Endpoint API

Jenis Request

**Query
Parameter**

GET https://reqres.in/api/users?page=2

Params • Authorization Headers (10) Body • Pre-request Script Tests Settings Cookies

Query Params

☒ Key	Value	Description	...	Bulk Edit
☒ page	2			
Key	Value	Description		

Body Cookies Headers (18) Test Results

Status: 200 OK Time: 541 ms Size: 1.92 KB Save as example

Pretty Raw Preview Visualize JSON

```
4 {
5   "total_pages": 2,
6   "data": [
7     {
8       "id": 7,
9       "email": "michael.lawson@reqres.in",
10      "first_name": "Michael",
11      "last_name": "Lawson",
12      "avatar": "https://reqres.in/img/faces/7-image.jpg"
13    },
14    {
15      "id": 8,
16      "email": "lindsay.ferguson@reqres.in",
```

URL Endpoint API *for* Login

Request URL Endpoint API: <https://reqres.in/api/login>

Parameter request: email & password

The screenshot displays the Reqres REST API client interface. A red circle highlights the 'LOGIN - SUCCESSFUL' endpoint in the left sidebar. A red arrow points from this endpoint to the 'Request' tab, which shows the following JSON payload:

```
{  "email": "eve.holt@reqres",  "password": "cityslicka"}
```

Another red arrow points from the 'Response' tab to the 'token' field in the JSON output:

```
{  "token": "QpwL5tke4Pnpja7X4"}
```

The status code for the response is 200. At the bottom left, there is a banner for Namecheap: 'Build your website for just \$3.88/mth. More value and performance with Namecheap.'

Jika success, maka output response yang diberikan berupa key session atau authentication key.

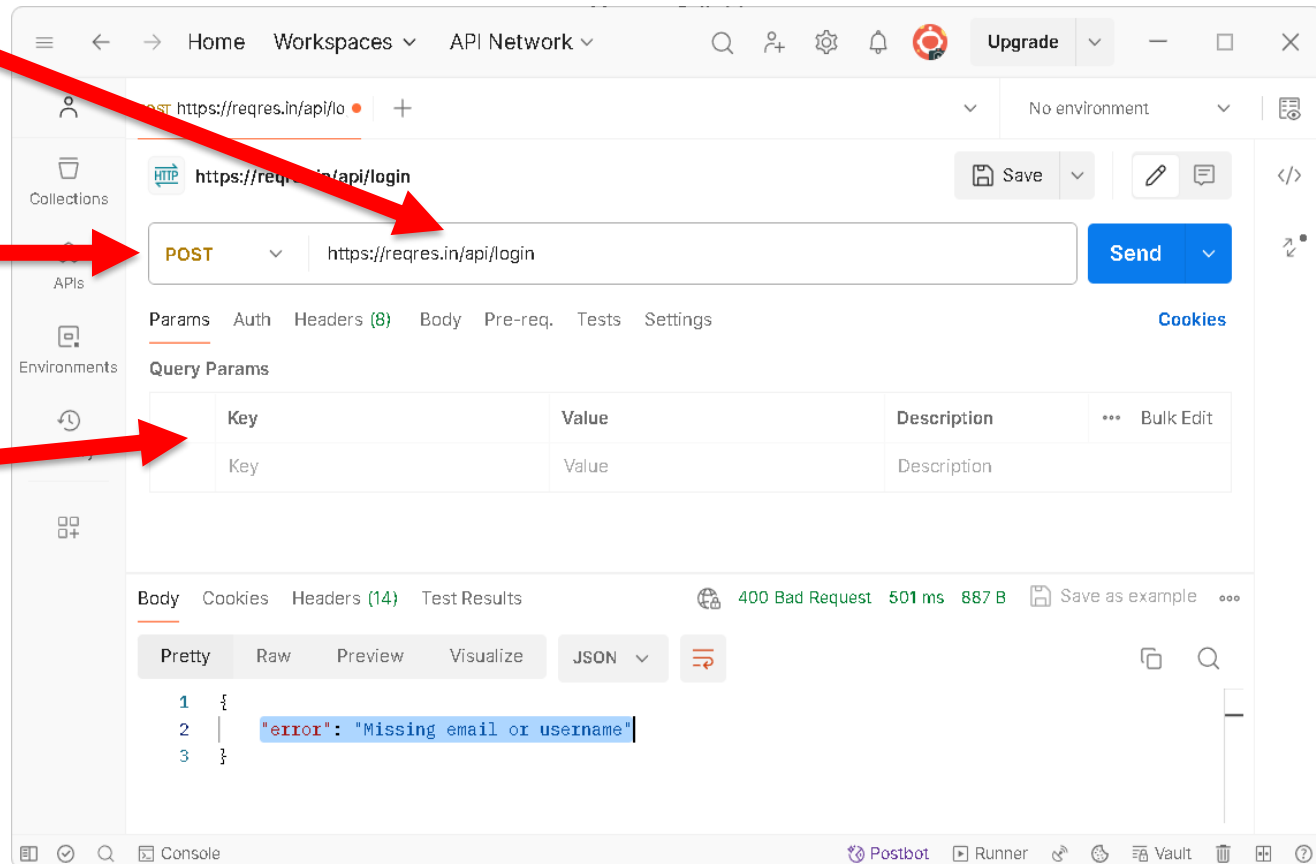
Login Postman Failed

Pilih jenis request, masukan URL endpoint API dan specify query parameter data yang akan di request ke server. Perhatikan bawah, request di bawah tidak ada email dan password yang diinputkan.

URL Endpoint API

Jenis Request

Query Parameter



Login Postman Failed

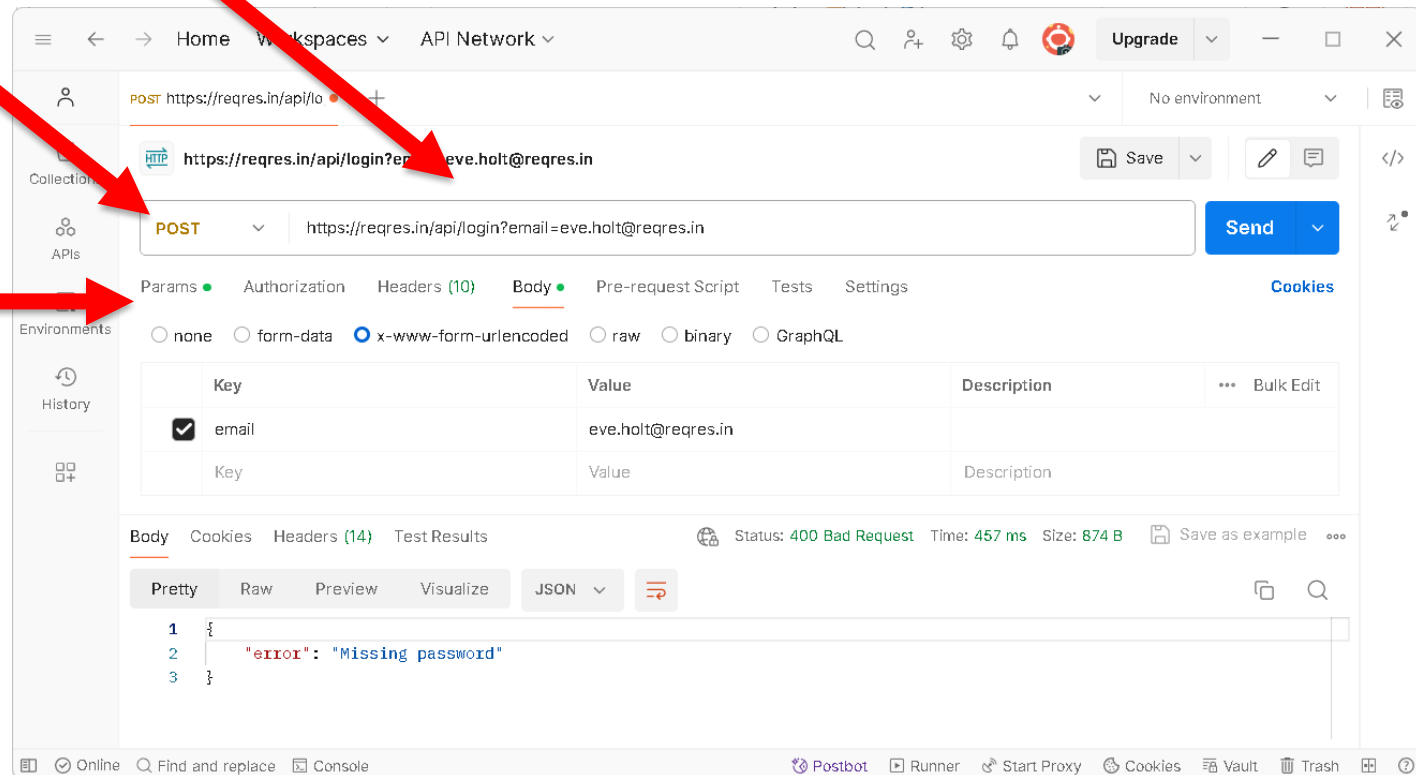
Pilih jenis request, masukan URL endpoint API dan specify query parameter data yang akan di request ke server. Perhatikan bawah, request di bawah tidak ada password yang diinputkan, hanya email saja.

URL Endpoint API

Jenis Request

Query Parameter

Di bagian query parameter pilih “**Body**”, selanjutnya “**x-www-form-urlencoded**”.



Login Postman Success

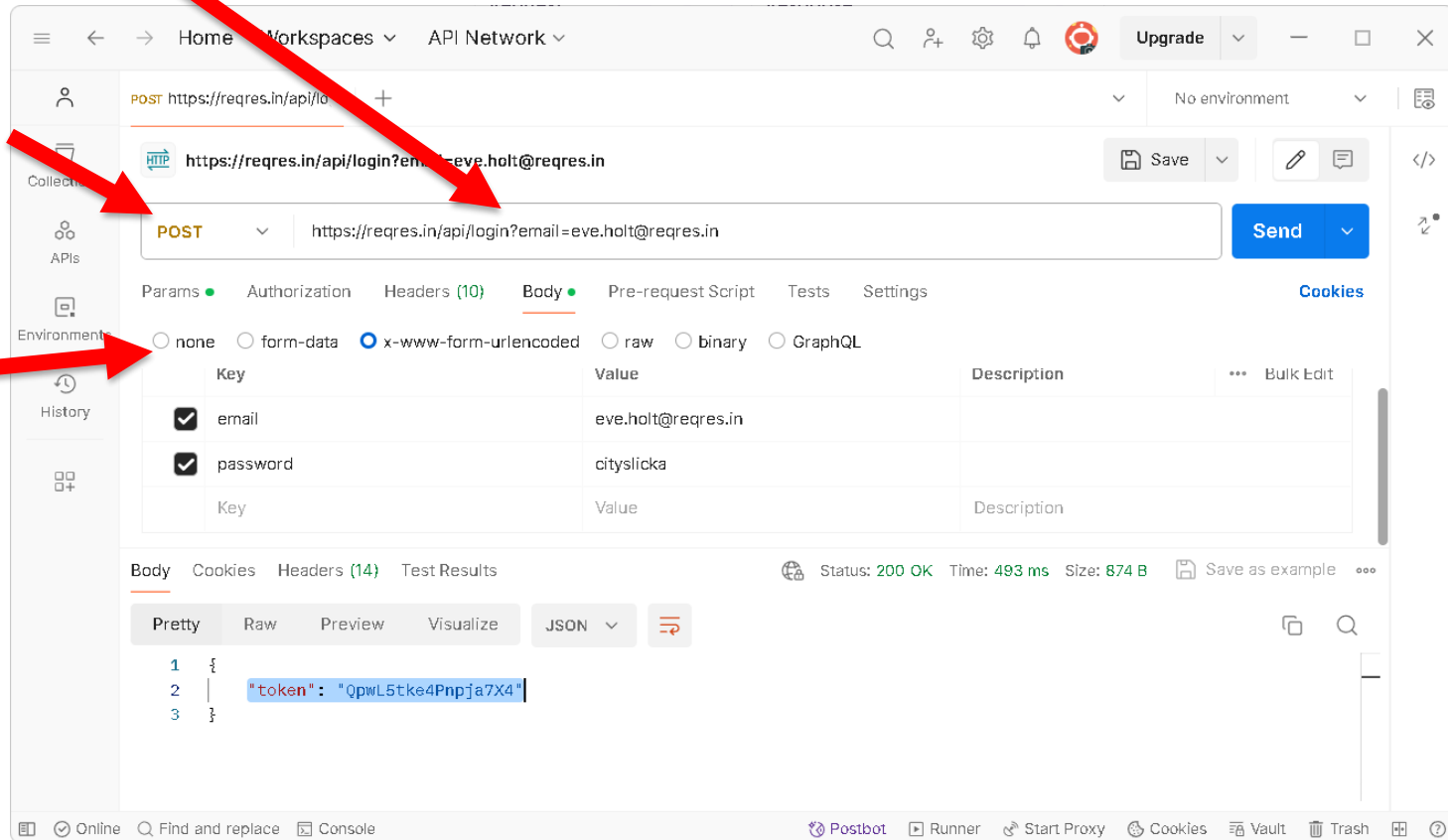
Pilih jenis request, masukan URL endpoint API dan specify query parameter data yang akan di request ke server. Perhatikan bawah, request di bawah sudah terdapat email dan password yang diinputkan.

URL Endpoint API

Jenis Request

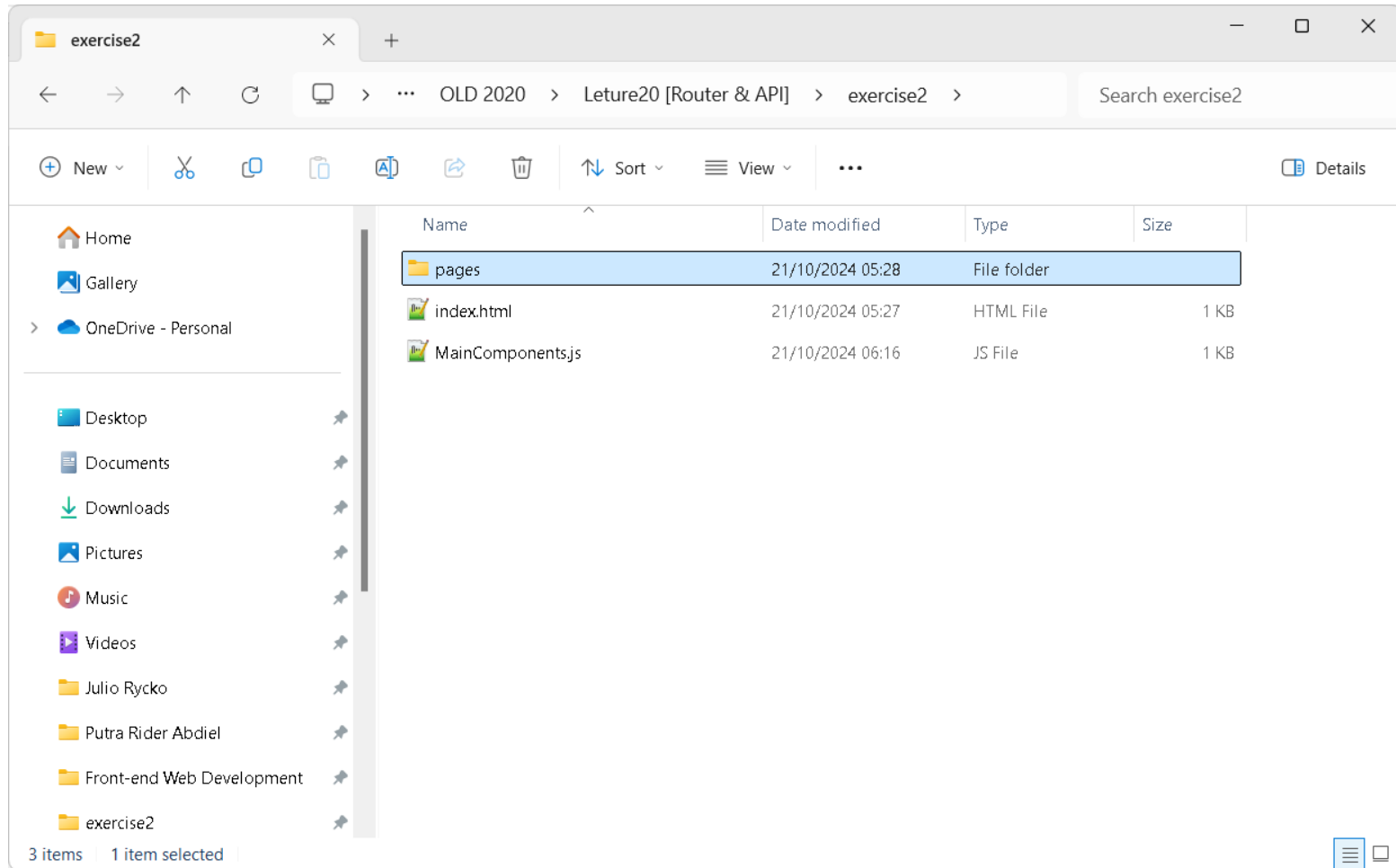
Query Parameter

Di bagian query parameter pilih “Body”, selanjutnya “x-www-form-urlencoded”.

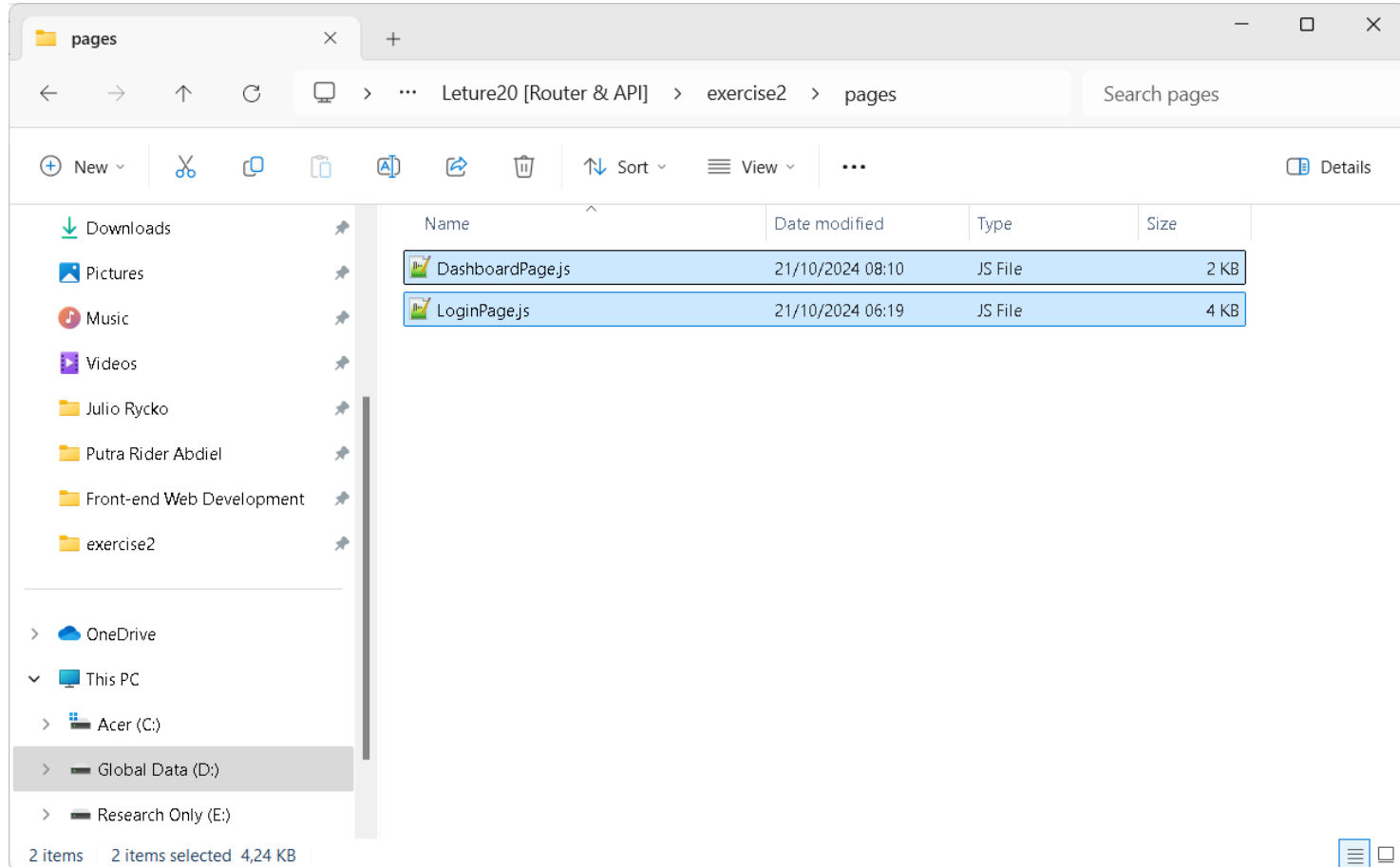


Bagaimana cara untuk
request URL Endpoint API
dari ReactJS?

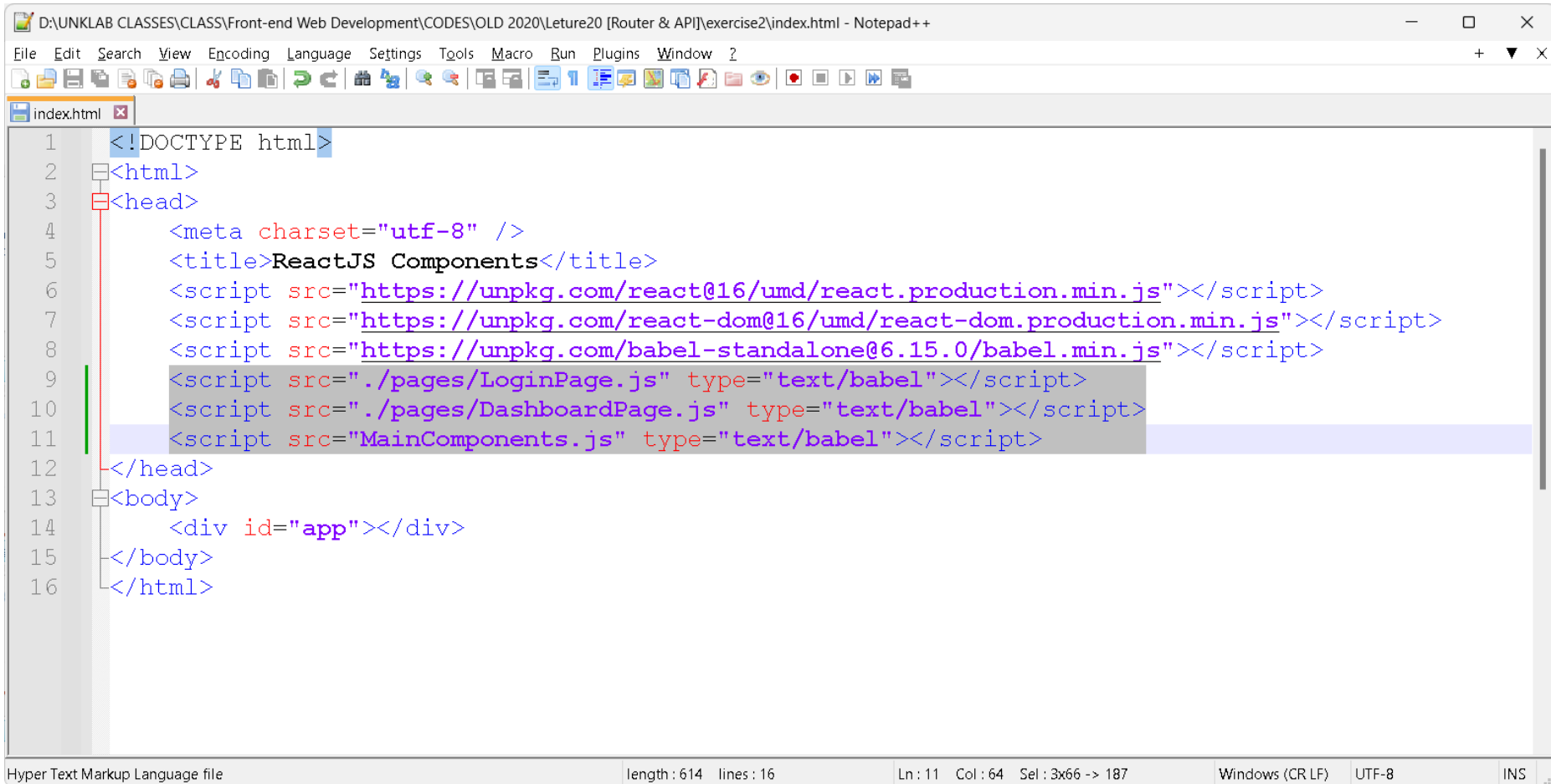
Working Directory



Login & Dashboard Pages



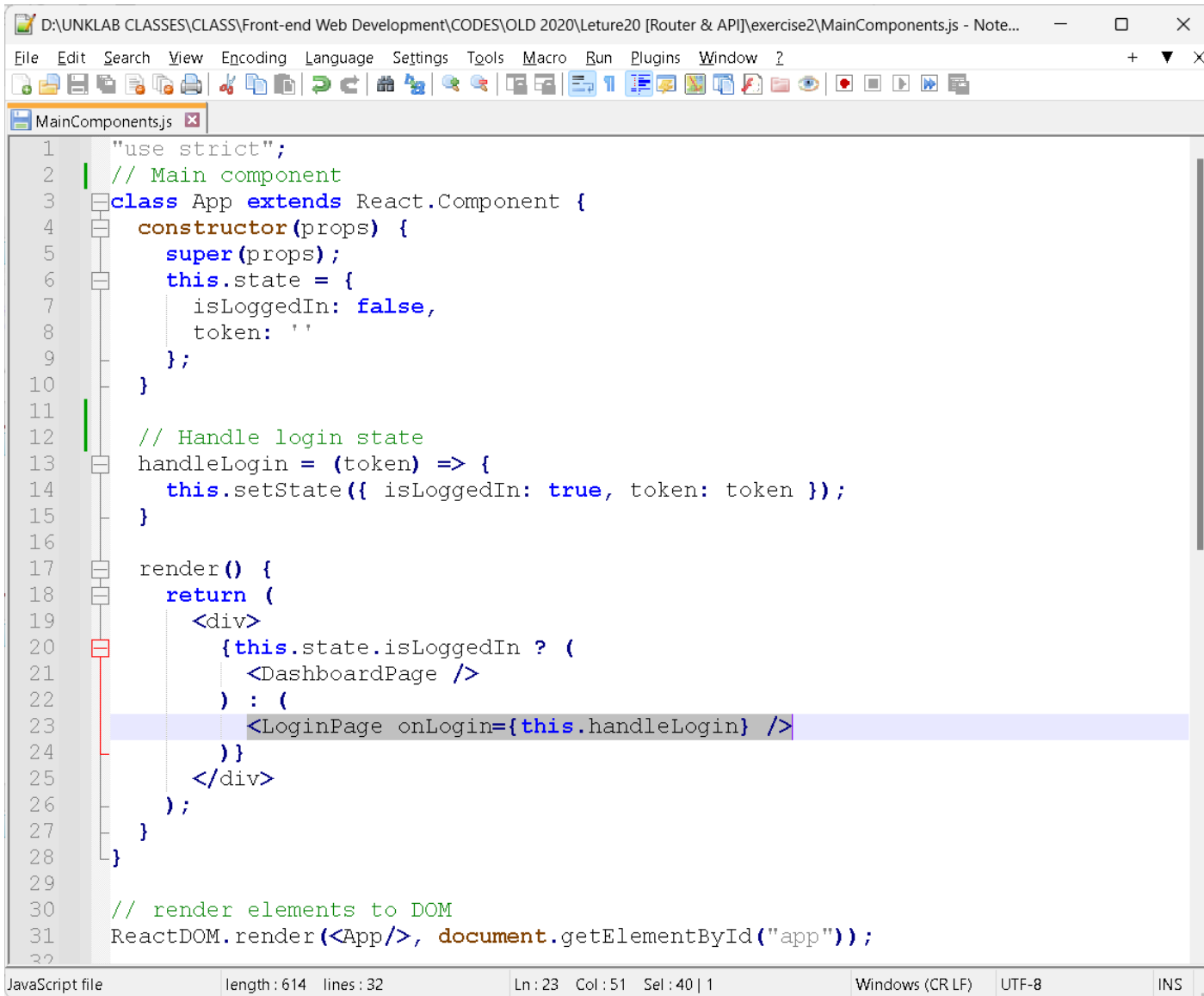
File index.html



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="utf-8" />
5   <title>ReactJS Components</title>
6   <script src="https://unpkg.com/react@16/umd/react.production.min.js"></script>
7   <script src="https://unpkg.com/react-dom@16/umd/react-dom.production.min.js"></script>
8   <script src="https://unpkg.com/babel-standalone@6.15.0/babel.min.js"></script>
9   <script src="./pages/LoginPage.js" type="text/babel"></script>
10  <script src="./pages/DashboardPage.js" type="text/babel"></script>
11  <script src="MainComponents.js" type="text/babel"></script>
12 </head>
13 <body>
14   <div id="app"></div>
15 </body>
16 </html>
```

Hyper Text Markup Language file length : 614 lines : 16 Ln : 11 Col : 64 Sel : 3x66 -> 187 Windows (CR LF) UTF-8 INS

File MainComponents.js



```
1  "use strict";
2  // Main component
3  class App extends React.Component {
4    constructor(props) {
5      super(props);
6      this.state = {
7        isLoggedIn: false,
8        token: ''
9      };
10   }
11
12   // Handle login state
13   handleLogin = (token) => {
14     this.setState({ isLoggedIn: true, token: token });
15   }
16
17   render() {
18     return (
19       <div>
20         {this.state.isLoggedIn ? (
21           <DashboardPage />
22         ) : (
23           <LoginPage onLogin={this.handleLogin} />
24         )}
25       </div>
26     );
27   }
28 }
29
30 // render elements to DOM
31 ReactDOM.render(<App />, document.getElementById("app"));
```

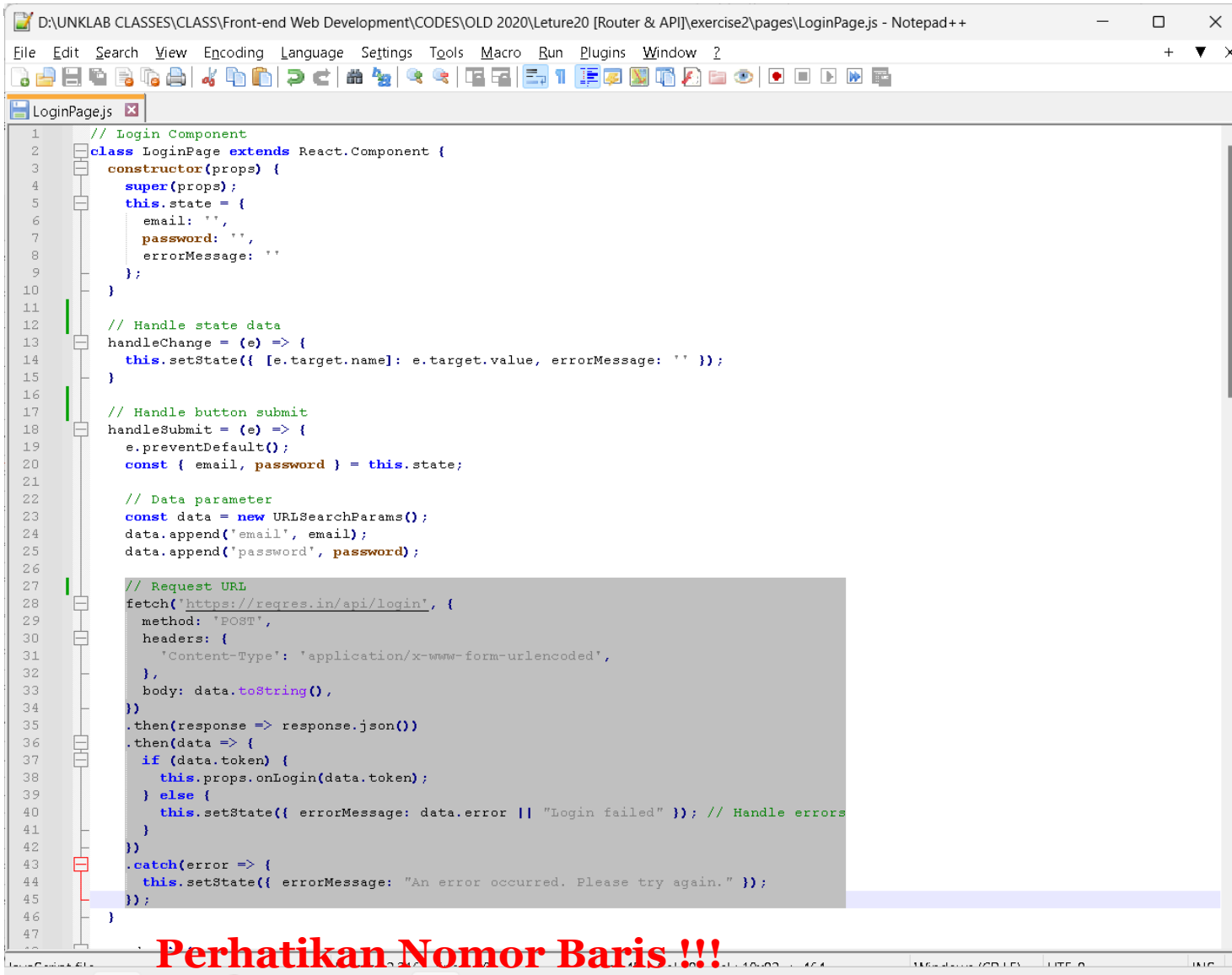
JavaScript file length : 614 lines : 32 Ln : 23 Col : 51 Sel : 40 | 1 Windows (CR LF) UTF-8 INS

File Dashboard.js

```
D:\UNKLAB CLASSES\CLASS\Front-end Web Development\CODS\OLD 2020\Lecture20 [Router & API]\exercise2\pages\DashboardPage.js - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
DashboardPage.js
1 // Dashboard component
2 class DashboardPage extends React.Component {
3   render() {
4     return (
5       <div style={{ display: "flex", flexDirection: "column", minHeight: "100vh", backgroundColor: "#f0f0f0" }}>
6         <header style={{ backgroundColor: "#4CAF50", padding: "10px", color: "white", textAlign: "center" }}>
7           <h1>Front End Web Development</h1>
8         </header>
9         <div style={{ flex: "1", display: "flex" }}>
10           <aside style={{ width: "200px", backgroundColor: "#ccc", padding: "10px" }}>
11             <h2>Sidebar</h2>
12             <ul>
13               <li>Home</li>
14               <li>Statistics</li>
15               <li>Profile</li>
16               <li>Projects</li>
17               <li>About</li>
18               <li>Help</li>
19               <li>Contacts</li>
20             </ul>
21           </aside>
22           <main style={{ flex: "1", padding: "20px" }}>
23             <h2>Welcome to the Dashboard!</h2>
24           </main>
25         </div>
26         <footer style={{ backgroundColor: "#4CAF50", padding: "10px", color: "white", textAlign: "center" }}>
27           <p>2024@Universitas Klabat</p>
28         </footer>
29       </div>
30     );
31   }
32 }
33
```

JavaScript file length: 1.181 lines: 33 Ln: 20 Col: 18 Sel: 9x21 -> 132 Windows (CR LF) UTF-8 INS

File LoginPage.js



```
1 // Login Component
2 class LoginPage extends React.Component {
3   constructor(props) {
4     super(props);
5     this.state = {
6       email: '',
7       password: '',
8       errorMessage: ''
9     };
10  }
11
12  // Handle state data
13  handleChange = (e) => {
14    this.setState({ [e.target.name]: e.target.value, errorMessage: '' });
15  }
16
17  // Handle button submit
18  handleSubmit = (e) => {
19    e.preventDefault();
20    const { email, password } = this.state;
21
22    // Data parameter
23    const data = new URLSearchParams();
24    data.append('email', email);
25    data.append('password', password);
26
27    // Request URL
28    fetch('https://regres.in/api/login', {
29      method: 'POST',
30      headers: {
31        'Content-Type': 'application/x-www-form-urlencoded',
32      },
33      body: data.toString(),
34    })
35      .then(response => response.json())
36      .then(data => {
37        if (data.token) {
38          this.props.onLogin(data.token);
39        } else {
40          this.setState({ errorMessage: data.error || "Login failed" }); // Handle errors
41        }
42      })
43      .catch(error => {
44        this.setState({ errorMessage: "An error occurred. Please try again." });
45      });
46  }
47 }
```

Perhatikan Nomor Baris !!!

File LoginPage.js

```
D:\UNKLAB CLASSES\CLASS\Front-end Web Development\CODS\OLD 2020\Lecture 20 [Router & API]\exercise2\pages\LoginPage.js - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
LoginPage.js
46 }
47
48 render() {
49   return (
50     <div style={{ display: "flex", justifyContent: "center", alignItems: "center", height: "100vh", backgroundColor: "#f0f0f0" }}>
51       <table style={{ border: "1px solid #ccc", backgroundColor: "rgba(255, 255, 255, 0.9)", borderRadius: "8px",
52         padding: "20px", width: "300px" }}>
53         <tbody>
54           <tr>
55             <td colspan="2" style={{ textAlign: "center" }}>
56               <h1>Login !!!</h1>
57             </td>
58           </tr>
59           <tr>
60             <td style={{ padding: "10px" }}>
61               <label style={{ width: "100%", display: "block" }}>Email:</label>
62             </td>
63             <td style={{ padding: "10px" }}>
64               <input type="email" name="email" value={this.state.email} onChange={this.handleChange}
65                 style={{ width: "100%" }}
66             </td>
67           </tr>
68           <tr>
69             <td style={{ padding: "10px" }}>
70               <label style={{ width: "100%", display: "block" }}>Password:</label>
71             </td>
72             <td style={{ padding: "10px" }}>
73               <input type="password" name="password" value={this.state.password} onChange={this.handleChange}
74                 style={{ width: "100%" }}
75             </td>
76           </tr>
77           <tr>
78             <td colspan="2" style={{ padding: "10px" }}>
79               <button type="submit" onClick={this.handleSubmit} style={{ width: "100%" }}>
80                 Login Student
81               </button>
82             </td>
83           </tr>
84           <tr>
85             <td colspan="2" style={{ padding: "10px" }}>
86               {this.state.errorMessage && ( // Display error message
87                 <div style={{ padding: "10px", color: "red", textAlign: "center" }}>
88                   {this.state.errorMessage}
89                 </div>
90               )}
91             </td>
92           </tr>
93         </tbody>
94       </table>
95     </div>
96   );
97 }
```

Perhatikan Nomor Baris !!!

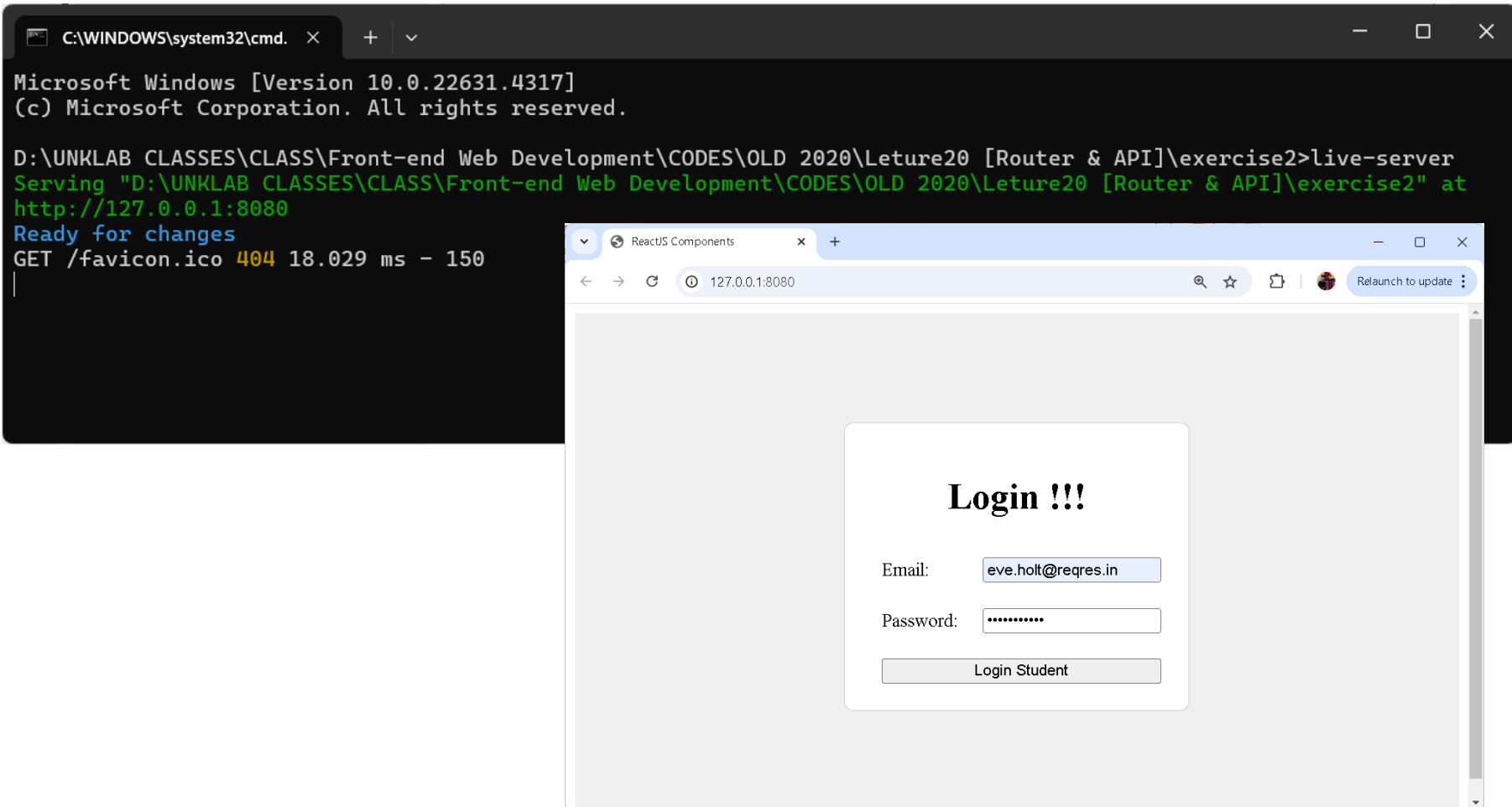
File LoginPage.js

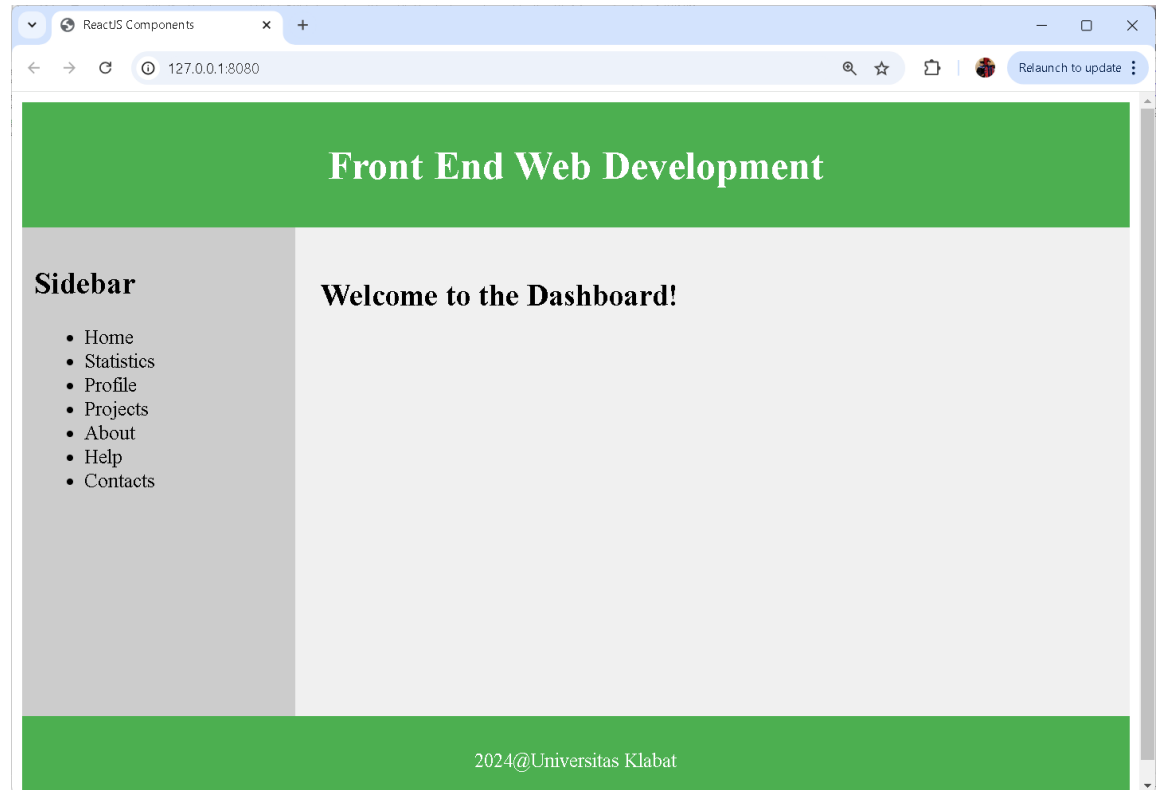
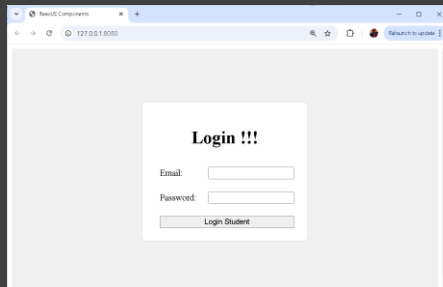
```
D:\UNKLAB CLASSES\CLASS\Front-end Web Development\CODES\OLD 2020\Lecture20 [Router & API]\exercise2\pages\LoginPage.js - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
LoginPage.js
53 <tbody>
54 <tr>
55 <td colspan="2" style={{ textAlign: "center" }}>
56 <h1>Login !!!</h1>
57 </td>
58 </tr>
59 <tr>
60 <td style={{ padding: "10px" }}>
61 <label style={{ width: "100%", display: "block" }}>Email:</label>
62 </td>
63 <td style={{ padding: "10px" }}>
64 <input type="email" name="email" value={this.state.email} onChange={this.handleChange}
65 style={{ width: "100%" }}
66 </td>
67 </tr>
68 <tr>
69 <td style={{ padding: "10px" }}>
70 <label style={{ width: "100%", display: "block" }}>Password:</label>
71 </td>
72 <td style={{ padding: "10px" }}>
73 <input type="password" name="password" value={this.state.password} onChange={this.handleChange}
74 style={{ width: "100%" }}
75 </td>
76 </tr>
77 <tr>
78 <td colspan="2" style={{ padding: "10px" }}>
79 <button type="submit" onClick={this.handleSubmit} style={{ width: "100%" }}>
80 Login Student
81 </button>
82 </td>
83 </tr>
84 <tr>
85 <td colspan="2" style={{ padding: "10px" }}>
86 {this.state.errorMessage && ( // Display error message
87 <tr>
88 <td colspan="2" style={{ padding: "10px", color: "red", textAlign: "center" }}>
89 {this.state.errorMessage}
90 </td>
91 </tr>
92 </td>
93 </tbody>
94 </table>
95 </div>
96 );
97 }
98 }
99
```

File yang sama, perhatikan Nomor Baris !!!

Expected Output

- *live-server*
- `live-server --port=8080`
- ~~`live-server index.html`~~







Your Turn !!!

Student Instructions

Students diharapkan dapat mengikuti *step by step* cara instalasi dan konfigurasi cloud server (*virtual server*) di atas. Adapun demo dan penilaian akan meliputi beberapa penguasaan *skill* dibawah ini:

1. **(20%)** Mampu melakukan instalasi dan konfigurasi (OS server, apache, PHP, .htaccess, MySQL).
2. **(10%)** Mampu membuat *table* (*tbl_lecturer*) di dalam *database* (*db_university*), serta melakukan *insert* data ke dalam *table* yang sesuai dengan simulasi database yang dilakukan di dalam class.
3. **(10%)** Mampu untuk mengakses *cloud server* (*virtual server*) melalui **Virtual Console** di IDCloudHost panel?
4. **(10%)** Mampu untuk mengakses *cloud server* (*virtual server*) melalui **Putty**?
5. **(10%)** Mampu untuk mengakses *cloud server* (*virtual server*) melalui **WinSCP**?
6. **(10%)** Mampu untuk mengakses *cloud server* (*virtual server*) melalui **CMD**?
7. **(20 %)** Mampu lakukan *remote server* jarak jauh dan melakukan edit/update pada **source code PHP** di server menggunakan salah satu text editor (nano/vim).
8. **(10%)** Demo Rest API menggunakan Postman.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

